

Sparse Optimization using Multiple Shooting and Guess Propagation

Revised manuscript with tightened error bounds and a numerical results section
Report R004 ([reports/2026-09-11-R004-revised-manuscript/](#))

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Abstract

We study the discovery of sparse polynomial governing equations $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}; p)$ from noisy trajectory samples by fitting the parameters p of a candidate library through simulation. Fitting by a single forward simulation from the first datum (single shooting) produces a cost landscape whose local minima and blow-up regions multiply with the length of the record. We instead restart the simulation at shooting nodes placed on the data (multiple shooting without continuity constraints) and *propagate the guess*: the minimiser found with short windows becomes the starting point for longer windows, ending with a full single-shooting fit. The analysis bounds the sensitivity of the flow to the initial state and to the parameters, the cost at the true parameter, the change of the cost when shooting nodes are removed, and the resulting displacement of the minimiser. This revision replaces the Lipschitz constant L of the earlier draft by the logarithmic norm $\mu \leq L$ throughout, removes an attractor-size constant from the node-removal bound so that it is quadratic in the small quantities, states the noise terms in expectation, corrects a missing per-window count, and adds a trajectory-dependent refinement in a weighted norm. Each change is local to one proof step, and the assumptions under which each statement holds are now stated in full: a fixed admissible region, a closure condition that confines every admissible trajectory to a tube around the true orbit, a noise model with the event on which the deterministic bounds hold, and a constrained optimality condition in place of a vanishing gradient. A numerical section on the FitzHugh–Nagumo system compares every bound with measurement. For the amplification of an initial-state error the draft’s a priori factor is $e^{L\Delta T} = 1.8 \times 10^{13}$ at $\Delta T = 10$ and the revised one is $e^{\mu\Delta T} = 1.2 \times 10^5$; along the true orbit the *linearised endpoint* per-window factors are 34.8 at $\Delta T = 5$ and 139 at $\Delta T = 10$ in the Euclidean norm, and 8.04 at $\Delta T = 5$ in the weighted norm $\|Dx\|$, which corresponds to the Euclidean statement $8.04 \text{ cond}(D) = 28.5$; the largest amplification *observed* over the 24 probes of Figure 7 is 8.5. The section also maps the cost landscape as the window grows (local minima per slice from 1.25 to 7.9), and reports the headline experiment over 8 optimiser seeds: with guess propagation the median parameter error at full single shooting is 0.32 against 1.65 without it (paired difference -1.14 , 95 % bootstrap CI $[-1.43, -0.77]$, better in 7 of 8 seeds), with the same behaviour on Lotka–Volterra and, with a gradient optimiser, on Lorenz. The three hand-drawn figures of the draft are replaced by figures computed from the data. All numbers are inherited from reports R002 and R003, whose result files are copied here with hashes; ten inheritance gates pass (§8). Two theoretical-premise checks are *not evaluated*: the closure condition (A1) of §2, and the reconnection hypothesis (R) relating the constrained minimisers of the theory to the unconstrained iterates reported in §6. Inserting the indicative constants carried over from the R002 probe box into (A1) produces enormous factors for long windows, which motivates — but does not establish — the infeasibility of a useful admissible region. Every statement of §3–§4 is therefore conditional, at every stage of the schedule.

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1 Multiple shooting and guess propagation: a greedy approach

Problem to solve. Discover the system of governing equations

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}; p), \quad \mathbf{x} \in C^1(\mathbb{R})^n, \quad (1)$$

such that the trajectory matches the observations $\{\mathbf{y}_0, \mathbf{y}_1, \dots, \mathbf{y}_{N-1}\}$ at the time points $\{t_0, t_1, \dots, t_{N-1}\}$. Suppose there is a library of candidate functions $\Theta = \{\Theta_1(\mathbf{x}), \dots, \Theta_J(\mathbf{x})\}$ such

$$\frac{dx_i}{dt} = \sum_{j=1}^J p_{ij} \Theta_j(\mathbf{x}), \quad i = 1, \dots, n, \quad (2)$$

a

--- true trajectory • v samples y_i • w samples y_i

one data interval $\Delta t = 1$

$N + 1 = 101$ samples, $\Delta t = 1$, 5 % relative noise

state

time t

b

grey arrows: vector field of the true FHN model

--- true orbit

• noisy samples

w

v

Data-generating model (FitzHugh-Nagumo)

$$\dot{v} = v - v^3/3 - w + 0.5$$

$$\dot{w} = (v + 0.7 - 0.8w) / 12.5$$

Ansatz $\dot{x}_i = \sum_j p_{ij} \Theta_j(v, w)$, where

$\Theta =$ every monomial in (v, w) of degree ≤ 3
 (10 terms per equation $\Rightarrow$ 20 unknowns p_{ij}).

c

unknown sparse coefficients p (20 entries, 7 non-zero)

$\dot{v}$	0.5	-1			1														
$\dot{w}$	0.056	-0.064			0.08														-0.333
	1	w	w^2	w^3	v	vw	vw^2	v^2	v^2w	v^3									

library monomial $\Theta_j(v, w)$

blank cell = true zero; cell colour = sign of p_{ij} , shaded by $\sqrt{|p_{ij}|}$

Loss functions. Assuming all states are measured, the model-only loss is

3

where $x_m(t; p)$ is the model trajectory. A variant that also accounts for errors in the observations adds a data term,

$$\mathbf{L}(p, \mathbf{x}) = R_1 \frac{1}{N} \sum_{k=0}^{N-1} \|\mathbf{x}_k - \mathbf{y}_k\|^2 + R_2 \frac{1}{N-1} \sum_{k=0}^{N-2} \|\mathbf{x}_k - \mathbf{x}_{m,k}\|^2, \quad (4)$$

which is useful for sparse and noisy data. The analysis below uses (3).

Integration and the difficulty of single shooting. Let δt be the smallest time step of the integration scheme and $F_{\delta t}$ its one-step operator, $x_m(t + \delta t; p) = F_{\delta t}(x_m(t; p))$. Tracing the trajectory from $\mathbf{x}_0 \approx \mathbf{y}_0$ to t_k composes the operator $S = (t_k - t_0)/\delta t$ times, $x_{m,k} = F_{\delta t}^{(S)}(x_0; p)$. Many compositions introduce local minima into the loss landscape, make the gradient unstable, and let the trajectory blow up for unstable parameter values. Figure 2 shows the effect on FHN: a parameter vector only 0.038 away from the truth (in the Euclidean norm on the 20 coefficients) slips phase after two periods and then pays the full size of the attractor in every residual, so the single-shooting cost is 2.60 against 0.0099 at the truth.

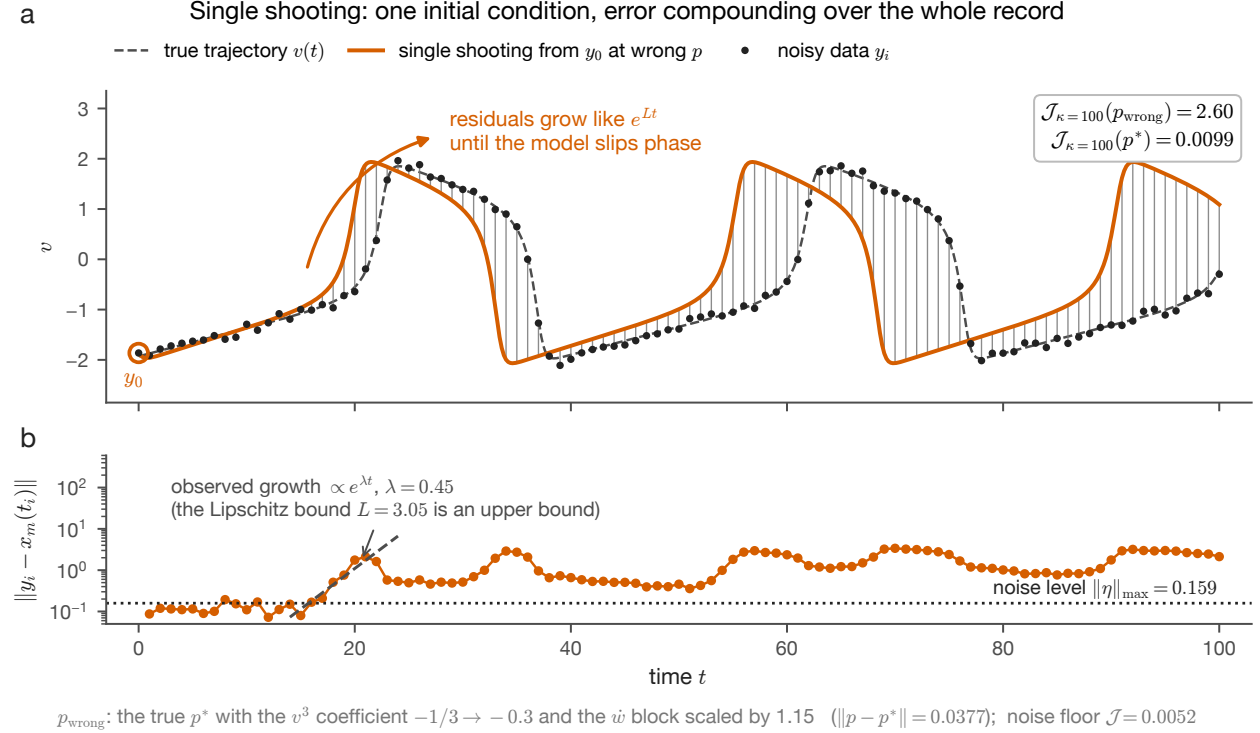


Figure 2: Single shooting. The model is integrated once from the first datum y_0 over the whole record with a wrong parameter p_{wrong} (cubic coefficient -0.30 instead of $-1/3$, w time-scale multiplied by 1.15; $\|p_{\text{wrong}} - p^*\| = 0.038$); thin vertical lines are the residuals. The residual (right, log scale) sits at the noise level until the model slips phase near $t \approx 18$, then grows and saturates at the size of the attractor; the cost is $\mathcal{J}_{100}(p_{\text{wrong}}) = 2.60$ against 0.0099 at p^* . Inherited from R002 Figure 2 (figures/R002_fig02_single_shooting.pdf).

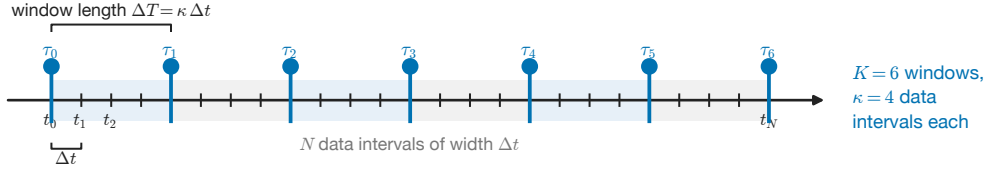
Multiple shooting. One way to tackle this is to restart the trajectory from several points of the record. Partition $[t_0, t_{N-1}]$ into K windows $[\tau_{k-1}, \tau_k]$ with nodes $t_0 = \tau_0 < \tau_1 < \dots < \tau_K = t_{N-1}$;

for simplicity we take $\{\tau_k\} \subset \{t_0, \dots, t_{N-1}\}$ (Figure 3). Each node is the initial condition of the model trajectory on its window, and the cost is

$$J_K(p) = \sum_{k=1}^K \sum_{t_i \in (\tau_{k-1}, \tau_k]} \|x_m(t_i; p, y_{\tau_{k-1}}) - y_i\|^2, \quad x_m(t_i; p, y_{\tau_{k-1}}) = F_{\delta t}^{(S_i)}(y_{\tau_{k-1}}; p). \quad (5)$$

The node states are the noisy data themselves; they are *not* optimisation variables and no continuity is enforced at the window ends (this differs from textbook multiple shooting). If t_i is not an integrator step point the solver interpolates. As the error analysis of §3 shows, restarting resets the propagation of the initial-condition error at every node, so the amplification is paid over one window instead of over the whole record (Figure 4). Our goal is not to replace a solver that starts at the first observation but to provide it with a good initial guess: the parameters found with windows are handed to the full single-shooting problem, which avoids its stability problems and lands near its global minimum.

^a Fine partition: a shooting node τ_k at every κ -th datum; each window restarts from the datum $y_{\tau_{k-1}}$



^b Node removal: delete every other node $\Rightarrow$ half as many windows, each twice as long

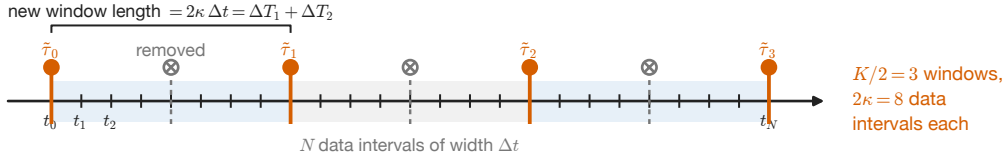


Figure 3: The shooting partition. Top: the data times $t_0, \dots, t_{N-1}$ (ticks, spacing Δt) and the shooting nodes $\tau_0 < \tau_1 < \dots < \tau_K$ placed at every κ -th datum; the windows $[\tau_{k-1}, \tau_k]$ of length $\Delta T = \kappa \Delta t$ are shaded alternately. Bottom: the coarser partition obtained by removing every other node ($K/2$ windows of length $2\Delta T$), the elementary step of guess propagation. Schematic; replaces the hand-drawn `shooting_partition.png` of the draft. Inherited from R002 Figure 3.

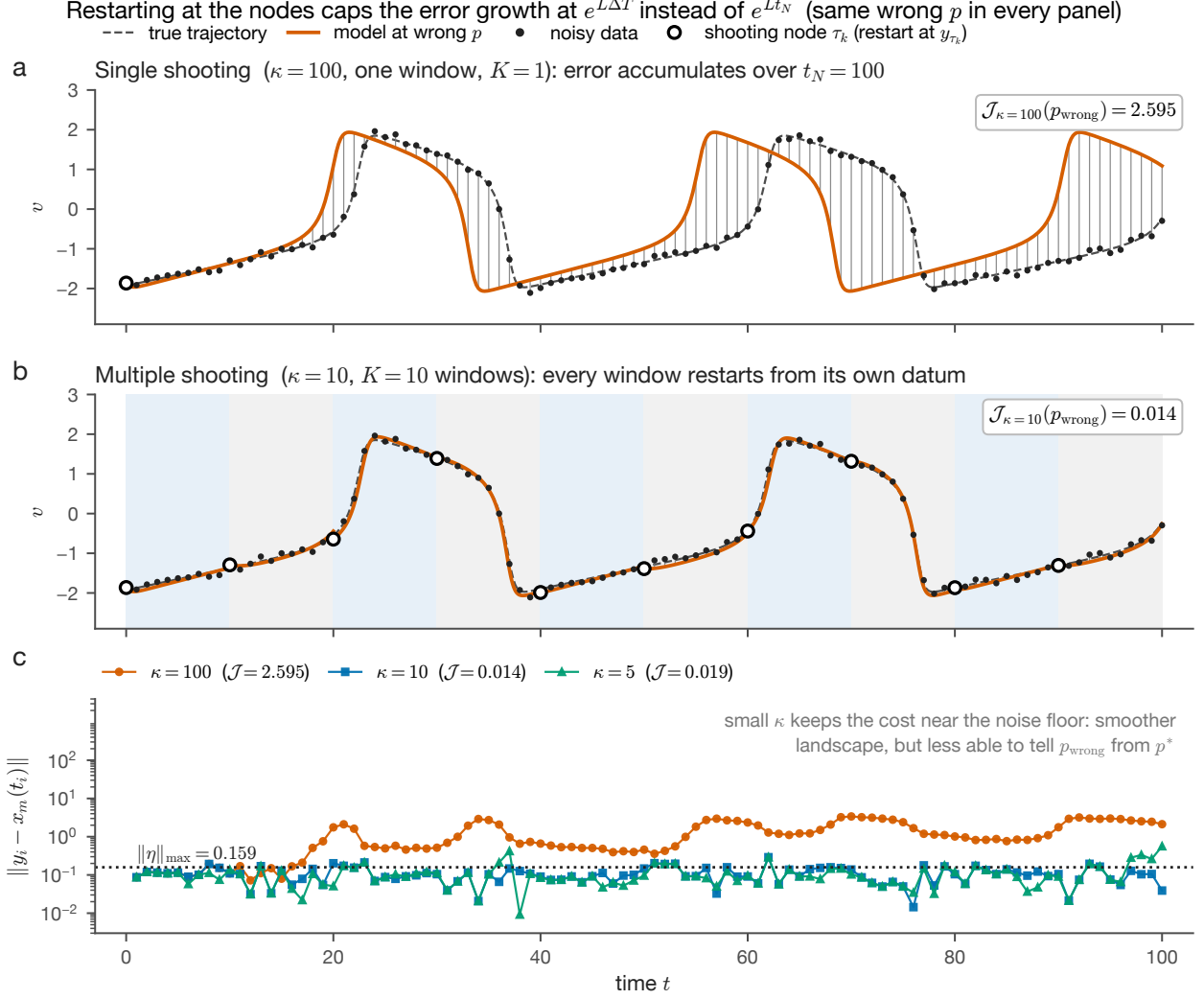


Figure 4: Multiple shooting. The same wrong parameter as Figure 2, integrated (a) as single shooting ($\kappa = 100$, $J = 2.60$) and (b) restarting from the datum at every 10th sample ($\kappa = 10$, $J = 0.0140$); windows shaded, node markers highlighted; (c) the per-datum residual magnitude for $\kappa = 100, 10, 5$. Restarting resets the accumulated error at every node. The flip side, visible in (c): short windows keep even a wrong model near the noise floor, i.e. they give a smooth landscape at the price of less power to discriminate p from p^* , which is why the sweep must end at single shooting. Replaces `multiple_shooting.png`. Inherited from R002 Figure 4.

Guess propagation. Start with many shooting nodes, solve, remove nodes, and restart the optimisation from the previous optimum (Algorithm 1, Figure 5). With one window per data interval and $\delta t = \Delta t$ the problem is close to a finite-difference (SINDy-type) regression and nearly convex; as nodes are removed the model is enforced over longer and longer horizons and the landscape acquires local minima, but the optimiser is already inside the right basin. In the later sections we show that, under the assumptions collected in §2, the optimum moves by a bounded amount at each node removal.

Algorithm 1 Multiple shooting with guess propagation

- 1: $p_{\text{guess}} \leftarrow \text{random}$
 - 2: **for** K in a decreasing schedule of numbers of windows (window size κ increasing, ending at single shooting) **do**
 - 3: $p^{(K)} \leftarrow \arg \min_p J_K(p)$ started from p_{guess}
 - 4: $p_{\text{guess}} \leftarrow p^{(K)}$
 - 5: **end for**
 - 6: **return** $p^{(1)}$ (the single-shooting fit)
-

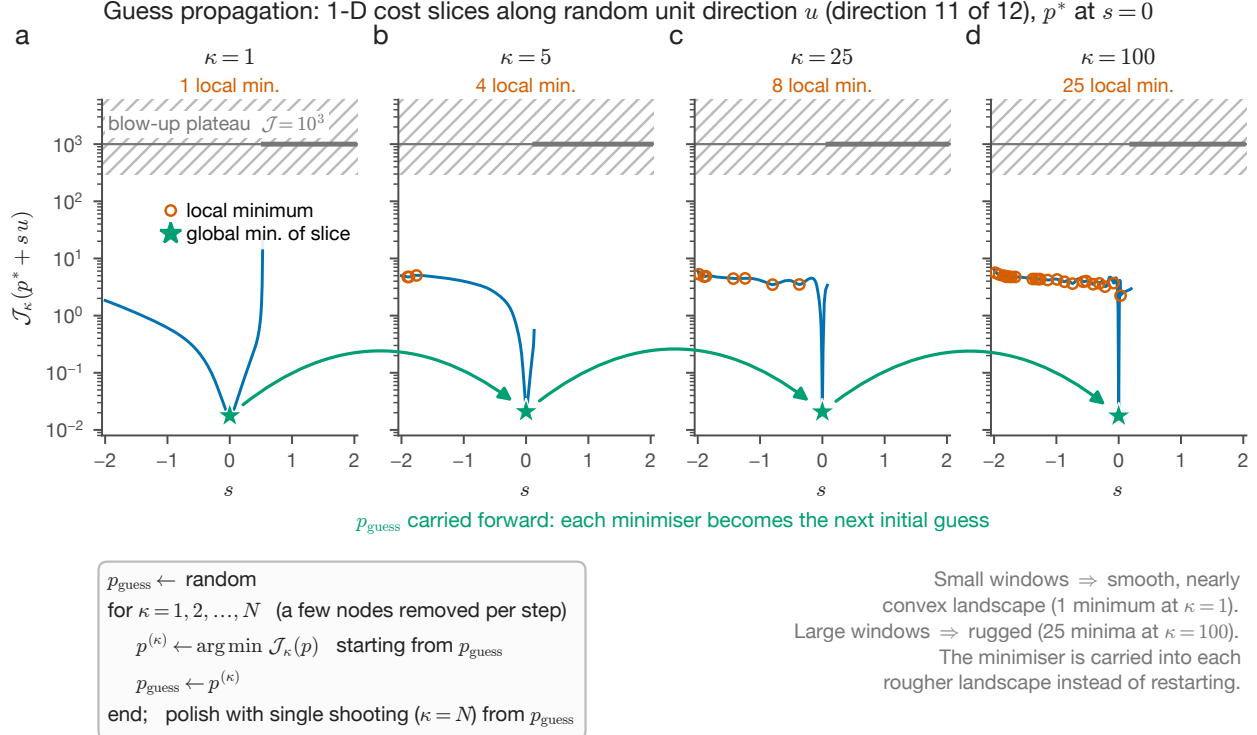


Figure 5: The guess-propagation algorithm. One-dimensional slices $J_{\kappa}(p^* + s u)$ along one random unit direction u (the one with the most local minima at $\kappa = 100$: 1, 4, 8, 25 local minima at $\kappa = 1, 5, 25, 100$, circled) on a log scale; hatched band = blow-up plateau; star = global minimum at $s = 0$. Small windows give a smooth, nearly convex landscape; large windows add local minima and a wide plateau. The algorithm solves the $\kappa = 1$ problem from a random start and carries each minimiser (arrow) into the next, more rugged landscape, ending with a single-shooting polish. Inherited from R002 Figure 6.

2 Notation and definitions

Everything used in the statements, captions and tables is defined here, including what is inherited from the earlier draft and from reports R002 and R003; the provenance is given with each item. Nothing below is decorative: the assumption labels (A0)–(A3), (B0)–(B2), (C1)–(C2), (R) and the scope tags (S1)–(S4) are cited by name in every statement of §3–§5.

2.1 Model, flow, admissible region, data and costs

Consider $\dot{x} = f(x; p)$ with $x \in \mathbb{R}^d$ and $p \in \mathbb{R}^{n_p}$ (the earlier draft wrote $p \in \mathbb{R}$; nothing below uses a scalar parameter, and $n_p = 20$ in the numerical section). p^* is the true parameter and $x^*(t)$ the true trajectory. The flow $\varphi_f(t; p, x_0)$ is the solution of the initial-value problem at time t ; it satisfies $\varphi_f(t_1 + t_2; p, x_0) = \varphi_f(t_2; p, \varphi_f(t_1; p, x_0))$. In the analysis the flow replaces the numerical integrator, i.e. discretisation error is ignored. The data are N samples $y_0, \dots, y_{N-1}$ at times $t_i = t_0 + i \Delta t$; the indexing and the noise model are stated in full below.

The labels used below, in one place. Every label is defined where it is introduced; this list exists so that a reader meeting one in a later section can find it.

- **(A0)** regularity and admissible region $(X, P, \mathcal{T}, r_X, \Delta T_{\max})$; **(A1)** closure of the admissible region, with the two necessary conditions (A1-i) and (A1-ii); **(A2)** noise model, in the bounded variant (A2-a) and the Gaussian variant (A2-b), together with the admissible event E_X ; **(A3)** the *uniform random strong-convexity assumption* of Proposition 2, under which the realisation-wise displacement bound may be integrated.
- **(S1)–(S4)** the four scopes of statement (local deterministic flow estimates; confinement; path-wise noisy-cost statements on E_X ; expectations). Every theorem environment in §3–§4 opens with its tag.
- **(B0)–(B2)** the hypotheses of the basin-retention Corollary 5; **(C1)–(C2)** the hypotheses of the penalty-bias bound of Remark 3(ii); **(R)** the reconnection hypothesis of Remark 1, which is what would relate the constrained minimisers of the theory to the unconstrained iterates of §6.

Assumptions (A0)–(A2) are fixed *before the data are seen*; nothing in them is chosen a posteriori, so none of $X, P, L, \tilde{L}, \mu, \mu_-$ is a random object. (A3) and (R), by contrast, are assumptions *about* noise-dependent objects and are flagged as such where they are used.

(A0) Regularity and admissible region. Fix a compact convex parameter set $P \subset \mathbb{R}^{n_p}$ with $p^* \in P$; a tube radius $r_X > 0$; the closed tube around the true orbit

$$\mathcal{T} = \{x \in \mathbb{R}^d : \text{dist}(x, x^*([t_0, t_{N-1}])) \leq r_X\}; \quad (6)$$

a compact *convex* $X \subset \mathbb{R}^d$ with $\mathcal{T} \subseteq X$; and a horizon $\Delta T_{\max} > 0$ at least as large as the longest window of any partition under discussion — *the coarse partitions included*, i.e.

$$\Delta T_{\max} \geq \max_{\text{partitions under discussion}} \max_k (\tau_k - \tau_{k-1}), \quad (7)$$

where “partitions under discussion” means every fine partition, every coarse partition obtained from it by a node removal of §4, and every common refinement of Corollary 6. Assume f is C^1 on an open neighbourhood of $X \times P$, and that for every $(x, p) \in X \times P$ the solution of $\dot{\xi} = f(\xi; p)$, $\xi(0) = x$, exists for as long as it remains in X , up to time $\Delta T_{\max}$. All constants of §2.2 are suprema or infima over $X \times P$. For a removal set $\mathcal{I}_R$ write

$$\Lambda_R := \max_{k \in \mathcal{I}_R} (h_k + w_k) \quad (8)$$

for the longest elapsed time of a coarse predictor, from its retained launch node to the last datum it predicts, with h_k and w_k as in Proposition 3. **Read this:** Λ_R is *not* a new horizon: it is the longest *affected coarse window*, and (7) already covers it. Indeed, let $\tau_j, \dots, \tau_{j+n}$ be a maximal run

of consecutively removed nodes, with τ_{j-1} and τ_{j+n+1} retained. For every k in the run $\tau_k^- = \tau_{j-1}$, so $h_k + w_k = \tau_{k+1} - \tau_{j-1}$, which increases along the run and is maximal at $k = j + n$, where $\tau_{k+1} = \tau_{j+n+1}$ is retained and $h_k + w_k = \tau_{j+n+1} - \tau_{j-1}$ — the coarse window spanning the run. Hence $\Lambda_R \leq \max_k(\text{coarse window } k) \leq \Delta T_{\max}$ automatically. An earlier version of this paragraph claimed that $h_k + w_k$ is not a coarse window for a consecutive run and imposed $\Delta T_{\max} \geq \Lambda_R$ as a separate requirement, together with the statement that this makes (9) strictly harder; both are withdrawn. Λ_R is retained as notation because §2.3 contrasts it with the collapsed argument $\Delta T_1 + \Delta T_2$ of Corollary 4, which can be strictly larger. **Read this:** (A0) is not cosmetic. For the 20-coefficient cubic library the state Jacobian $J_f(x; p)$ depends on p — the coefficient of v in $\dot{v}$ sits on its diagonal and shifts μ one for one — so a constant measured at p^* certifies nothing at p_{wrong} , $p^{(K)}$ or $\hat{p}^{(K)}$. Worse, polynomial candidates with bounded coefficients can blow up in finite time. The analysis is a statement about $X \times P$, and the numerical plateau lies outside it. Where the implemented loss returns its flat blow-up value, the *numerical* trajectory produced by the integrator has exceeded 10^3 in some component or returned NaN; this manuscript’s bounds say nothing about such an evaluation, in either direction. In particular it is *not* claimed that the exact flow left X , nor that the candidate is outside P : the threshold 10^3 in `ms_loss` is an implementation constant chosen to keep the optimiser numerically alive, it has never been related to X , the observation is about a Tsit5 trajectory and not about the exact flow, and (A1) and the event E_X are unverified for the realised experiment (premise check P1, §8). Recovering *any* inference from the plateau would require all four: a verified E_X , a verified (A1), a proof that $X \subset \{\|x\|_\infty < 10^3\}$, and a controlled bound on the integrator’s error. None is available here. Finally, the constants quoted in §6 are *measurements along the true orbit at p^* over R002’s probe box*, i.e. indicative values, not certified suprema over $X \times P$.

(A1) Closure of the admissible region. With $\mu^+ = \max(\mu, 0)$ and Q as in (21), there is $r' \in (0, r_X]$ with

$$e^{\mu^+ \Delta T_{\max}} r' + Q(\Delta T_{\max}) \text{diam}(P) < r_X. \quad (9)$$

Since $r' > 0$, (9) is satisfiable *if and only if* both

$$(A1\text{-i}) \quad Q(\Delta T_{\max}) \text{diam}(P) < r_X \quad \text{and} \quad (A1\text{-ii}) \quad r' < e^{-\mu^+ \Delta T_{\max}} (r_X - Q(\Delta T_{\max}) \text{diam}(P)). \quad (10)$$

The consequence of (A0)–(A1) that the rest of the manuscript uses is Lemma 7 (*no escape from the tube*) and its Corollary 1; they are proved in §3.1, after the local flow lemmas whose hypotheses they discharge. **Read this:** (A1-i) is the binding one, and it contains no r' . (i) (A1-i) is a condition on the *parameter set*: if it fails, no positive noise radius whatever satisfies (9), and the confinement machinery is unavailable for that $\Delta T_{\max}$ at any noise level. Describing the long-window failure as “ r' must be exponentially small” — as an earlier version of this paragraph did — misplaces it. (ii) Whether enlarging r_X helps cannot be settled in general, and is not settled here. Feasibility of (A1-i) must be determined *jointly* from $\mu(X \times P)$, $\tilde{L}(X \times P)$, r_X and P : μ and $\tilde{L}$ are suprema over $X \times P$ with $X \supseteq \mathcal{T}(r_X)$, hence non-decreasing in r_X , and $Q(\Delta T_{\max})$ depends exponentially on $\mu(X \times P)\Delta T_{\max}$ while the right-hand side of (A1-i) grows only linearly in r_X ; but no general conclusion follows from monotonicity alone, and the constants must be *recomputed* over the larger region rather than carried over. **Read this:** in particular the values $\mu = 1.17$ and $\tilde{L} = 10.56$ *cannot be held fixed while X grows*. For the cubic candidate library $\partial f / \partial p$ contains monomials through degree three, so $\tilde{L}(X)$ grows without bound as X expands; and μ is not uniform over P , since candidate coefficients alter the Jacobian and may introduce positive growth that is unbounded on a large X . Two earlier versions of this item are withdrawn:

one asserted that the constants are strictly increasing in r_X and that (A1-i) is a competition a larger tube necessarily loses; the other asserted the opposite, that inflating the tube to a radius exceeding $Q(\Delta T_{\max}) \text{diam}(P)$ makes (A1-i) satisfiable for FHN with $\mu = 1.17$ and $\tilde{L} = 10.56$ held fixed. Both used constants that are not global for the candidate family the report actually fits. (iii) Orders of magnitude, *indicative only*, obtained by inserting the probe-region constants $\mu = 1.17$ and $\tilde{L} = 10.56$ into $Q(s) = \frac{\tilde{L}}{\mu}(e^{\mu s} - 1)$: $Q(1) = 20.1$, $Q(2) = 84.7$, $Q(5) = 3.13 \cdot 10^3$, $Q(10) = 1.09 \cdot 10^6$, $Q(75) = 1.16 \cdot 10^{39}$, $Q(95) = 1.69 \cdot 10^{49}$, $Q(100) = 5.86 \cdot 10^{51}$. Read with those constants, (A1-i) would require $\text{diam}(P) < r_X/20$ at $\Delta T_{\max} = 1$ and $\text{diam}(P) < r_X/(5.9 \cdot 10^{51})$ at $\Delta T_{\max} = 100$. (iv) **Read this:** nothing in this section is *certified* at any window length. The constants above were not computed over the $X \times P$ of (A0), so (A1) has not been evaluated for any $\Delta T_{\max}$, short or long; what the orders of magnitude show is that short windows are the only regime in which (A1-i) is not already excluded by many orders of magnitude, and whether it holds there is unknown. An earlier version of this item said that the machinery “certifies short windows only” and that “the theory supports the beginning of the schedule and is silent at its end”; both are withdrawn as unearned. The correct reading is: the theory is conditional throughout the schedule, and the indicative constants motivate — without establishing — the conclusion that the condition is hopeless at its end, the terminal single-shooting stage, which is the stage the headline experiment of §6 reports. (v) These constants were measured over the probe box of R002/R003, not over the $X \times P$ of (A0), which has never been instantiated numerically. The table above is therefore indicative of magnitude and is *not* a check of (A1); see the premise check P1 in §8.

(A2) Noise model and the admissible event. $y_i = x_i + \eta_i$, $x_i = x^*(t_i)$, with $\eta_0, \dots, \eta_{N-1}$ i.i.d., $\mathbb{E}\eta_i = 0$ and $\text{Cov}(\eta_i) = \Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$, so $\mathbb{E}\|\eta_i\|^2 = \text{tr } \Sigma$. Two variants are used:

(A2-a) *bounded*: $\|\eta_i\| \leq r'$ almost surely;

(A2-b) *Gaussian*: $\eta_i \sim \mathcal{N}(0, \Sigma)$, the law the data generator of R002 samples.

Define the *admissible event*

$$E_X = \left\{ \max_{0 \leq i \leq N-1} \|\eta_i\| \leq r' \right\}. \quad (11)$$

Under (A2-a), $\mathbb{P}(E_X) = 1$. Under (A2-b), for $r' > \sqrt{\text{tr } \Sigma}$,

$$\mathbb{P}(E_X^c) \leq N \min \left\{ \frac{\text{tr } \Sigma}{r'^2}, \exp \left(- \frac{(r' - \sqrt{\text{tr } \Sigma})^2}{2\|\Sigma\|_2} \right) \right\} \quad (12)$$

(Markov on $\|\eta_i\|^2$; Gaussian concentration of the $\|\Sigma\|_2^{1/2}$ -Lipschitz map $\xi \mapsto \|\Sigma^{1/2}\xi\|$ about its mean, together with $\mathbb{E}\|\eta_i\| \leq \sqrt{\text{tr } \Sigma}$). **Read this:** the noise is *anisotropic*. R002 draws component j with standard deviation $\sigma_j = 0.05 \times \text{std}_t x_j^*$, so $\Sigma \neq \sigma^2 I_d$ and $\mathbb{E}\|\eta_i\|^2 = \text{tr } \Sigma$, not $d\sigma^2$. The isotropic model is the specialisation $\text{tr } \Sigma = d\sigma^2$; all statements below are written with $\text{tr } \Sigma$. On E_X every shooting node satisfies $\|y_i - x_i\| \leq r'$, which is exactly the hypothesis Lemma 7 needs at every launch. Expectations are unconditional under (A2-a) and *conditional on E_X* under (A2-b); Lemma 1 is what makes the latter carry the same constants. No unconditional expectation is claimed under (A2-b): off E_X the constants of (A0) do not apply, a polynomial right-hand side can blow up in finite time, and $\mathbb{E}J_K(p^*)$ is not asserted to be finite.

Lemma 1 (conditioning on E_X preserves the noise structure). *Assume (A2-b). Conditionally on E_X , the vectors $\eta_0, \dots, \eta_{N-1}$ are independent, $\mathbb{E}[\eta_i | E_X] = 0$, and $\mathbb{E}[\|\eta_i\|^2 | E_X] \leq \text{tr } \Sigma$.*

Proof. $E_X = \bigcap_i A_i$ with $A_i = \{\|\eta_i\| \leq r'\}$ is a product event, so the conditional joint law is the product of the conditional marginals: independence is preserved. Each A_i and the density

of $\mathcal{N}(0, \Sigma)$ are invariant under $\eta \mapsto -\eta$, so each conditional law is centrally symmetric and its mean is 0. Finally $\|\eta_i\|^2$ is non-decreasing and $\mathbf{1}_{A_i}$ non-increasing in the same scalar $\|\eta_i\|$, so $\text{Cov}(\|\eta_i\|^2, \mathbf{1}_{A_i}) \leq 0$ by Chebyshev’s association inequality, i.e. $\mathbb{E}[\|\eta_i\|^2 \mathbf{1}_{A_i}] \leq \text{tr } \Sigma \cdot \mathbb{P}(A_i)$. $\square$

(A3) Uniform random strong convexity (used only for expectations). There exist a *deterministic* $m > 0$ and a *deterministic* convex $U \subseteq P$ with $p^* \in U$ such that, almost surely on E_X , J_K is m -strongly convex on U and $p^{(K)} \in U$, and $p^{(K)}$ is measurable. (A3) is invoked in exactly one place, the expectation form of Proposition 2; it is listed here because it is an assumption of the same standing as (A0)–(A2) and is named in the scope paragraph below. **Read this:** what the previous revision called “(A2)” in Proposition 2 is this (A3); (A2) is the noise model. (A3) is an assumption about the *noise-dependent* objects J_K , $p^{(K)}$, m and U , and it is *not* verified for the FHN data (Figures 11 and 20).

Scope of statements. Four kinds of statement appear below and they are not interchangeable. Each theorem environment in §3–§4 opens with its tag.

- (S1) Local deterministic flow estimates.** Hypothesis: (A0), together with the requirement that the two trajectories named in the statement, *and the chord joining them*, remain in X on the interval in question — formally $t \leq T_X$ of Definition 1. No noise model, no event, no (A1). *Members:* Lemmas 5 and 6, Corollary 7(i)–(ii).
- (S2) Confinement under (A1).** Hypothesis: (A0)–(A1) and a launch within r' of the true orbit. Conclusion: T_X is the whole window, so (S1) applies on the full window to every such pair. Still deterministic. *Members:* Lemma 7 and Corollary 1.
- (S3) Noisy-cost statements, pathwise on E_X .** Hypothesis: (A0)–(A2), and the event E_X of (11), which is exactly what makes every shooting node an admissible launch for (S2). The conclusions are deterministic inequalities, asserted realisation by realisation on E_X and nowhere else. *Members:* Lemma 8, Proposition 1(a), the pathwise bound of Proposition 2, Proposition 3 and Corollaries 2–4, Theorem 1, Corollaries 5 and 6.
- (S4) Expectations.** Unconditional under the bounded model (A2-a), where $\mathbb{P}(E_X) = 1$; *conditional* on E_X under the Gaussian model (A2-b), with the same constants by Lemma 1 and $\mathbb{P}(E_X^c)$ bounded by (12). The displacement expectation additionally requires (A3). No unconditional expectation is claimed under (A2-b). *Members:* Proposition 1(b) and the expectation form of Proposition 2.

Read this: the distinction between (S1) and (S3) is the one a previous revision collapsed, by declaring that “every inequality in §3–§4 holds on E_X ”. Lemmas 5 and 6 contain no noise: their hypothesis is a *geometric* containment condition, and stamping a probabilistic event on them hides the dependency that Lemma 7 exists to establish — it is that hiding which made an earlier version of Lemma 7 circular, since it invoked globally-quantified versions of the very lemmas whose hypotheses its own conclusion supplies. The order is: (S1) is proved from (A0) alone; (S2) uses (S1) to prove confinement; (S3) uses (S2) on the event where the launches are admissible; (S4) integrates (S3).

Data and indexing (one convention throughout). There are N samples $y_0, \dots, y_{N-1}$ at times $t_i = t_0 + i \Delta t$, $i = 0, \dots, N-1$; hence $N-1$ data intervals and $N-1$ residuals. (In §6, $N = 101$, $\Delta t = 1$, $t_0 = 0$, $t_{N-1} = 100$.) The observation model is (11)’s (A2); we write $\|\eta\|_{\max} = \max_i \|\eta_i\|$.

Partitions. Shooting nodes are data times $\tau_0 = t_0 < \tau_1 < \dots < \tau_{K-1}$, from which windows are launched, together with the right end of the record $\tau_K = t_{N-1}$, which is not a launching node. Window k is $(\tau_{k-1}, \tau_k]$, $k = 1, \dots, K$; $\tau(i)$ is the last node at or before t_i ; $n_k = \#\{i : t_i \in (\tau_{k-1}, \tau_k]\}$, so $\sum_{k=1}^K n_k = N-1$ exactly; $n_{\max} = \max_k n_k$ and $\Delta T = \max_k (\tau_k - \tau_{k-1})$. For the uniform partition used in §6, $\tau_k = t_{k\kappa}$ for $k \leq K-1$ with $K = \lceil (N-1)/\kappa \rceil$; e.g. $\kappa = 3$, $N = 101$ gives $K = 34$, nodes at data indices $0, 3, \dots, 99$ and a last window $(t_{99}, t_{100}]$ holding one datum. The costs are

$$J_K(p) = \sum_{k=1}^K \sum_{t_i \in (\tau_{k-1}, \tau_k]} \|\varphi_f(t_i - \tau_{k-1}; p, y_{\tau_{k-1}}) - y_i\|^2 \quad (K \text{ windows; } N-1 \text{ terms}), \quad (13)$$

$$J_1(p) = \sum_{i=1}^{N-1} \|\varphi_f(t_i - t_0; p, y_0) - y_i\|^2 \quad (\text{single shooting}), \quad (14)$$

$$J_K^*(p) = \sum_{k=1}^K \sum_{t_i \in (\tau_{k-1}, \tau_k]} \|\varphi_f(t_i - \tau_{k-1}; p, x_{\tau_{k-1}}) - x_i\|^2 \quad (\text{noise-free windowed cost}). \quad (15)$$

J_K^* launches every window from the *true* state and compares with the true state, so $J_K^*(p^*) = 0$ and hence $p^* \in \arg \min J_K^*$ for every partition. **Read this:** membership, not equality. $J_K^*(p^*) = 0$ says p^* is a minimiser; with $n_p = 20$ library coefficients fitted from one 101-sample record of a two-state system there may be others, and no persistence-of-excitation or identifiability condition is assumed or verified anywhere in this report. Every statement below is about the *displacement between minimisers*, not about recovering a unique p^* . Note also that the earlier draft compared J_K with the noise-free *single-shooting* cost $J^* = J_1^*$; the decomposition used in its proof actually bounds the distance to J_K^* , which is the comparator used here.

Minimisers, and the only optimality condition used. All minimisers are taken over the compact set P of (A0):

$$p^{(K)} \in \arg \min_{p \in P} J_K(p), \quad \hat{p}^{(K)} \in \arg \min_{p \in P} \hat{J}_K(p), \quad p^{\text{pen}} \in \arg \min_{p \in P} (J_K/N + \text{penalty}), \quad (16)$$

each of which exists because the objective is continuous and P compact — the draft's $\arg \min_p$ over $\mathbb{R}^{n_p}$ guaranteed neither existence nor validity of the constants.

Lemma 2 (minimiser inequality). *Let P be convex, F differentiable on a neighbourhood of P , $p \in \arg \min_P F$, and let $U \subseteq P$ be convex with $p \in U$ and F m -strongly convex on U . Then*

$$F(q) \geq F(p) + \frac{m}{2} \|q - p\|^2 \quad \text{for all } q \in U. \quad (17)$$

Proof. First-order optimality on a convex set gives $\langle \nabla F(p), q - p \rangle \geq 0$ for every $q \in P$, hence for every $q \in U$. Strong convexity on U gives $F(q) \geq F(p) + \langle \nabla F(p), q - p \rangle + \frac{m}{2} \|q - p\|^2$. Add. $\square$

Read this: (17) is the *only* consequence of optimality used in Proposition 2, Theorem 1, Corollaries 5 and 6 and Remark 3(ii). The draft, and the previous revision, wrote $\nabla J_K(p^{(K)}) = 0$ in each of those proofs, which is false at a minimiser on ∂P . With (17) every one of those proofs holds verbatim for minimisers on ∂P and no minimiser is required to be interior. This is adopted as *insurance*, not as a response to a boundary phenomenon known to occur. An earlier version of this paragraph asserted that the $\gamma = 0.05$ shrinkage makes boundary and near-boundary solutions typical; that claim is withdrawn. The shrinkage pulls coefficients towards 0, which is a statement about coordinates, not about ∂P ; P has never been instantiated, so nothing is known about where

its boundary is or whether any reported solution is near it. The variational inequality is used because it is never weaker than $\nabla J_K = 0$ and costs nothing, and interiority is needed in exactly one place — the stationarity sentence of Corollary 5 — where it is obtained from (B0) rather than assumed.

Lemma 3 (local minimiser inequality). *Let $U \subseteq \mathbb{R}^{n_p}$ be convex, F differentiable on a neighbourhood of U , and let $p \in U$ satisfy either $\nabla F(p) = 0$, or $\langle \nabla F(p), q - p \rangle \geq 0$ for all $q \in U$. If F is m -strongly convex on U then*

$$F(q) \geq F(p) + \frac{m}{2} \|q - p\|^2 \quad \text{for all } q \in U. \quad (18)$$

Proof. Strong convexity on U gives $F(q) \geq F(p) + \langle \nabla F(p), q - p \rangle + \frac{m}{2} \|q - p\|^2$; the hypothesis makes the middle term non-negative. $\square$

Remark 1 (reconnection to the unconstrained computation). The statements of §3–§4 concern minimisers of the *constrained* problems (16), taken over the compact convex P of (A0). The runs reported in §6 solve *unconstrained* problems: Nelder–Mead and BFGS over $p \in \mathbb{R}^{n_p}$, with no projection, no feasibility test, and no instantiation of P anywhere in the implementation. A point p_{num} returned by such a run is therefore *not*, without further hypotheses, one of the objects any theorem below bounds.

The hypothesis that would reconnect them is

(R) There exist a compact convex P satisfying (A0)–(A1) and a convex $U \subseteq P$ with $p_{\text{num}} \in \text{int } U$, such that $\nabla J_K(p_{\text{num}}) = 0$ and J_K is m -strongly convex on U for some $m > 0$.

It is *not* verified anywhere in this report (premise check P2, §8). Under (R), Lemma 3 gives (18) at $p = p_{\text{num}}$ on U , which is the *only* consequence of optimality used in Proposition 2, Theorem 1 and Corollaries 5 and 6; and p_{num} is a local minimiser of $\min_{p \in P} J_K$. Consequently, under (R):

- (i) Corollary 5 holds verbatim with $p^{(K)}$ replaced by p_{num} , provided $\bar{B} \subseteq U$: its proof uses $p^{(K)} \in \arg \min_P J_K$ only through Lemma 2 on $\bar{B}$, and (18) supplies exactly that. This is the *only* one of the three that runs under (R) as stated.
- (ii) Proposition 2’s displacement bound holds with p_{num} in place of $p^{(K)}$ *provided* $p^* \in U$ as well — the proposition applies the minimiser inequality at $q = p^*$, so without $p^* \in U$ neither $J_K(p^*) \geq J_K(p_{\text{num}}) + \frac{m}{2} \|p^* - p_{\text{num}}\|^2$ nor the parameter-displacement bound follows. We therefore state the result-specific variant **(R)^{p*}** = (R) together with $p^* \in U$, and use it for Proposition 2 only; basin retention (item (i)) keeps the weaker (R).
- (iii) Theorem 1 holds in its *restricted* form only. Its proof also requires $\hat{J}_K(\hat{p}) \leq \hat{J}_K(p_{\text{num}})$, i.e. that the coarse minimiser be taken over a set containing p_{num} ; with p_{num} merely a local minimiser the usable statement is (49) of Corollary 5, over $\bar{B}$.

Read this: (R) alone does *not* reconnect all aspects of the numerical experiment, and this remark does not claim that it does. Three further qualifications are needed, each stated elsewhere in this manuscript and none of them verified here. (a) *The event.* Every statement of §3–§4 is asserted only on the realised event E_X of (11); a reported run is covered only if its realised noise satisfies $\max_i \|\eta_i\| \leq r'$ for the r' of (A1), which premise check P1 records as unevaluated. (b) *The penalty.* The objective actually minimised in §6 is the penalised (24), not J_K . Stationarity of the *penalised* numerical objective does *not* imply the stationarity of J_K that (R) asserts: the two gradients differ by ∇G , and Remark 3(ii) bounds the resulting displacement only under the further hypotheses (C1)–(C2). (c) *The discretisation.* The runs integrate with a fixed-step Tsit5 scheme,

whereas every bound below is written for the exact flow; no bound on the integrator's error is available here. Finally, no claim is made that p_{num} is a *global* minimiser of J_K over P , nor that the minimisers of $\hat{J}_K$ found in §6 are global. (R) is stronger than what §6 establishes and weaker than what a fully rigorous reconnection would need: it does not require exhibiting P explicitly, only its existence together with the local convexity, but it does require both, and neither has been checked. The displacement, basin-retention and penalty-bias statements below should therefore be read as *conditional on (R), on E_X , and on (b)–(c)* whenever they are applied to a number reported in §6.

Node removal. $\mathcal{I}_R \subset \{1, \dots, K-1\}$ indexes the removed nodes (τ_0 and τ_K are never removed) and $R = \{0, 1, \dots, K\} \setminus \mathcal{I}_R$ the retained ones. For $r \in R \setminus \{K\}$ let $r^+ = \min\{q \in R : q > r\}$ be the next retained index, and for $k \in \mathcal{I}_R$ let τ_k^- and τ_k^+ be the nearest retained nodes to the left and to the right of τ_k . $\hat{J}_K$ is the cost of the coarse partition (39) and $\hat{p}^{(K)}$ its minimiser. $\Delta T_1 = \max_k(\tau_k - \tau_{k-1})$ is the longest window of the *fine* partition and $\Delta T_2 = \max_{k \in \mathcal{I}_R}(\tau_k - \tau_k^-)$ is the largest distance from a removed node to the nearest node retained to its left. $D_k = \{t_i : t_i \in (\tau_k, \tau_{k+1}]\}$ is the fine window launched at the removed node τ_k . **Read this:** ΔT and ΔT_1 denote the same object — the longest window of the partition under discussion. The subscript is used only in §4, where two partitions are in play: ΔT_1 is the *fine* one's longest window and ΔT_2 is the removed-node offset, so that $\Delta T_1 + \Delta T_2$ bounds a coarse window. The draft gave ΔT and ΔT_1 textually identical definitions, which made ΔT_2 look like a second window length; it is not.

2.2 Constants of the right-hand side

$J_f(x; p) = \partial f / \partial x$. Over the admissible region of (A0),

$$\begin{aligned} L &= \sup_{X \times P} \|J_f(x; p)\|_2, & \tilde{L} &= \sup_{X \times P} \|\partial f / \partial p\|_2, \\ \mu &= \sup_{X \times P} \lambda_{\max}\left(\frac{1}{2}(J_f + J_f^\top)\right), & \mu_- &= \inf_{X \times P} \lambda_{\min}\left(\frac{1}{2}(J_f + J_f^\top)\right), \end{aligned} \quad (19)$$

equivalently $\mu_- \|x - z\|^2 \leq \langle f(x; p) - f(z; p), x - z \rangle \leq \mu \|x - z\|^2$ for $x, z \in X$, $p \in P$. Always $-L \leq \mu_- \leq \mu \leq L$; unlike L , μ can be zero or negative. We write $\mu^+ = \max(\mu, 0)$. **Read this:** μ_- is *region-restricted by necessity*, not by convention. For the true FHN field the symmetric part of J_f is $\begin{pmatrix} 1-v^2 & -0.46 \\ -0.46 & -0.064 \end{pmatrix}$, whose smallest eigenvalue tends to $-\infty$ as $|v| \rightarrow \infty$: the global lower logarithmic norm is $-\infty$ and no finite global μ_- exists. The value -2.96 quoted in §6 is an infimum over the sampled true orbit and certifies the lower bound of Lemma 5 only for perturbations that remain in the region where it was measured. By contrast $\lambda_{\max}$ of the same matrix is maximised at $v = 0$ with value 1.17 and decreases thereafter, so the *upper* constant $\mu = 1.17$ is global in x for the true FHN field — at the single parameter p^* , and *not* over a set P of candidate coefficient vectors.

For a weighted norm $\|x\|_D = \|Dx\|$ with D invertible, μ_D is the logarithmic norm of $DJ_f D^{-1}$ and $\text{cond}(D) = \|D\| \|D^{-1}\|$ its condition number (written cond , not κ , because κ is the window size throughout this report). The *local* logarithmic norm along a pair of trajectories is $\mu(t) = \lambda_{\max}(\frac{1}{2}(A(t) + A(t)^\top))$ for the mean-value matrix $A(t)$ of (28); $\Lambda_k = \int_{\tau_{k-1}}^{\tau_k} \mu(s) ds$ is the window *endpoint* exponent and

$$M_k = \sup_{\tau_{k-1} \leq a \leq b \leq \tau_k} \exp\left(\int_a^b \mu(r) dr\right) \geq \max\{1, e^{\Lambda_k}\} \quad (20)$$

the window *transition* bound (Corollary 7, Remark 5). Write

$$Q(s) = \frac{\tilde{L}}{\mu}(e^{\mu s} - 1), \quad \text{read as } \tilde{L}s \text{ when } \mu = 0, \quad (21)$$

the continuous extension $\lim_{\mu \rightarrow 0}(e^{\mu s} - 1)/\mu = s$. For every sign of μ , $Q(s) \geq 0$ and $s \mapsto Q(s)$ is non-decreasing, with $Q(s) \leq \tilde{L}s$ when $\mu \leq 0$ and $Q(s) \leq \tilde{L}/|\mu|$ when $\mu < 0$. The residual abbreviations are

$$\rho_s(p) = (1 + e^{\mu s})\|\eta\|_{\max} + Q(s)\|p - p^*\|, \quad \bar{\rho}_s(p) = \max\{2\|\eta\|_{\max}, \rho_s(p)\}, \quad s \geq 0, \quad (22)$$

and, for a launch time σ and a datum time $t \geq \sigma$ with $t - \sigma \leq \Delta T_{\max}$,

$$r_{\sigma,t}(p) = e^{\mu(t-\sigma)}\|\eta_\sigma\| + Q(t-\sigma)\|p - p^*\| + \|\eta_t\|, \quad (23)$$

the *per-datum residual bound*: it carries the noise realised at the launch node and at the datum, not the global $\|\eta\|_{\max}$. Replacing both by $\|\eta\|_{\max}$ gives $r_{\sigma,t}(p) \leq \rho_{t-\sigma}(p)$, and Lemma 8 is exactly the statement $\|\varphi_f(t - \sigma; p, y_\sigma) - y_t\| \leq r_{\sigma,t}(p) \leq \rho_{t-\sigma}(p) \leq \bar{\rho}_s(p)$ for $t - \sigma \leq s$.

Lemma 4 (properties of the envelope). *For every $s \geq 0$, every $p \in P$ and every sign of μ : (i) $\bar{\rho}_s(p) = \sup_{0 \leq r \leq s} \rho_r(p)$, so $\rho_\sigma \leq \bar{\rho}_s$ whenever $0 \leq \sigma \leq s$; (ii) $s \mapsto \bar{\rho}_s(p)$ is non-decreasing; (iii) $\delta \mapsto \bar{\rho}_s$ is non-decreasing, where $\delta = \|p - p^*\|$, and $E \mapsto \bar{\rho}_s$ is non-decreasing, where $E = \|\eta\|_{\max}$; (iv) $\bar{\rho}_s(p^*) = (1 + e^{\mu^+ s})\|\eta\|_{\max}$; (v) if $\mu \geq 0$ then $\bar{\rho}_s = \rho_s$ for every $s \geq 0$; if $\mu < 0$ then $\bar{\rho}_0 = \rho_0 = 2\|\eta\|_{\max}$ for every p , while for every $s > 0$*

$$\bar{\rho}_s = 2\|\eta\|_{\max} \iff \tilde{L}\|p - p^*\| \leq |\mu|\|\eta\|_{\max}, \quad \text{and} \quad \bar{\rho}_s = \rho_s \text{ otherwise,}$$

the branch being independent of s on $s > 0$.

Proof. (i) $\partial_r \rho_r = e^{\mu r}(\mu\|\eta\|_{\max} + \tilde{L}\|p - p^*\|)$ has a sign independent of r , so $\sup_{0 \leq r \leq s} \rho_r = \max\{\rho_0, \rho_s\}$ and $\rho_0 = 2\|\eta\|_{\max}$. (ii) is immediate from (i). (iii) A maximum of two functions non-decreasing in δ (resp. E) is non-decreasing. (iv) At $p = p^*$, $\bar{\rho}_s = \max\{2, 1 + e^{\mu s}\}\|\eta\|_{\max} = (1 + e^{\mu^+ s})\|\eta\|_{\max}$. (v) For $\mu \geq 0$, $\rho_s \geq \rho_0 = 2\|\eta\|_{\max}$ by (i). For $\mu < 0$ write $Q(s) = \frac{\tilde{L}}{|\mu|}(1 - e^{-|\mu|s})$ and $e^{\mu s} - 1 = -(1 - e^{-|\mu|s})$, so

$$\rho_s - 2\|\eta\|_{\max} = (1 - e^{-|\mu|s})\left(\frac{\tilde{L}}{|\mu|}\|p - p^*\| - \|\eta\|_{\max}\right).$$

For $s > 0$ the first factor is strictly positive, so the sign is that of $\tilde{L}\|p - p^*\| - |\mu|\|\eta\|_{\max}$ and does not depend on s ; at $s = 0$ the first factor vanishes and $\rho_0 = 2\|\eta\|_{\max}$ regardless of p . $\square$

Read this: item (v) is a statement about $s > 0$. An earlier version asserted it for all $s \geq 0$, which is false in the “only if” direction at $s = 0$, where $\bar{\rho}_0 = 2\|\eta\|_{\max}$ for *every* parameter distance. Note also that ρ_s is *not* monotone in s : for $\mu < 0$ and $p = p^*$ it strictly *decreases*, so every step that bounds a shorter window’s residual by a longer window’s uses $\bar{\rho}$, never ρ .

Remark 2 (the smooth quadratic majorant, and why it is not the primary bound). Put $\rho_s^+(p) = (1 + e^{\mu^+ s})\|\eta\|_{\max} + Q(s)\|p - p^*\|$. Then $\bar{\rho}_s \leq \rho_s^+$ for every $s \geq 0$ and every p , with equality *exactly* when

$$\mu \geq 0, \quad \text{or} \quad s = 0, \quad \text{or} \quad \|\eta\|_{\max} = 0, \quad \text{or} \quad Q(s)\|p - p^*\| = 0 \quad (\text{i.e. } \tilde{L} = 0 \text{ or } p = p^*),$$

and strict inequality otherwise. [*Proof.* For $\mu \geq 0$, $\mu^+ = \mu$ and $\bar{\rho}_s = \rho_s = \rho_s^+$ by Lemma 4(v). For $\mu < 0$, $\rho_s^+ = 2\|\eta\|_{\max} + Q(s)\|p - p^*\|$ and $\bar{\rho}_s = \max\{2\|\eta\|_{\max}, (1 + e^{\mu s})\|\eta\|_{\max} + Q(s)\|p - p^*\|\}$:

if $s = 0$, or $\tilde{L} = 0$, or $p = p^*$ — equivalently $Q(s)\|p - p^*\| = 0$ — both sides equal $2\|\eta\|_{\max}$; if $\|\eta\|_{\max} = 0$ both equal $Q(s)\|p - p^*\|$; otherwise $s > 0$, $\|\eta\|_{\max} > 0$ and $Q(s)\|p - p^*\| > 0$, and either $\bar{\rho}_s = (1 + e^{\mu s})\|\eta\|_{\max} + Q(s)\|p - p^*\| < \rho_s^+$ because $e^{\mu s} < 1$, or $\bar{\rho}_s = 2\|\eta\|_{\max} < \rho_s^+$ because $Q(s)\|p - p^*\| > 0$.] So every bound below remains true with $\bar{\rho}$ replaced by ρ^+ ; ρ^+ is a single affine form in $(\|\eta\|_{\max}, \|p - p^*\|)$ with non-negative deterministic coefficients $A_s = 1 + e^{\mu^+ s}$, $B_s = Q(s)$, which makes Δ_K of (44) a homogeneous quadratic form rather than a piecewise quadratic one. **Read this:** ρ^+ is a *convenience*, not the bound, and the regime where the difference matters is *not* exhibited anywhere in this report. (i) *Where $\bar{\rho}$ is strictly better.* Let the system be uniformly contractive on $X \times P$, i.e. $\lambda_{\max}(\text{sym } J_f(x; p)) \leq \mu < 0$ for all $(x, p) \in X \times P$ — for instance a linear field $\dot{x} = Ax + B(p)$ with $\text{sym } A \prec 0$, or the gradient flow of a potential that is uniformly strongly convex on X . Then for every $s > 0$ and every p with $0 < \tilde{L}\|p - p^*\| \leq |\mu|\|\eta\|_{\max}$, Lemma 4(v) gives $\bar{\rho}_s = 2\|\eta\|_{\max}$ while $\rho_s^+ = 2\|\eta\|_{\max} + Q(s)\|p - p^*\|$, larger by up to $\tilde{L}\|p - p^*\|/|\mu|$. Since the whole point of replacing L by a one-sided constant is to exploit contraction, the primary statements carry $\bar{\rho}$, and Δ_K is described as *piecewise* quadratic — the truthful description — with the branch switching on the sign of $\tilde{L}\|q - p^*\| - |\mu|\|\eta\|_{\max}$. This is an illustration by construction, not an instance drawn from §6. (ii) *FHN exhibits none of this.* The global constant measured in §6 is $\mu = 1.17 > 0$, and for $\mu \geq 0$ Lemma 4(v) makes ρ , $\bar{\rho}$ and ρ^+ *identically equal*. So no number in §6 — at p^* , at the off-truth perturbation, or at p_{wrong} — distinguishes the three, and this report contains *no* numerical evidence bearing on the choice; no experiment here exhibits the $\bar{\rho}$ branch at all. The negative per-window exponents Λ_k^D reported in §6 are *not* instances of $\mu < 0$: they are linearised, weighted, endpoint exponents along the true orbit at the true parameter, and Remark 5 gives four independent reasons why they may not be substituted for the uniform μ of $X \times P$. An earlier version of this remark cited them as the regime where the $\bar{\rho}/\rho^+$ distinction “lives”, which commits exactly the substitution Remark 5 forbids; that citation is withdrawn. (iii) *A still earlier version* made ρ^+ primary and justified it by claiming the choice “costs nothing where every number in §6 is evaluated”. That claim was doubly wrong: Figure 8 evaluates off-truth as well as at p^* , and, more fundamentally, §6 cannot distinguish the alternatives at all.

$m > 0$ denotes a strong-convexity constant of a cost near its minimiser (the draft wrote μ for it, and wrote m for the parameter dimension; here n_p is the parameter dimension, m is the strong-convexity constant, μ is the logarithmic norm and κ is the window size).

2.3 Numerical protocol (R002)

The target is FHN, $\dot{v} = v - \frac{1}{3}v^3 - w + 0.5$, $\dot{w} = \frac{1}{12.5}(v + 0.7 - 0.8w)$, written over the cubic library $\Theta = \{1, w, w^2, w^3, v, vw, vw^2, v^2, v^2w, v^3\}$ ($J = 10$, $p \in \mathbb{R}^{20}$, 7 non-zeros). Data: fixed-step Tsit5 at 0.01 from $(1, 1)$ to $T = 156$, every 100th point kept, cropped to $t \in [50, 150]$ and shifted to $[0, 100]$ ($N = 101$, $\Delta t = 1$), Gaussian noise of standard deviation $0.05 \times$ each component’s own standard deviation; $\|\eta\|_{\max} = 0.159$, noise floor $\frac{1}{N-1} \sum \|\eta_i\|^2 = 0.0052$. The optimised cost is the normalised, sparsity-penalised version of (13). With $z = [x_0; p] \in \mathbb{R}^{d+n_p}$ and $\tau(i) = t_{\kappa[i/\kappa]}$,

$$J_{\kappa}(z) = \frac{1}{N} \left[\|x_0 - y_0\|^2 + \sum_{i=0}^{N-2} \|\hat{x}(t_{i+1} - \tau(i); p, y_{\tau(i)}) - y_{i+1}\|^2 \right] + \frac{\gamma}{n_p} \sum_{j=1}^{n_p} \text{smooth } \ell_1(p_j), \quad (24)$$

where $\hat{x}(s; p, x)$ is the S -substep fixed-step Tsit5 approximation of $\varphi_f(s; p, x)$, κ is the window size in data intervals ($K = \lceil (N-1)/\kappa \rceil$ windows launched at the data indices $0, \kappa, 2\kappa, \dots$; $\kappa = 1$ is one window per interval, $\kappa = N-1 = 100$ is single shooting), each interval is integrated with $S = 10$ substeps of length $\delta t = \Delta t/S$, $\text{smooth } \ell_1(x) = \alpha^{-1}[\log(1 + e^{-\alpha x}) + \log(1 + e^{\alpha x})]$ with

$\alpha = 500$, and $\gamma = 0.05$. If a state exceeds 10^3 in magnitude or becomes NaN the loss returns the constant 10^3 (*flat blow-up penalty*, a plateau with zero gradient); the *graded* penalty returns $10^3(1 + \frac{N-1-i}{N-1}) + J_{\text{partial}}/N$ when the blow-up occurs at index i . The theory checks use $\gamma = 0$.

Read this: x_0 is a *dummy* in (24). The implementation overwrites the running state with the datum $y_{\tau(i)}$ whenever $i \equiv 0 \pmod{\kappa}$, and $i = 0$ satisfies this for *every* κ , so the first integration step already starts from y_0 and x_0 never enters an integrator — not even at $\kappa = N - 1$. Its only appearance is the term $\|x_0 - y_0\|^2/N$, whose minimiser is $x_0 = y_0$ and whose Hessian block is $(2/N)I_d$ *independently of* κ . Consequently (i) the optimisation is effectively over p with d inert coordinates carried along; (ii) the Hessian of (24) in z is block diagonal, $\text{diag}((2/N)I_d, \nabla_p^2 J_\kappa)$, so strong convexity in z is equivalent to strong convexity in p ; and (iii) the “sensitivity to v_0 ” panel of Figure 11 is exactly the parabola $\|x_0 - y_0\|^2/N$, which is why it is the same shallow parabola at every window size. The multiple-shooting formulation of (13) — node states fixed at the data, not optimised — is what the code implements; the free x_0 is vestigial.

Remark 3 (what transfers from J_K to the implemented objective). Write $J_\kappa^{\text{impl}}(z) = \frac{1}{N}[\|x_0 - y_0\|^2 + J_K(p)] + \frac{\gamma}{n_p} \sum_j \text{smooth } \ell_1(p_j)$, the exact content of (24) up to discretisation. The first and third terms do not depend on the partition. Hence:

- (i) $\hat{J}_\kappa^{\text{impl}} - J_\kappa^{\text{impl}} = \frac{1}{N}(\hat{J}_K - J_K)$, so Proposition 3 holds verbatim for the implemented objective with every bound divided by N , and Theorem 1 and Corollaries 5–6 hold with the same substitution *provided m is a strong-convexity constant of the penalised cost*.
- (ii) Proposition 2 does *not* transfer. Its comparator $p^\star$ minimises $J_K^\star$ but is not a minimiser of $J_K/N + \text{penalty}$. Write $F = J_K/N$ and $G(p) = \frac{\gamma}{n_p} \sum_j \text{smooth } \ell_1(p_j)$, and *assume*:
 - (C1) $p^{(K)} \in \arg \min_P F$ and $p^{\text{pen}} \in \arg \min_P (F + G)$ (both exist, P compact);
 - (C2) there is a convex $U \subseteq P$ containing $p^{(K)}$, p^{pen} and the segment joining them, on which F is m -strongly convex.

Then

$$\|p^{\text{pen}} - p^{(K)}\| \leq \frac{1}{m} \|\nabla G(p^{\text{pen}})\| \leq \frac{\gamma}{m\sqrt{n_p}}. \quad (25)$$

Proof. Adding the two first-order optimality inequalities $\langle \nabla F(p^{(K)}), p^{\text{pen}} - p^{(K)} \rangle \geq 0$ and $\langle \nabla F(p^{\text{pen}}) + \nabla G(p^{\text{pen}}), p^{(K)} - p^{\text{pen}} \rangle \geq 0$ gives $\langle \nabla F(p^{\text{pen}}) - \nabla F(p^{(K)}), p^{\text{pen}} - p^{(K)} \rangle \leq \|\nabla G(p^{\text{pen}})\| \|p^{\text{pen}} - p^{(K)}\|$; strong monotonicity of ∇F on U (equivalent to m -strong convexity there, and requiring the segment to lie in U , whence (C2)) bounds the left side below by $m\|p^{\text{pen}} - p^{(K)}\|^2$. For the gradient, $\frac{d}{dx} \text{smooth } \ell_1(x) = 2\sigma(\alpha x) - 1 = \tanh(\alpha x/2) \in (-1, 1)$, so $\|\nabla G\|_2 \leq (\gamma/n_p)\sqrt{n_p} = \gamma/\sqrt{n_p}$, uniformly in p and independently of α . $\square$ **Read this:** (C1)–(C2) are *assumptions*, not facts about the FHN runs, and (25) is therefore conditional — and, when applied to a reported number, conditional also on (R) of Remark 1. “ F is strongly convex near its own minimiser” is *not* enough: the penalised minimiser may lie outside that neighbourhood, which is the same circularity flagged for Theorem 1. Note also that the smoothing sharpness $\alpha = 500$ does not appear in (25) at all. This shrinkage bias must be added to any statement about $\|p^\star - \cdot\|$; it is the shrinkage visible in Figure 15.

- (iii) Curvature diagnostics computed with $\gamma = 0$ are diagnostics of the *data term*, not of the optimised objective; see Figures 11 and 20.
- (iv) The blow-up replacement is outside the analysis entirely: where it fires, the loss returned is not $\sum_i \|\varphi_f(\cdot; p, \cdot) - y_i\|^2$ for any flow this manuscript bounds, so no statement of §3–§4 applies to that evaluation. No conclusion about p , P or ∂P is drawn from it; see the (A0) discussion above.

Arms. **GP:** sweep κ upward through the FULL schedule $\{1, 2, 3, 4, 5, 6, 8, 10, 12,$

15, 20, 25, 33, 50, 75, 100}, each stage started from the previous minimiser. **best-GP**: as GP but started from the lowest-cost minimiser seen so far. **Control** (no propagation): every stage started from the same random seed vector. Optimiser: Nelder–Mead, 2500 iterations per stage, unless stated; BFGS/L-BFGS use the exact automatic-differentiation gradient. Seeds $s = 1, \dots, 8$: $z_{\text{seed}} = [y_0; 0.1\xi]$, $\xi \sim \mathcal{N}(0, I_{20})$, giving $\|p_0 - p^*\| \in [1.50, 1.84]$. Arms are compared at equal budgets (same iterations, schedule and seeds) unless a table says otherwise. Other schedules: SHORT $\{1, 2, 5, 10, 25, 50, 100\}$ (variations), DENSE (44 stages), COARSE $\{1, 5, 25, 100\}$, JUMP $\{1, 100\}$.

Read this: the sweep schedules are in general *not* nested, so the stages of Algorithm 1 are *not* node removals. The node set at window size κ is $S_\kappa = L_\kappa \cup \{M\}$ with $L_\kappa = \{j\kappa : j\kappa < M\}$ the launch nodes and $M = N - 1 = 100$ the common terminal node, which belongs to every partition. Hence the stage $\kappa_a \rightarrow \kappa_b$ is a node removal iff $S_{\kappa_b} \subseteq S_{\kappa_a}$, i.e. iff $L_{\kappa_b} \subseteq L_{\kappa_a}$, and for increasing $\kappa_a < \kappa_b \leq M$ this holds

$$\text{iff} \quad \kappa_a \mid \kappa_b \quad \text{or} \quad \kappa_b = M \quad (26)$$

(for $\kappa_b < M$ the element $\kappa_b \in L_{\kappa_b}$ forces $\kappa_a \mid \kappa_b$, and conversely divisibility gives inclusion; for $\kappa_b = M$, $L_M = \{0\}$ and inclusion is automatic). Counting transitions with (26): FULL $\{1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 25, 33, 50, 75, 100\}$ has *two* of 15 nested ($1 \rightarrow 2$ and $75 \rightarrow 100$); DENSE (44 stages) *two* of 43 ($1 \rightarrow 2$ and $95 \rightarrow 100$); SHORT $\{1, 2, 5, 10, 25, 50, 100\}$ four of six ($1 \rightarrow 2$, $5 \rightarrow 10$, $25 \rightarrow 50$, $50 \rightarrow 100$; the failures are $2 \rightarrow 5$ and $10 \rightarrow 25$); COARSE $\{1, 5, 25, 100\}$ three of three and JUMP $\{1, 100\}$ one of one. The earlier version of this manuscript asserted that the sub-partition hypothesis “is what the algorithm does”; that assertion was false and is withdrawn, and so is a later version’s criterion “ κ_a divides κ_b ” with one nested transition in FULL and one in DENSE, which omitted the $\kappa_b = M$ case of (26). The general case is covered by Corollary 6, which compares two arbitrary partitions through their common refinement.

Read this: the second nested transition of FULL and of DENSE is in each case the terminal $\kappa_b = M$ stage, i.e. the collapse to single shooting, which is a node removal only in the degenerate sense that *every* interior node is removed. The exact quantities, from the definitions of Proposition 3 and Corollary 4:

	FULL $75 \rightarrow 100$	DENSE $95 \rightarrow 100$
fine node set $S_\kappa = L_\kappa \cup \{M\}$	$\{0, 75, 100\}$	$\{0, 95, 100\}$
coarse node set	$\{0, 100\}$	$\{0, 100\}$
removed set $\mathcal{I}_R$	$\{75\}, \mathcal{I}_R = 1 = K - 1$	$\{95\}, \mathcal{I}_R = 1 = K - 1$
$\Delta T_1 = \max_k(\tau_k - \tau_{k-1})$	$\max\{75, 25\} = 75$	$\max\{95, 5\} = 95$
$\Delta T_2 = \max_{k \in \mathcal{I}_R} w_k$	75	95
$h_k, D_k $	25, 25	5, 5
collapsed argument $\Delta T_1 + \Delta T_2$	150	190
actual coarse span $\Lambda_R = \max_k(h_k + w_k)$	100	100

So Δ_K of (44) carries $e^{\mu^+ \Delta T_1}$ times two envelopes with arguments 75 and 150, i.e. a leading factor $e^{380\mu^+}$ for FULL and $e^{380\mu^+}$ for DENSE. Proposition 3 therefore covers two of FULL’s fifteen stages, one of which it covers vacuously. **Read this:** two corrections to an earlier version of this paragraph, which reported $\Delta T_2 = 100$ and a factor $e^{100\mu^+}$. ΔT_2 is the offset of the removed node to its retained left neighbour, which is 75 (FULL) and 95 (DENSE), not 100; and the composite argument $\Delta T_1 + \Delta T_2$ is 150 and 190, so the earlier factor was understated, not overstated. The *actual* elapsed time in the primary bound (40) never exceeds $\Lambda_R = 100$ in either case, which is the coarse window $\{0, 100\}$ itself — so it is covered by (7) with no separate requirement, the coarse partitions being among the partitions under discussion. The gap between $\Lambda_R = 100$ and the

collapsed argument $\Delta T_1 + \Delta T_2 = 150$ (FULL) or 190 (DENSE) is entirely the price of collapsing h_k and w_k to independent maxima in Corollary 4. On these transitions the per-datum bound carries $e^{\mu(t_i - 75)} \leq e^{25\mu^+}$ and residual arguments at most 100, against the collapsed $e^{75\mu^+}$ and $\bar{\rho}_{150}$.

Metrics. J_κ at the returned minimiser (values $\geq 10^3$ flag the plateau); the *parameter error* $\|p - p^*\|$ (Euclidean, 20 coefficients; the identification metric and the primary one here); the *recovery score* in $[0, 1]$: each coefficient earns credit 1 if it is correctly zero ($|p_j| < 0.05$ when $p_j^* = 0$) or correctly non-zero within 25 % of the truth, averaged over the 20 coefficients; the *paired difference* of $\|p - p^*\|$ between two arms sharing a seed, reported as the mean with a 95 % percentile bootstrap CI over 2000 resamples of the seeds (the resampling unit is the optimiser seed), with *cellwise* counts of seeds in which one arm is strictly better.

2.4 Symbols and frozen constants

symbol	meaning	value	where frozen
$N, \Delta t$	samples, spacing ($N - 1$ residuals)	101, 1	R002 data recipe
d, n_p	state dimension, parameter dimension	2, 20	§2.1
κ, K	window size (data intervals), number of windows	swept, $\lceil 100/\kappa \rceil$	R002 schedules
$\Delta T, \Delta T_1, \Delta T_2$	longest window of the partition under discussion; of the fine partition; largest removed-node offset $\tau_k - \tau_k^-$	swept	draft, §2.1
$\Delta T_{\max}, \Lambda_R$	horizon of (A0); longest affected coarse span (8)	not instantiated	§2.1
$X, P, \mathcal{T}, r_X, r'$	state region, parameter set, tube, tube radius, noise radius (A0)–(A1)	not instantiated	§2.1
E_X	admissible event $\{\max_i \ \eta_i\ \leq r'\}$ (11)	not instantiated	§2.1
$S, \delta t$	integrator sub-steps per interval, step	10, 0.1	R002 driver
γ, α	sparsity weight, smooth- ℓ_1 sharpness	0.05, 500	R002 driver
$\ \eta\ _{\max}, \Sigma$	largest noise norm; noise covariance (anisotropic)	0.159	R002 fhn_data_meta
L	Lipschitz constant in x along the orbit / on $[-2.5, 2.5] \times [-1, 2]$	3.05 / 5.34	R002 concept_meta
μ, μ_-, μ^+	logarithmic norm along the orbit, lower counterpart, $\max(\mu, 0)$	1.17, -2.96	R002 / R003
$\tilde{L}$	Lipschitz constant in p along the orbit	10.56	R002 concept_meta
$Q(s), \rho_s, \bar{\rho}_s, \rho_s^+, r_{\sigma,t}$	parameter-sensitivity factor (21); residual bound, its monotone envelope, its smooth majorant, its per-datum form (23)	symbolic	§2.2
Λ_k, M_k	window endpoint exponent; window transition bound (20)	Table 2	R003
$D, \text{cond}(D)$	weighting $\text{diag}(1, s)$, $s = \sqrt{ J_{12} / J_{21} }$; its condition number	$s = 3.54$	R003
m	strong-convexity constant near a minimiser	assumed	draft (as “ μ ”)
blow-up threshold	$ x > 10^3$ or NaN	10^3	R002 loss

Read this: L and μ are measured along the true orbit at p^* over R002’s probe box, not assumed — and not certified as the suprema over $X \times P$ that (A0) requires, since $X \times P$ has never been instantiated (premise check P1, §8). The numerical section shows that the amplification of an

initial-state error on the FHN limit cycle never exceeds 8.5; bounds carrying $e^{L\Delta T}$ are loose by orders of magnitude for $\Delta T \gtrsim 2$, bounds carrying $e^{\mu\Delta T}$ by much less, and the linearised per-window endpoint exponents of §5 are close to the measurement.

3 Convergence analysis of multiple shooting

Throughout, f satisfies (A0) and the constants of §2.2 are finite on $X \times P$. Every statement opens with its scope tag (S1)–(S4) of §2. Statements marked [\(revised\)](#) differ from the earlier draft; each is followed by the change and its effect.

3.1 Sensitivity of the flow to the initial state and to the parameters

Definition 1 (confinement time of a pair). *For $x_1, x_2 \in X$ and $p_1, p_2 \in P$ write $\xi_j(s) = \varphi_f(s; p_j, x_j)$ and let $[u, v] = \{u + \theta(v - u) : \theta \in [0, 1]\}$. Set*

$$T_X((x_1, p_1), (x_2, p_2)) = \sup \left\{ t \in [0, \Delta T_{\max}] : \xi_1, \xi_2 \text{ exist on } [0, t] \text{ and } [\xi_1(s), \xi_2(s)] \subset X \ \forall s \in [0, t] \right\}.$$

Three remarks. The chord contains its endpoints, so the displayed condition already requires $\xi_1(s), \xi_2(s) \in X$. $T_X \geq 0$ always, since $x_1, x_2 \in X$ and X is convex. And the defining set is closed, so the condition holds on all of $[0, T_X]$: by (A0) each ξ_j exists while it remains in the compact set X , hence extends continuously to T_X with $\xi_j(T_X) \in X$, and X closed and convex gives $[\xi_1(T_X), \xi_2(T_X)] \subset X$.

Lemma 5 (flow sensitivity to the initial state, local; [revised](#)). *Scope (S1). Assume (A0). Let $p \in P$, $x_1, x_2 \in X$, and let $T = T_X((x_1, p), (x_2, p))$. Then for every $t \in [0, T]$,*

$$e^{\mu-t}\|x_1 - x_2\| \leq \|\varphi_f(t; p, x_1) - \varphi_f(t; p, x_2)\| \leq e^{\mu t}\|x_1 - x_2\|. \quad (27)$$

In particular, since $\mu \leq L$ and $\mu_- \geq -L$, the draft's bounds $e^{-Lt}\|x_1 - x_2\| \leq \|\cdot\| \leq e^{Lt}\|x_1 - x_2\|$ hold on $[0, T]$.

Proof. Let $\hat{x}_j(t) = \varphi_f(t; p, x_j)$ and $\delta = \hat{x}_1 - \hat{x}_2$, so $\dot{\delta} = f(\hat{x}_1; p) - f(\hat{x}_2; p)$ with $\delta(0) = x_1 - x_2$. For $t < T$ the chord $\hat{x}_2(t) + s\delta(t)$, $s \in [0, 1]$, lies in X by Definition 1, so the mean-value form $f(\hat{x}_1; p) - f(\hat{x}_2; p) = A(t)\delta$ with $A(t) = \int_0^1 J_f(\hat{x}_2 + s\delta; p) ds$ is valid and every J_f in the integrand is evaluated at a point of X ; hence

$$\frac{1}{2} \frac{d}{dt} \|\delta\|^2 = \langle \delta, A(t)\delta \rangle = \langle \delta, \frac{1}{2}(A + A^\top)\delta \rangle \in [\mu_- \|\delta\|^2, \mu \|\delta\|^2] \quad (28)$$

on $[0, T)$. If $x_1 = x_2$ both sides of (27) vanish. If $x_1 \neq x_2$ then $\delta(t) \neq 0$ on $[0, T)$: two solutions of the same initial-value problem with the same parameter that agree at one time agree at all times by uniqueness (guaranteed by (A0)), so $\delta(t_*) = 0$ would force $\delta \equiv 0$ and contradict $\delta(0) \neq 0$. On $\{\delta \neq 0\}$, $\|\delta\|$ is differentiable and (28) gives $\mu_- \|\delta\| \leq \frac{d}{dt} \|\delta\| \leq \mu \|\delta\|$; Grönwall in both directions gives (27) on $[0, T)$, and both sides are continuous at $t = T$. $\square$

Read this: the quantifier. This lemma is *not* asserted for all $t \leq \Delta T_{\max}$: under (A0) alone a trajectory launched in X may leave X , after which the mean-value matrix integrates J_f over a chord outside X and $\lambda_{\max}(\text{sym } A) \leq \mu$ has no justification — and for a polynomial f the solution may cease to exist. A previous revision stated (27) for all $x_1, x_2 \in X$ and all $t \leq \Delta T_{\max}$ while

simultaneously removing forward invariance from (A0), which made the statement false and the proof of Lemma 7 circular. Full-window versions are Corollary 1, obtained *after* Lemma 7. *What changed.* The draft bounded $\|\dot{\delta}\| \leq L\|\delta\|$ and integrated $\frac{d}{dt} \ln\|\delta\|$. That step applies Cauchy–Schwarz to $\langle \delta, \dot{\delta} \rangle$ and discards the sign information in the inner product; (28) keeps it. On FHN the Jacobian is $\begin{pmatrix} 1-v^2 & -1 \\ 0.08 & -0.064 \end{pmatrix}$ and $1-v^2 \leq 1$ for every v , so $\mu = 1.17$ on all of $\mathbb{R}^2$ at p^* (attained at $v = 0$), whereas L grows like v^2 and is 3.05 on the orbit but 5.34 on the box. At $t = \Delta T = 2$ the bound falls from $e^{L\Delta T} = 447$ to $e^{\mu\Delta T} = 10.4$ against a measured peak of 6.9 (Table 1).

Lemma 6 (flow sensitivity to the parameters, local; *revised*). Scope (S1). *Assume (A0). Let $x_0 \in X$, $p_1, p_2 \in P$, and let $T = T_X((x_0, p_1), (x_0, p_2))$. Then for every $t \in [0, T]$,*

$$\|\varphi_f(t; p_1, x_0) - \varphi_f(t; p_2, x_0)\| \leq Q(t) \|p_1 - p_2\|, \quad Q(t) = \frac{\tilde{L}}{\mu}(e^{\mu t} - 1) \quad (29)$$

with the conventions of (21).

Proof. Let $\hat{x}_j(t) = \varphi_f(t; p_j, x_0)$ and $\delta = \hat{x}_1 - \hat{x}_2$, so $\delta(0) = 0$ and

$$\dot{\delta} = \underbrace{[f(\hat{x}_1; p_1) - f(\hat{x}_2; p_1)]}_{\text{state difference, equal parameter}} + \underbrace{[f(\hat{x}_2; p_1) - f(\hat{x}_2; p_2)]}_{\text{parameter difference, equal state}}.$$

For $t < T$ the chord $[\hat{x}_1(t), \hat{x}_2(t)]$ lies in X by Definition 1, so the first bracket equals $A(t)\delta$ with A as in (28) and $\lambda_{\max}(\text{sym } A) \leq \mu$; and $\hat{x}_2(t) \in X$, $p_1, p_2 \in P$, so the second bracket has norm at most $\tilde{L}\|p_1 - p_2\|$ by the definition of $\tilde{L}$ as a supremum over $X \times P$.

The differential inequality, including at the zeros of δ . Here $\delta(0) = 0$ and, unlike in Lemma 5, uniqueness does *not* exclude later zeros: two trajectories with different parameters may meet at $t > 0$ without coinciding. We therefore work with the upper right Dini derivative $D^+\phi(t) = \limsup_{h \downarrow 0} (\phi(t+h) - \phi(t))/h$ of $\phi = \|\delta\|$.

- If $\delta(t) \neq 0$ then ϕ is differentiable at t and $D^+\phi = \dot{\phi} = \langle \delta, \dot{\delta} \rangle / \|\delta\| \leq \mu\|\delta\| + \tilde{L}\|p_1 - p_2\|$ by (28) and Cauchy–Schwarz on the second bracket.
- If $\delta(t) = 0$ then $\hat{x}_1(t) = \hat{x}_2(t) =: \bar{x}$ and $\phi(t+h) = \|h\dot{\delta}(t) + o(h)\|$, so $D^+\phi(t) = \|\dot{\delta}(t)\| = \|f(\bar{x}; p_1) - f(\bar{x}; p_2)\| \leq \tilde{L}\|p_1 - p_2\| = \mu\phi(t) + \tilde{L}\|p_1 - p_2\|$.

Hence $D^+\|\delta\| \leq \mu\|\delta\| + \tilde{L}\|p_1 - p_2\|$ on $[0, T]$. The comparison lemma for Dini derivatives (any continuous ϕ with $D^+\phi \leq a\phi + b$ and $\phi(0) = \phi_0$ satisfies $\phi(t) \leq e^{at}\phi_0 + \frac{b}{a}(e^{at} - 1)$ with $\phi_0 = 0$ gives (29) on $[0, T]$; both sides are continuous at $t = T$. $\square$

Alternative (regularisation). With $\phi_\varepsilon = (\|\delta\|^2 + \varepsilon^2)^{1/2} \geq \varepsilon > 0$ one has $\dot{\phi}_\varepsilon = \langle \delta, \dot{\delta} \rangle / \phi_\varepsilon \leq \mu\|\delta\|^2 / \phi_\varepsilon + \tilde{L}\|p_1 - p_2\|$, and since $\|\delta\|^2 / \phi_\varepsilon = \phi_\varepsilon - \varepsilon^2 / \phi_\varepsilon \in [\phi_\varepsilon - \varepsilon, \phi_\varepsilon]$, $\dot{\phi}_\varepsilon \leq \mu\phi_\varepsilon + \tilde{L}\|p_1 - p_2\| + |\mu|\varepsilon$; Grönwall and $\varepsilon \downarrow 0$ give the same conclusion. (The remainder $|\mu|\varepsilon$ is needed only when $\mu < 0$.)

What changed. The first bracket's bound, $L \rightarrow \mu$, as in Lemma 5; the removal of the division by $\|\delta\|$ at $\delta = 0$, which the draft's argument performed silently at $t = 0$; and the restriction to $t \leq T_X$. The consequence of $L \rightarrow \mu$ is qualitative: for a dissipative system ($\mu < 0$) the revised bound *saturates* at $\tilde{L}/|\mu|$, which is the shape the measurement has (Figure 7b), whereas $\frac{\tilde{L}}{L}(e^{Lt} - 1)$ can never saturate because a Lipschitz constant is non-negative.

Lemma 7 (no escape from the tube). Scope (S2). *Assume (A0)–(A1). Let $\tau \in [t_0, t_{N-1}]$, $x_\tau = x^*(\tau)$, let $z \in \mathbb{R}^d$ with $\|z - x_\tau\| \leq r'$, and put $S = \min\{\Delta T_{\max}, t_{N-1} - \tau\}$. Then for every $p \in P$ and every $s \in [0, S]$,*

$$\varphi_f(s; p, x_\tau) \in \mathcal{T} \subseteq X \quad \text{and} \quad \varphi_f(s; p, z) \in \mathcal{T} \subseteq X.$$

Proof. Two exit-time bootstraps; the first is needed to license the second. Both use only Lemmas 5 and 6, and only at times at which the relevant chord has already been placed in X .

Step 1 (clean launch). Let $T = \sup\{t \in [0, S] : \varphi_f(s; p, x_\tau) \in \mathcal{T} \ \forall s \leq t\}$; $T \geq 0$ since $x_\tau \in \mathcal{T}$. Fix $s < T$. Then $\varphi_f(s; p, x_\tau) \in \mathcal{T} \subseteq X$, and $x^*(\tau + s) = \varphi_f(s; p^*, x_\tau) \in \mathcal{T} \subseteq X$ because it lies on the orbit; X is convex, so the chord between them lies in X . Hence $T \leq T_X((x_\tau, p), (x_\tau, p^*))$ and Lemma 6 applies on $[0, T]$, giving

$$\begin{aligned} \text{dist}(\varphi_f(s; p, x_\tau), x^*([t_0, t_{N-1}])) &\leq \|\varphi_f(s; p, x_\tau) - x^*(\tau + s)\| \\ &\leq Q(s)\|p - p^*\| \leq Q(\Delta T_{\max}) \text{diam}(P) < r_X \end{aligned} \quad (30)$$

for all $s \leq T$, the last inequality being *strict* by (9) (which subtracts a positive $e^{\mu^+ \Delta T_{\max} r'}$). Suppose $T < S$. The trajectory remains in the compact set $\mathcal{T} \subseteq X$ on $[0, T]$, so by (A0) it extends to T with $\varphi_f(T; p, x_\tau) \in X$, and (30) holds at $s = T$ with strict inequality; by continuity of $s \mapsto \text{dist}(\varphi_f(s; p, x_\tau), x^*([t_0, t_{N-1}]))$ there is $\varepsilon > 0$ with the distance $< r_X$, hence the trajectory in $\mathcal{T}$, on $[0, T + \varepsilon] \cap [0, S]$ — contradicting the definition of T as a supremum. Therefore $T = S$, and (30) holds on $[0, S]$.

Step 2 (perturbed launch). Let $T' = \sup\{t \in [0, S] : \varphi_f(s; p, z) \in \mathcal{T} \ \forall s \leq t\}$; $T' \geq 0$ since $\|z - x_\tau\| \leq r' \leq r_X$. Fix $s < T'$. Then $\varphi_f(s; p, z) \in \mathcal{T} \subseteq X$ and, by Step 1, $\varphi_f(s; p, x_\tau) \in \mathcal{T} \subseteq X$; X convex puts the chord in X , so $T' \leq T_X((z, p), (x_\tau, p))$ and Lemma 5 applies on $[0, T']$:

$$\|\varphi_f(s; p, z) - \varphi_f(s; p, x_\tau)\| \leq e^{\mu s} \|z - x_\tau\| \leq e^{\mu^+ \Delta T_{\max} r'},$$

using $e^{\mu s} \leq e^{\mu^+ \Delta T_{\max}}$ for both signs of μ . With (30),

$$\text{dist}(\varphi_f(s; p, z), x^*([t_0, t_{N-1}])) \leq e^{\mu^+ \Delta T_{\max} r'} + Q(\Delta T_{\max}) \text{diam}(P) < r_X$$

by (9), again *strictly*. The same extension-and-continuity argument as in Step 1 gives $T' = S$. $\square$

Corollary 1 (full-window estimates for admissible launches). Scope (S2). Assume (A0)–(A1). Let $\tau \in [t_0, t_{N-1}]$, $S = \min\{\Delta T_{\max}, t_{N-1} - \tau\}$, and let z, z_1, z_2 satisfy $\|\cdot - x^*(\tau)\| \leq r'$. Then for all $p, p_1, p_2 \in P$ and all $s \in [0, S]$: all the trajectories below lie in $\mathcal{T} \subseteq X$, every chord between two of them lies in X , and

$$e^{\mu-s} \|z_1 - z_2\| \leq \|\varphi_f(s; p, z_1) - \varphi_f(s; p, z_2)\| \leq e^{\mu s} \|z_1 - z_2\|, \quad (31)$$

$$\|\varphi_f(s; p_1, z) - \varphi_f(s; p_2, z)\| \leq Q(s) \|p_1 - p_2\|. \quad (32)$$

If moreover $\sigma + s \leq S$, then $\varphi_f(s; p, \varphi_f(\sigma; p, z)) = \varphi_f(\sigma + s; p, z)$ is one of the trajectories already confined above, so (31) applies to any pair of such same-parameter continuations. No statement is made about (32) for launches from a reachable state.

Proof. Lemma 7 places each trajectory in $\mathcal{T} \subseteq X$ for all $s \leq S$; X is convex, so every chord lies in X ; hence $T_X = S$ for each pair, and Lemmas 5 and 6 apply on the whole of $[0, S]$. The final sentence is the flow property together with the confinement already established: the continuation is the same trajectory, re-based. $\square$

Read this: §3–§4 cite *this corollary*, never Lemmas 5/6 directly; the lemmas are local and their hypothesis $t \leq T_X$ is discharged once and for all here. An earlier version of this corollary added a blanket clause asserting that all conclusions — including the parameter-sensitivity estimate (32) — survive replacing z by any reachable state $\varphi_f(\sigma; p_0, z)$. That is *false as stated*: the

flow property confines a continuation under the *same* parameter, but $\varphi_f(s; p_1, \varphi_f(\sigma; p_0, z))$ and $\varphi_f(s; p_2, \varphi_f(\sigma; p_0, z))$ for $p_1 \neq p_2$ are launches from a state that Lemma 7 places within r_X — not within r' — of the orbit, which is not a hypothesis this corollary has. The clause is withdrawn; Proposition 3 does not need it, and argues directly instead (see its proof).

Lemma 8 (largest residual of a window; [revised](#)). Scope (S3). *Assume (A0)–(A2) and work on the event E_X of (11). Let τ be a node, let $s \in (0, \Delta T_{\max}]$ and let $t_i \in (\tau, \tau + s]$. Then, with $\sigma_i = t_i - \tau$ and every $p \in P$,*

$$\|\varphi_f(\sigma_i; p, y_\tau) - y_i\| \leq r_{\tau, t_i}(p) = e^{\mu\sigma_i} \|\eta_\tau\| + Q(\sigma_i) \|p - p^*\| + \|\eta_i\| \leq \rho_{\sigma_i}(p) \leq \bar{\rho}_s(p). \quad (33)$$

Proof. On E_X , $\|\eta_\tau\| \leq r'$, so Lemma 7 places $\varphi_f(r; p, y_\tau)$, $\varphi_f(r; p, x_\tau)$ and $x^*(\tau + r)$ in $\mathcal{T} \subseteq X$ for all $r \leq \sigma_i$, and all chords in X by convexity; Corollary 1 therefore applies to every step below. Split $\varphi_f(\sigma_i; p, y_\tau) - y_i = [\varphi_f(\sigma_i; p, y_\tau) - \varphi_f(\sigma_i; p, x_\tau)] + [\varphi_f(\sigma_i; p, x_\tau) - x_i] - \eta_i$. Estimate (31) bounds the first bracket by $e^{\mu\sigma_i} \|\eta_\tau\|$; since $x_i = \varphi_f(\sigma_i; p^*, x_\tau)$, (32) bounds the second by $Q(\sigma_i) \|p - p^*\|$. This is $r_{\tau, t_i}(p)$ of (23); with $\|\eta_\tau\|, \|\eta_i\| \leq \|\eta\|_{\max}$ it is at most $\rho_{\sigma_i}(p)$, and $\rho_{\sigma_i} \leq \bar{\rho}_s$ by Lemma 4(i). $\square$

Read this: the last step is *not* monotonicity of ρ , which fails for $\mu < 0$; it is Lemma 4(i), i.e. the definition of the envelope $\bar{\rho}$.

3.2 The windowed cost near the truth

Proposition 1 (cost at the true parameter; [revised](#)). Scope (S3) for (a) and (c), (S4) for (b). *Assume (A0)–(A2). Consider a partition with longest window $\Delta T \leq \Delta T_{\max}$ and n_k data in window k , $\sum_k n_k = N - 1$.*

(a) Realisation-wise, on E_X : for every $p \in P$,

$$J_K(p) \leq \sum_{k=1}^K \sum_{t_i \in (\tau_{k-1}, \tau_k]} \left(e^{\mu(t_i - \tau_{k-1})} \|\eta_{\tau_{k-1}}\| + Q(t_i - \tau_{k-1}) \|p - p^*\| + \|\eta_i\| \right)^2, \quad (34)$$

$$J_K(p^*) \leq \|\eta\|_{\max}^2 \sum_{k=1}^K \sum_{j=1}^{n_k} (1 + e^{\mu j \Delta t})^2 \leq (N - 1) \|\eta\|_{\max}^2 (1 + e^{\mu^+ \Delta T})^2. \quad (35)$$

Moreover, with $\beta := Q(\Delta T) \text{diam}(P)$,

$$\sup_{p \in P} |J_K(p) - J_K^*(p)| \leq (N - 1) (1 + e^{\mu^+ \Delta T}) \|\eta\|_{\max} (2\beta + (1 + e^{\mu^+ \Delta T}) \|\eta\|_{\max}), \quad (36)$$

so $J_K \rightarrow J_K^*$ uniformly on P as $\|\eta\|_{\max} \rightarrow 0$, and $J_K^*(p^*) = 0$.

(b) In expectation. Under (A2-a),

$$\mathbb{E} J_K(p^*) \leq \text{tr } \Sigma \sum_{k=1}^K \sum_{j=1}^{n_k} (1 + e^{2\mu j \Delta t}) \leq (N - 1) \text{tr } \Sigma (1 + e^{2\mu^+ \Delta T}). \quad (37)$$

Under (A2-b) the same two bounds hold for $\mathbb{E}[J_K(p^*) \mid E_X]$, and $\mathbb{P}(E_X^c)$ is bounded by (12). No unconditional expectation is claimed under (A2-b).

(c) Single shooting. On E_X , $J_1(p) \leq \sum_{i=1}^{N-1} (e^{\mu(t_i - t_0)} \|\eta_0\| + Q(t_i - t_0) \|p - p^*\| + \|\eta_i\|)^2$, in which the initial-condition error is amplified over the whole record.

Proof. (a) (34) is (33) squared and summed. At p^* the middle term vanishes and the j -th datum of a window sits at $t_i - \tau_{k-1} = j\Delta t$, giving the first inequality of (35); for the second, $e^{\mu j\Delta t} \leq e^{\mu^+ \Delta T}$ in both cases ($\mu \geq 0$: $\mu j\Delta t \leq \mu \Delta T$; $\mu < 0$: $e^{\mu j\Delta t} \leq 1 = e^{\mu^+ \Delta T}$), and $\sum_k n_k = N - 1$. For (36), write $a = \|\varphi_f(\sigma; p, y_\tau) - y_i\|$ and $b = \|\varphi_f(\sigma; p, x_\tau) - x_i\|$ for the same index; then $|a^2 - b^2| \leq |a - b|(a + b)$, $|a - b| \leq e^{\mu\sigma} \|\eta_\tau\| + \|\eta_i\| \leq (1 + e^{\mu^+ \Delta T}) \|\eta\|_{\max}$ and $b \leq Q(\sigma) \|p - p^*\| \leq \beta$ by (32), so $a + b \leq 2\beta + (1 + e^{\mu^+ \Delta T}) \|\eta\|_{\max}$; sum over the $N - 1$ indices. The bound is uniform in $p \in P$ and vanishes linearly in $\|\eta\|_{\max}$.

(b) Expand $\|u - \eta_i\|^2 = \|u\|^2 - 2\langle u, \eta_i \rangle + \|\eta_i\|^2$ with $u = \varphi_f(t_i - \tau_{k-1}; p^*, y_{\tau_{k-1}}) - x_i$, a function of $\eta_{\tau_{k-1}}$ alone. Since $t_i \in (\tau_{k-1}, \tau_k]$ we have $t_i \neq \tau_{k-1}$, so η_i and $\eta_{\tau_{k-1}}$ are distinct members of the i.i.d. family; independence and $\mathbb{E}\eta_i = 0$ kill the cross term *termwise* (unaffected by η_{τ_k} appearing both as a residual noise in window k and as a launch noise in window $k + 1$: those are different summands). Then $\mathbb{E}\|\eta_i\|^2 = \text{tr } \Sigma$ and $\mathbb{E}\|u\|^2 \leq e^{2\mu j\Delta t} \mathbb{E}\|\eta_{\tau_{k-1}}\|^2$ by (31), which under (A2-a) applies surely because $\mathbb{P}(E_X) = 1$. Under (A2-b) run the identical computation under $\mathbb{P}(\cdot | E_X)$: by Lemma 1 the η_i are still independent and mean zero, so the cross term still vanishes, and $\mathbb{E}[\|\eta_i\|^2 | E_X] \leq \text{tr } \Sigma$ gives the same right-hand side. $\square$

What changed. (i) $L \rightarrow \mu$. (ii) The draft applied $e^{L\Delta T}$ to every datum in a window; (35) keeps the offset $j\Delta t$, and the geometric sum $\sum_{j \leq n_k} e^{2\mu j\Delta t}$ saves a factor of order $\mu\Delta T$ relative to $n_k e^{2\mu\Delta T}$ when $\mu > 0$. (iii) $\|\eta\|_{\max} \rightarrow \text{tr } \Sigma$ in expectation. On the FHN data the realised ratio $\|\eta\|_{\max}^2 / (\frac{1}{N-1} \sum_i \|\eta_i\|^2)$ is 4.9 (a direct measurement from the residual array; the isotropic Gaussian heuristic $1 + 2 \log N/d$ gives 5.6, a different quantity that the data happen to sit near and that does not apply to the anisotropic Σ). (iv) μ^+ and $N - 1$ in the collapsed bounds: for $\mu \leq 0$ they read $4(N - 1)\|\eta\|_{\max}^2$ and $2(N - 1)\text{tr } \Sigma$, whereas the draft's $e^{\mu\Delta T}$ form reversed the inequality. (v) The scope is now explicit: (a) is pathwise on E_X , (b) is unconditional only under bounded noise. (vi) The draft's “as $\Sigma \rightarrow 0$ ” has been replaced by the deterministic limit $\|\eta\|_{\max} \rightarrow 0$ with the explicit rate (36); a covariance does not index a pathwise sequence without a coupling.

3.3 Bound on the error $\|p^* - p^{(K)}\|$

Proposition 2 (displacement of the minimiser from the truth). Scope (S3) for (38), (S4) for the expectation form. Assume (A0)–(A2) and $p^{(K)} \in \arg \min_P J_K$ as in (16). Suppose J_K is m -strongly convex, $m > 0$, on a convex $U \subseteq P$ containing $p^{(K)}$ and p^* . Then, on E_X , realisation by realisation,

$$\|p^* - p^{(K)}\| \leq \sqrt{\frac{2}{m} (J_K(p^*) - J_K(p^{(K)}))} \leq \sqrt{\frac{2}{m} ((N - 1)\|\eta\|_{\max}^2 (1 + e^{\mu^+ \Delta T})^2 - J_K(p^{(K)}))}. \quad (38)$$

If in addition (A3) holds, then, under (A2-a), $\mathbb{E}\|p^* - p^{(K)}\|^2 \leq \frac{2}{m} ((N - 1)\text{tr } \Sigma(1 + e^{2\mu^+ \Delta T}) - \mathbb{E}J_K(p^{(K)}))$, and under (A2-b) the same bound holds for $\mathbb{E}[\|p^* - p^{(K)}\|^2 | E_X]$ with $\mathbb{E}[J_K(p^{(K)}) | E_X]$ on the right. For single shooting, $\|p^* - p^{(1)}\| \leq \sqrt{\frac{2}{m} (\sum_{i=1}^{N-1} (e^{\mu(t_i - t_0)} \|\eta_0\| + \|\eta_i\|)^2 - J_1(p^{(1)}))}$.

Proof. Lemma 2 with $F = J_K$, $p = p^{(K)}$, $q = p^* \in U$ gives the first bound; Proposition 1(a) gives the second. Under (A3) the pointwise inequality holds on a set of probability one (respectively, of full conditional probability given E_X) with a common constant and a common neighbourhood, so expectations may be taken on both sides, and Proposition 1(b) supplies the right-hand side. $\square$

Read this: four caveats. *First*, what was called “(A2)” in a previous revision of this proposition is the assumption listed as (A3) in §2; (A2) is the noise model. *Second*, (A3) is an assumption

about the *noise-dependent* objects J_K , $p^{(K)}$, m and U , and it is *not* verified for the FHN data: Figure 11 shows the noisy data term is indefinite at p^* for every $\kappa \geq 2$, and Figure 20(a) shows the points a derivative-free optimiser returns are positive definite in only 36% of the (seed, κ) cells. Without (A3) the expectation form is a modelling statement, not a consequence of (38). *Third*, the proof no longer uses $\nabla J_K(p^{(K)}) = 0$ — which is false when the minimiser lies on ∂P — but Lemma 2. The hypothesis $p^* \in U$ is essential and is stated: the inequality is applied at $q = p^*$. When this proposition is transferred to an unconstrained iterate through Remark 1, the variant needed is (R^{p^*}) , i.e. (R) together with $p^* \in U$; the weaker (R) suffices only for basin retention. *Fourth*, see Remark 3(ii): this proposition compares with p^* and therefore does not apply to the penalised objective actually optimised, whose minimiser is displaced from $p^{(K)}$ by up to $\gamma/(m\sqrt{n_p})$ under the containment hypotheses stated there. The bound is sharp in kind, not in size: the first inequality in (38) is an identity for a quadratic cost, so the whole slack is in the estimate of $J_K(p^*)$, i.e. in the exponential.

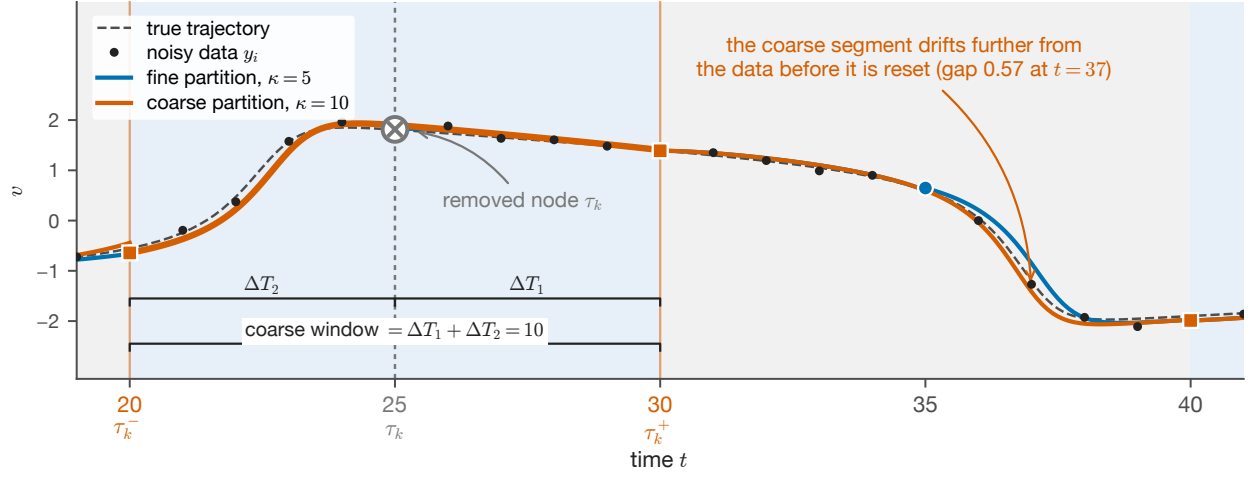
4 Propagating the best guess across shooting-node partitions

We now bound how the optimum moves when shooting nodes are removed. With $R = \{0, \dots, K\} \setminus \mathcal{I}_R$ and $r^+ = \min\{q \in R : q > r\}$ as in §2, the coarse cost is

$$\hat{\mathcal{J}}_K(p) = \sum_{r \in R \setminus \{K\}} \sum_{t_i \in (\tau_r, \tau_{r^+}]} \|\varphi_f(t_i - \tau_r; p, y_{\tau_r}) - y_i\|^2. \quad (39)$$

The intervals $(\tau_r, \tau_{r^+}]$, $r \in R \setminus \{K\}$, partition $(\tau_0, \tau_K]$, so (39) has the same $N - 1$ residual indices as (13); only the launching node of some of them changes. Every datum in a fine window D_k launched at a removed node τ_k is now predicted from the nearest retained node to its left, τ_k^- , through τ_k (Figure 6). Throughout this section the coarse partition is a *sub-partition* of the fine one. **Read this:** that hypothesis is *not* satisfied by most stages of Algorithm 1: by (26), only two of FULL’s fifteen transitions are node removals, and one of those is the degenerate terminal stage. The earlier version of this manuscript claimed the sub-partition hypothesis “is what the algorithm does”; it is withdrawn (§2.3). The general case is Corollary 6.

Removing the node τ_k : the segment launched from τ_k^- must now cover $[\tau_k^-, \tau_k^+]$



$$\|\hat{\mathcal{J}}_K(p) - \mathcal{J}_K(p)\| \leq 2 |\mathcal{I}_R| C_{\max} \left(e^{L(\Delta T_1 + \Delta T_2)} \|\eta\|_{\max} + e^{L\Delta T_1} \left[\frac{\tilde{L}}{L} (e^{L\Delta T_2} - 1) \|p - p^*\| + \|\eta\|_{\max} \right] \right)$$

$\Delta T_1 = \max_k (\tau_k - \tau_{k-1})$ = fine window length; $\Delta T_2 = \max_k (\tau_k - \tau_k^-)$ = distance from the removed node to its left neighbour (both = 5 here); $|\mathcal{I}_R|$ = number of removed nodes; $C_{\max}$ = largest segment/data magnitude over the affected windows
 $L = 3.05$, $\tilde{L} = 10.56$ (Lipschitz in x , in p); $\|\eta\|_{\max} = 0.159$. Background shading: the coarse ($\kappa = 10$) windows.

Figure 6: Node removal. Zoom on $t \in [19, 41]$ with the wrong parameter of Figure 2: the fine partition ($\kappa = 5$, blue segments) and the coarse one after removing the node at $t = 25$ (open marker; $\kappa = 10$, vermilion), with $\tau_k^- = 20$ and $\tau_k^+ = 30$; the measured gap between the fine and coarse segments at $t = 37$ is printed. After removal the segment launched at τ_k^- must cover $[\tau_k^-, \tau_k^+]$; the cost changes only on the windows adjacent to removed nodes. Replaces `comparison_shooting_nodes.png`. Inherited from R002 Figure 5; the bound printed in the R002 caption is the draft's version, superseded by Proposition 3.

Proposition 3 (cost change under node removal, per datum; revised). Scope (S3). Assume (A0)–(A2) and work on E_X . Let the coarse partition be a sub-partition of the fine one, with removed indices $\mathcal{I}_R \subset \{1, \dots, K-1\}$. For $k \in \mathcal{I}_R$ put

$$h_k = \tau_{k+1} - \tau_k, \quad w_k = \tau_k - \tau_k^-, \quad D_k = \{t_i \in (\tau_k, \tau_{k+1}]\}, \quad g_k(p) = \|\varphi_f(w_k; p, y_{\tau_k^-}) - y_{\tau_k}\|,$$

where τ_k^- is the nearest retained node to the left of τ_k . Then for every $p \in P$

$$|\hat{\mathcal{J}}_K(p) - \mathcal{J}_K(p)| \leq \sum_{k \in \mathcal{I}_R} \sum_{t_i \in D_k} e^{\mu(t_i - \tau_k)} g_k(p) \left[r_{\tau_k^-, t_i}(p) + r_{\tau_k, t_i}(p) \right], \quad (40)$$

and the node factor obeys

$$g_k(p) \leq r_{\tau_k^-, \tau_k}(p) = e^{\mu w_k} \|\eta_{\tau_k^-}\| + Q(w_k) \|p - p^*\| + \|\eta_{\tau_k}\| \leq \rho_{w_k}(p). \quad (41)$$

Proof. Which data change predictor. In $\mathcal{J}_K$ the datum $t_i \in (\tau_{k-1}, \tau_k]$ is launched from τ_{k-1} ; in $\hat{\mathcal{J}}_K$ from the last retained node at or before τ_{k-1} . So t_i changes predictor iff its fine launching node is removed, i.e. exactly for $t_i \in \bigcup_{k \in \mathcal{I}_R} D_k$. The half-open convention handles the two apparent

off-by-one cases: with τ_k removed and τ_{k-1} retained, the datum at time τ_k lies in $(\tau_{k-1}, \tau_k]$, keeps its predictor, and is correctly *not* in D_k ; with τ_{k-1} and τ_k both removed, the datum at τ_k lies in D_{k-1} , does change predictor, and is correctly counted there. The D_k are pairwise disjoint.

Admissibility of every flow evaluation, argued pairwise. Fix $k \in \mathcal{I}_R$ and $t_i \in D_k$, and put

$$u_i = \varphi_f(t_i - \tau_k^-; p, y_{\tau_k^-}), \quad v_i = \varphi_f(t_i - \tau_k; p, y_{\tau_k}), \quad a_i = \|u_i - y_i\|, \quad b_i = \|v_i - y_i\|.$$

On E_X we have $\|y_{\tau_k^-} - x^*(\tau_k^-)\| \leq r'$ and $\|y_{\tau_k} - x^*(\tau_k)\| \leq r'$, so Lemma 7 applies to each launch *separately*: the trajectory $s \mapsto \varphi_f(s; p, y_{\tau_k^-})$ lies in $\mathcal{T} \subseteq X$ for $s \leq t_i - \tau_k^- \leq \Delta T_{\max}$, and $s \mapsto \varphi_f(s; p, y_{\tau_k})$ lies in $\mathcal{T} \subseteq X$ for $s \leq t_i - \tau_k$. By the flow property the first of these is the coarse continuation $\varphi_f(s - \tau_k; p, \varphi_f(w_k; p, y_{\tau_k^-}))$ for $s \geq \tau_k$ — a *same-parameter* continuation, which is why no statement about launches from reachable states is needed. Both trajectories therefore lie in X on $[\tau_k, t_i]$ and, X being convex, so does the chord joining them at each time; hence $T_X((\varphi_f(w_k; p, y_{\tau_k^-}), p), (y_{\tau_k}, p)) \geq t_i - \tau_k$ and Lemma 5 applies to that pair on $[0, t_i - \tau_k]$.

The per-datum estimate. $\hat{J}_K - J_K = \sum_{k \in \mathcal{I}_R} \sum_{t_i \in D_k} (a_i^2 - b_i^2)$ and $|a_i^2 - b_i^2| \leq \|u_i - v_i\|(a_i + b_i)$ by the reverse triangle inequality. Lemma 5 at the *actual* elapsed time gives

$$\|u_i - v_i\| \leq e^{\mu(t_i - \tau_k)} \|\varphi_f(w_k; p, y_{\tau_k^-}) - y_{\tau_k}\| = e^{\mu(t_i - \tau_k)} g_k(p),$$

with the actual μ and no enlargement of the exponent, hence no monotonicity requirement and no μ^+ . Lemma 8 applied at node τ_k^- and datum t_i gives $a_i \leq r_{\tau_k^-, t_i}(p)$, and at node τ_k and datum t_i gives $b_i \leq r_{\tau_k, t_i}(p)$ — each at its own elapsed time and with its own realised noise, so neither an envelope nor the global $\|\eta\|_{\max}$ is needed. Summing gives (40). (41) is Lemma 8 applied to the single datum y_{τ_k} launched from $y_{\tau_k^-}$ over elapsed time w_k . $\square$

Corollary 2 (uniform-noise form). *Under the hypotheses of Proposition 3, replacing every realised noise norm by $\|\eta\|_{\max}$,*

$$|\hat{J}_K(p) - J_K(p)| \leq \sum_{k \in \mathcal{I}_R} \sum_{t_i \in D_k} e^{\mu(t_i - \tau_k)} g_k(p) [\rho_{t_i - \tau_k^-}(p) + \rho_{t_i - \tau_k}(p)]. \quad (42)$$

Proof. $r_{\sigma, t} \leq \rho_{t - \sigma}$ by (23) and $\|\eta_\bullet\| \leq \|\eta\|_{\max}$. $\square$

Corollary 3 (per-node form). *Under the same hypotheses,*

$$|\hat{J}_K(p) - J_K(p)| \leq \sum_{k \in \mathcal{I}_R} |D_k| e^{\mu^+ h_k} g_k(p) [\bar{\rho}_{h_k + w_k}(p) + \bar{\rho}_{h_k}(p)]. \quad (43)$$

Proof. For $t_i \in D_k$: $t_i - \tau_k \leq h_k$, so $e^{\mu(t_i - \tau_k)} \leq e^{\mu^+ h_k}$ (with μ^+ , because for $\mu < 0$ the correct majorant of $e^{\mu(t_i - \tau_k)}$ over $t_i - \tau_k \in [0, h_k]$ is $e^0 = 1$); $t_i - \tau_k^- \leq h_k + w_k$ and $t_i - \tau_k \leq h_k$, so $\rho_{t_i - \tau_k^-} \leq \bar{\rho}_{h_k + w_k}$ and $\rho_{t_i - \tau_k} \leq \bar{\rho}_{h_k}$ by Lemma 4(i). The inner sum has $|D_k|$ terms. $\square$

Corollary 4 (collapsed form). *Under the same hypotheses, with $n_{\max} = \max_{k \in \mathcal{I}_R} |D_k|$, $\Delta T_1 = \max_k (\tau_k - \tau_{k-1})$ over the fine partition and $\Delta T_2 = \max_{k \in \mathcal{I}_R} w_k$,*

$$|\hat{J}_K(p) - J_K(p)| \leq n_{\max} |\mathcal{I}_R| e^{\mu^+ \Delta T_1} \bar{\rho}_{\Delta T_2}(p) [\bar{\rho}_{\Delta T_1 + \Delta T_2}(p) + \bar{\rho}_{\Delta T_1}(p)] =: \Delta_K(p). \quad (44)$$

At $p = p^*$ this is $n_{\max} |\mathcal{I}_R| e^{\mu^+ \Delta T_1} (1 + e^{\mu^+ \Delta T_2}) [(1 + e^{\mu^+ (\Delta T_1 + \Delta T_2)}) + (1 + e^{\mu^+ \Delta T_1})] \|\eta\|_{\max}^2$, quadratic in the noise. Moreover Δ_K is non-decreasing in $\delta = \|p - p^*\|$, so $\sup_{q \in U} \Delta_K(q) = \Delta_K$ evaluated at $\delta = R_U := \sup_{q \in U} \|q - p^*\|$.

Proof. $h_k \leq \Delta T_1$, $w_k \leq \Delta T_2$, $|D_k| \leq n_{\max}$, $g_k \leq \rho_{w_k} \leq \bar{\rho}_{\Delta T_2}$ by (41) and Lemma 4(i), and $\bar{\rho}_{h_k+w_k} \leq \bar{\rho}_{\Delta T_1+\Delta T_2}$, $\bar{\rho}_{h_k} \leq \bar{\rho}_{\Delta T_1}$ by Lemma 4(ii); the sum has $|\mathcal{I}_R|$ terms. The value at p^* is Lemma 4(iv) and the monotonicity in δ is Lemma 4(iii). $\square$

Read this: four levels, each obtained from the one above by a single named enlargement, and each reversible by stepping back. (a) (40) is the sharpest statement the Grönwall machinery of §3 supports: the actual μ , the actual per-datum offsets, the two residual terms kept separately, and the *realised* noise at each launch node and each datum rather than the global $\|\eta\|_{\max}$. It is a theoretical bound, not an evaluable one: $\|\eta_\sigma\|$ and p^* are unknown in a discovery problem. The evaluable object is the score of (45) below, which is a different thing and is not claimed to be this one. (b) (42) substitutes $\|\eta\|_{\max}$ for realised noise; (43) enlarges the elapsed times; (44) maximises over k . A previous revision stated (42) as the primary bound, which discarded level (a) silently, and a still earlier one bounded $a_i + b_i$ by $2\bar{\rho}_{h_k+w_k}$, discarding the shorter of the two envelopes for a factor of two. (c) (44) is the form used in Theorem 1 and Corollaries 5–6, because those need a bound *uniform over q in a set*; it contains $|\mathcal{I}_R|$ but not *which* nodes, and the global $\|\eta\|_{\max}$ but not the noise at any particular node. (d) By Lemma 4(v), Δ_K is *piecewise* quadratic in $(\|\eta\|_{\max}, \delta)$, the branch switching on the sign of $\tilde{L}\delta - |\mu|\|\eta\|_{\max}$ when $\mu < 0$ and being the single quadratic branch throughout when $\mu \geq 0$; see (47). *What changed relative to the draft.* (i) $L \rightarrow \mu$, and $\mu \rightarrow \mu^+$ wherever an exponent is enlarged to a window length. (ii) The draft bounded $a_i + b_i$ through $\|u_i + v_i - 2y_i\| \leq \dots + 2\max\|y_i\|$, which injects the size of the data (2.34 on FHN) into the constant; bounding $a_i + b_i$ by two residuals makes the bound quadratic in $(\|\eta\|, \|p - p^*\|)$, as the left-hand side is, and removes a factor $\|y\|_{\max}/\|\eta\|_{\max} \approx 15$. (iii) The draft collapsed the sum over t_i to $|\mathcal{I}_R|$ without the number of data per window; $n_{\max}$ restores it. (iv) The draft's intermediate inequality carried $e^{2L(\Delta T_1+\Delta T_2)}$ while its statement had $e^{L(\Delta T_1+\Delta T_2)}$. (v) $\rho \rightarrow \bar{\rho}$ wherever an argument is enlarged, without which the steps bounding a shorter window's residual by a longer window's are false for $\mu < 0$.

4.1 Node selection: what is computable, and what it costs

For a proposed removal set R (a subset of the interior fine nodes) write $\tau_k^-(R)$ for the nearest node of $S \setminus R$ to the left of τ_k , $w_k(R) = \tau_k - \tau_k^-(R)$, $g_k(R; p) = \|\varphi_f(w_k(R); p, y_{\tau_k^-(R)}) - y_{\tau_k}\|$, and, for $t_i \in D_k$,

$$a_i(R; p) = \|\varphi_f(t_i - \tau_k^-(R); p, y_{\tau_k^-(R)}) - y_i\|, \quad b_i(p) = \|\varphi_f(t_i - \tau_k; p, y_{\tau_k}) - y_i\|,$$

the *proposed coarse* and *current fine* residuals. The proof of Proposition 3 bounds $|a_i^2 - b_i^2|$ by $e^{\mu(t_i - \tau_k)} g_k(R; p)(a_i + b_i)$ before any residual is bounded by r , ρ or $\bar{\rho}$, so

$$S(R; p) = \sum_{k \in R} \sum_{t_i \in D_k} e^{\mu(t_i - \tau_k)} g_k(R; p) (a_i(R; p) + b_i(p)) \geq |\hat{\mathcal{J}}_K(p) - J_K(p)| \quad (45)$$

is a valid upper bound every factor of which is computed from quantities the integrator returns, given a numerical value for μ (any upper bound on μ may be used, since $t_i \geq \tau_k$; μ is a constant of (A0) and is itself not instantiated in this report, see P1 in §8). Two rules: (i) score complete proposed removal sets by $S(R; p)$ at the current iterate and take the smallest; (ii) *sequential greedy*: $R_0 = \emptyset$ and $R_j = R_{j-1} \cup \{\arg \min_c S(R_{j-1} \cup \{c\}; p)\}$ over retained interior candidates c , *recomputing* τ_k^- , w_k , g_k and the a_i of the whole current set after each removal. Qualitatively, (45) prefers nodes that launch few data ($|D_k|$ small), sit close to their retained left neighbour (w_k small), and above all have a small coarse residual g_k .

Read this: three things this paragraph says that a previous revision did not. (1) *The score a previous revision proposed was not observable.* It read $\sum \sum e^{\mu(t_i - \tau_k)} g_k(R; p) [\rho_{t_i - \tau_k^-}(R)(p) + \rho_{t_i - \tau_k}(p)]$, and $\rho_s(p) = (1 + e^{\mu s}) \|\eta\|_{\max} + Q(s) \|p - p^*\|$ contains the noise sup-norm and the distance to the truth, neither of which exists in a discovery problem. Making g_k observable does not make the product observable. That form is withdrawn in favour of (45). (2) *The exact gap is observable at the same cost, and dominates the score for ranking — in absolute value.* The integration that produces a_i produces $a_i^2 - b_i^2$, so $\hat{J}_K(p) - J_K(p) = \sum_{k \in R} \sum_{t_i \in D_k} (a_i^2 - b_i^2)$ is available at exactly the cost of (45), with no slack and no μ . *Rank by $|\hat{J}_K(p) - J_K(p)|$, not by the signed gap.* The perturbation theory controls the absolute value, and every consumer of it — Theorem 1, Corollary 5 — consumes a two-sided bound; ranking by the signed quantity would prefer a large *negative* change of objective, which is a different and much larger perturbation, not a small one. If a large negative change of objective is deliberately wanted (a heuristic that pursues descent rather than partition continuity), that is a separate rule, unrelated to minimising the perturbation bound, and must be stated as such. (45) and, above it, Δ_K of (44) earn their place only where a bound *uniform over q in a set* is required — which is what Theorem 1 and Corollary 5 consume and what a pointwise evaluation cannot supply. (3) *There is no set-independent per-node score.* τ_k^- is the nearest *retained* node, so removing one node changes w_k , τ_k^- , g_k and every a_i for the already-removed nodes in the same consecutive run to its right; a per-node term is a contribution *given the current retained set*, not a marginal cost that can be ranked once. Ranking once and removing the top $|\mathcal{I}_R|$ can select a set whose bound is far from the sum of its pre-removal scores, most severely for block removals — the case of Figure 8(b).

Cost. Let R be the current removal set and C the number of retained interior candidates. Evaluating (45) for a given R costs one coarse-window integration per $k \in R$, over $[\tau_k^-(R), \max D_k]$ (length $w_k + h_k$), each of which yields g_k and all the a_i of that node as by-products; the b_i are residuals the fine objective already computes. Hence:

- **per candidate c :** $O(1)$ coarse-window integrations for c itself, *plus* a re-integration for every already-removed $k \in R$ whose retained left neighbour changes when c is removed — none if c is isolated from R , and the whole consecutive run to the right of c in the worst case, i.e. $O(|R|)$;
- **per greedy step:** $O(C |R|)$ window integrations in the worst case, $O(C)$ when removals stay isolated;
- **whole greedy schedule** of $|\mathcal{I}_R|$ removals: $O(|\mathcal{I}_R| C |R|) = O(C |\mathcal{I}_R|^2)$ in the worst case.

Read this: caching, which is a remark and not the headline figure. Removing c changes τ_k^- only for nodes in the removed run immediately to its right, so contributions cached by the key $(\tau_k^-(R), \tau_k)$ remain valid for every other candidate and a greedy step re-evaluates only the candidates whose left retained neighbour changed at the previous step — $O(1)$ candidates for isolated removals. The worst-case figures above stand. Standing caveat: all of this minimises a *bound* on the cost gap (or, for the exact gap, the absolute cost gap itself at one p), which is not the same as minimising the displacement of the minimiser; that step needs the strong-convexity premise of Theorem 1 and the reconnection hypothesis (R) of Remark 1. The design rule of the draft survives in this weakened form: remove few nodes at a time, keep w_k small, prefer nodes with a small measured g_k (Figure 8). “Prefer low-noise nodes” does *not* survive as an implementable rule, because $\|\eta_{\tau_k}\|$ in (41) is not observable.

4.2 Displacement of the minimiser

Theorem 1 (a posteriori, conditional displacement of the minimiser under node removal; *revised*). Scope (S3). Assume (A0)–(A2) and work on E_X . Let $p^{(K)} \in \arg \min_P J_K$ and $\hat{p}^{(K)} \in \arg \min_P \hat{J}_K$ as in (16), and suppose J_K is m -strongly convex, $m > 0$, on a convex $U \subseteq P$ containing both. Then

$$\|\hat{p}^{(K)} - p^{(K)}\| \leq \sqrt{\frac{2}{m}(\Delta_K(p^{(K)}) + \Delta_K(\hat{p}^{(K)}))} \leq \sqrt{\frac{4}{m} \sup_{q \in U} \Delta_K(q)}, \quad (46)$$

with Δ_K from (44).

Proof. Lemma 2 with $F = J_K$, $p = p^{(K)}$, $q = \hat{p}^{(K)} \in U$ gives $\frac{m}{2}\|\hat{p}^{(K)} - p^{(K)}\|^2 \leq J_K(\hat{p}^{(K)}) - J_K(p^{(K)})$. Insert $\pm \hat{J}_K(\hat{p}^{(K)})$ and $\pm \hat{J}_K(p^{(K)})$:

$$J_K(\hat{p}^{(K)}) - J_K(p^{(K)}) = [J_K(\hat{p}^{(K)}) - \hat{J}_K(\hat{p}^{(K)})] + [\hat{J}_K(\hat{p}^{(K)}) - \hat{J}_K(p^{(K)})] + [\hat{J}_K(p^{(K)}) - J_K(p^{(K)})].$$

The middle bracket is ≤ 0 because $\hat{p}^{(K)}$ minimises $\hat{J}_K$ over P and $p^{(K)} \in P$; the outer brackets are bounded by Corollary 4. The second inequality of (46) replaces each Δ_K by its supremum over U . $\square$

Writing $E = \|\eta\|_{\max}$, $\delta = \|q - p^*\|$ and $c_0 = n_{\max}|\mathcal{I}_R|e^{\mu^+ \Delta T_1}$, the bound of (44) is

$$\Delta_K(q) = c_0 \bar{\rho}_{\Delta T_2}(q) [\bar{\rho}_{\Delta T_1 + \Delta T_2}(q) + \bar{\rho}_{\Delta T_1}(q)], \quad (47)$$

a *piecewise* homogeneous quadratic in (E, δ) with non-negative deterministic coefficients on each branch. By Lemma 4(v), and assuming $\Delta T_1, \Delta T_2 > 0$:

- if $\mu \geq 0$, or if $\mu < 0$ and $\tilde{L}\delta > |\mu|E$, every $\bar{\rho}$ equals the corresponding ρ and $\Delta_K = c_0 \rho_{\Delta T_2}[\rho_{\Delta T_1 + \Delta T_2} + \rho_{\Delta T_1}]$;
- if $\mu < 0$ and $\tilde{L}\delta \leq |\mu|E$, then $\bar{\rho}_s = 2E$ for *all three* arguments, so

$$\Delta_K = c_0 (2E) [2E + 2E] = 8c_0 E^2. \quad (48)$$

Read this: the factor in (48) is 8, not 4: the bracket of (47) holds *two* envelopes, each equal to $2E$ on that branch, so the product is $2E \cdot 4E$. An earlier version wrote $4c_0 E^2$, having collapsed the bracket as if it held one. The check that settles it is the agreement with Corollary 4 evaluated at $p = p^*$, which must lie on this branch when $\mu < 0$ since $\delta = 0$: there $\Delta_K(p^*) = c_0(1 + e^{\mu^+ \Delta T_2})[(1 + e^{\mu^+(\Delta T_1 + \Delta T_2)}) + (1 + e^{\mu^+ \Delta T_1})]E^2 = c_0 \cdot 2 \cdot 4 E^2 = 8c_0 E^2$ for $\mu < 0$. The two routes must agree, and any future change to (44) must be checked against both. On every branch Δ_K is non-decreasing in δ , so $\sup_{q \in U} \Delta_K(q)$ is Δ_K evaluated at $\delta = R_U := \sup_{q \in U} \|q - p^*\|$: the supremum in (46) is computable, not merely finite. On the branch (48) the displacement bound (46) reads $\|\hat{p}^{(K)} - p^{(K)}\| \leq \sqrt{32c_0 E^2/m} = 4E\sqrt{2c_0/m}$ and the smallness condition (B2) of Corollary 5 reads $8c_0 E^2 < mr^2/4$, i.e. $E < (r/4)\sqrt{m/(2c_0)}$. A smooth quadratic majorant $\Delta_K \leq \Delta_K^+$, obtained by replacing each $\bar{\rho}$ with ρ^+ , is given in Remark 2; it is convenient (for instance for solving (B2) for r in closed form) and strictly loose for $\mu < 0$ away from p^* . **Read this:** Theorem 1 is a *conditional a posteriori* estimate, not a proof that guess propagation stays in the right basin. It assumes what a displacement theorem is supposed to help establish — that $\hat{p}^{(K)}$ already lies in the old strong-convexity neighbourhood — and in its first form the right-hand side contains $\hat{p}^{(K)}$, so the inequality is implicit; the second form is explicit but needs U to be named, and R_U finite. Corollary 5 supplies the smallness condition under which the premise can be dispensed with, at the

price of speaking about the minimiser of $\hat{J}_K$ *restricted to a ball* rather than its global minimiser. The proof no longer uses $\nabla J_K(p^{(K)}) = 0$, so the statement covers minimisers on ∂P ; when it is applied to a number of §6 it is conditional on (R) of Remark 1, and then only in its restricted form. *What changed.* The structure of the proof is the draft's; the bound inherits the improvements of Proposition 3, so the displacement now scales like the noise (through $\sqrt{\Delta_K} \sim \|\eta\|_{\max}$), not like $\sqrt{\text{noise} \times \text{attractor size}}$. The draft mixed $L(p)$, $L(p^*)$ and $L_K = \max\{L(p^{(K)}), L(\hat{p}^{(K)})\}$ in one formula; with the constants taken over $X \times P$ the distinction disappears. A previous revision called Δ_K “a quadratic polynomial ... with coefficients ... and smaller”, which was wrong on both counts: with $\bar{\rho}$ it is only piecewise quadratic, and the third coefficient was left vague.

Corollary 5 (basin retention under node removal). Scope (S3). *Assume (A0)–(A2) and work on E_X . Let $r > 0$ and $\bar{B} = \{q : \|q - p^{(K)}\| \leq r\}$. Suppose*

(B0) $\bar{B} \subseteq P$;

(B1) J_K is m -strongly convex on $\bar{B}$ with $m > 0$, and $p^{(K)} \in \arg \min_P J_K$;

(B2) $\bar{\Delta} := \sup_{q \in \bar{B}} \Delta_K(q) < \frac{mr^2}{4}$.

Then the minimum of $\hat{J}_K$ over $\bar{B}$ is attained at an interior point $\hat{p}$ of $\bar{B}$, and

$$\|\hat{p} - p^{(K)}\| \leq 2\sqrt{\frac{\bar{\Delta}}{m}} < r. \quad (49)$$

Moreover $\hat{p}$ is a stationary point of $\hat{J}_K$, $\nabla \hat{J}_K(\hat{p}) = 0$, and hence a local minimiser of $\hat{J}_K$ on P .

Proof. $\hat{J}_K$ is continuous and $\bar{B}$ compact, so the minimum over $\bar{B}$ is attained. Let $q \in \partial \bar{B}$, i.e. $\|q - p^{(K)}\| = r$; by (B0) $q \in P$, so Lemma 2 with $U = \bar{B}$ gives $J_K(q) \geq J_K(p^{(K)}) + \frac{m}{2}r^2$. Corollary 4 gives $|\hat{J}_K - J_K| \leq \bar{\Delta}$ on $\bar{B}$ (legitimate by (B0), which puts $\bar{B}$ inside the region where the constants of (A0) hold), so

$$\hat{J}_K(q) \geq J_K(q) - \bar{\Delta} \geq J_K(p^{(K)}) + \frac{m}{2}r^2 - \bar{\Delta} \geq \hat{J}_K(p^{(K)}) + \frac{m}{2}r^2 - 2\bar{\Delta} > \hat{J}_K(p^{(K)})$$

by (B2). Hence every boundary point of $\bar{B}$ has strictly larger $\hat{J}_K$ than the interior point $p^{(K)}$, so the minimiser $\hat{p}$ over $\bar{B}$ is interior to $\bar{B}$. For the bound, $\hat{p}, p^{(K)} \in \bar{B}$ and, by Lemma 2 again,

$$\begin{aligned} \frac{m}{2}\|\hat{p} - p^{(K)}\|^2 &\leq J_K(\hat{p}) - J_K(p^{(K)}) = \underbrace{[J_K(\hat{p}) - \hat{J}_K(\hat{p})]}_{\leq \bar{\Delta}} + \underbrace{[\hat{J}_K(\hat{p}) - \hat{J}_K(p^{(K)})]}_{\leq 0} \\ &\quad + \underbrace{[\hat{J}_K(p^{(K)}) - J_K(p^{(K)})]}_{\leq \bar{\Delta}} \leq 2\bar{\Delta}, \end{aligned}$$

the middle bracket being ≤ 0 because $\hat{p}$ minimises $\hat{J}_K$ over $\bar{B} \ni p^{(K)}$. Finally, $\hat{p}$ is interior to $\bar{B}$ and $\bar{B} \subseteq P$ by (B0), so there is $\varepsilon > 0$ with $B(\hat{p}, \varepsilon) \subseteq \bar{B} \subseteq P$; hence $\hat{p} \in \text{int } P$, the unconstrained first-order condition applies, and $\nabla \hat{J}_K(\hat{p}) = 0$. $\square$

Read this: what this does and does not say, and what the hypotheses cost. (B0) is *not decorative*: without $\bar{B} \subseteq P$ the quantity $\sup_{\bar{B}} \Delta_K$ is a supremum of a bound whose constants are suprema over $X \times P$ and therefore need not apply on part of $\bar{B}$. *Interiority costs nothing extra here, but positive clearance from ∂P does.* Since $r > 0$, (B0) makes $\bar{B}$ a full-dimensional closed ball inside P , so every point interior to $\bar{B}$ is automatically interior to P and the stationarity conclusion

needs no hypothesis beyond (B0); an earlier version of this corollary added $\bar{B} \subseteq \text{int } P$, which is redundant. The genuine cost is that (B0) with $r > 0$ forces $p^{(K)} \in \text{int } P$ at distance at least r from ∂P , so *this corollary does not cover a minimiser on ∂P* , and the claim that Lemma 2 frees the theory from interiority assumptions must not be read as covering it. Proposition 2, Theorem 1 and the displacement bound of Corollary 6 do hold verbatim for boundary minimisers, because they use only (17). *It does not bound the global minimiser of $\hat{J}_K$* : the coarse landscape acquires minima elsewhere as the windows grow (Figures 5, 10), and no perturbation argument can exclude a lower value far away. It says the coarse problem *restricted to the ball* has its minimiser strictly inside, close to $p^{(K)}$ — which is the object guess propagation chases, since the next stage starts *at* $p^{(K)}$. A continuous descent path started at $p^{(K)}$ cannot leave $\bar{B}$, because $\hat{J}_K > \hat{J}_K(p^{(K)})$ on the whole boundary shell; Nelder–Mead’s iterates are not a continuous path and can in principle jump the shell, so for the runs of §6 this is a statement about the restricted problem, not a guarantee about the optimiser used — and, by Remark 1, applies to those runs only under (R). Condition (B2) is a genuine smallness requirement: $\bar{\Delta}$ grows with $|\mathcal{I}_R|$, $n_{\max}$, ΔT_1 and ΔT_2 , so it is the formal version of “remove few nodes at a time”. By (47), $\bar{\Delta} = \Delta_K$ evaluated at $\delta = \|p^{(K)} - p^*\| + r$.

Corollary 6 (re-partition with simultaneous node additions and removals). Scope (S3). *Assume (A0)–(A2) and work on E_X . Let A and B be two node sets, both containing t_0 and t_{N-1} , and let $C = A \cup B$ be their common refinement. Write J_A, J_B, J_C for the corresponding costs (13), let ΔT_1^C be the longest window of C , $n_{\max}^C$ the largest number of data in a window of C , and for $S \in \{A, B\}$ let ΔT^S be the longest window of S . Put*

$$\Delta^{C \rightarrow S}(p) = n_{\max}^C |C \setminus S| e^{\mu^+ \Delta T_1^C} \bar{\rho}_{\Delta T^S}(p) \left[\bar{\rho}_{\Delta T_1^C + \Delta T^S}(p) + \bar{\rho}_{\Delta T_1^C}(p) \right], \quad \Delta_{A,B} = \Delta^{C \rightarrow A} + \Delta^{C \rightarrow B}. \quad (50)$$

Then $|J_A(p) - J_B(p)| \leq \Delta_{A,B}(p)$ for every $p \in P$. At $p = p^$,*

$$\Delta^{C \rightarrow S}(p^*) = n_{\max}^C |C \setminus S| e^{\mu^+ \Delta T_1^C} (1 + e^{\mu^+ \Delta T^S}) \left[(1 + e^{\mu^+ (\Delta T_1^C + \Delta T^S)}) + (1 + e^{\mu^+ \Delta T_1^C}) \right] \|\eta\|_{\max}^2. \quad (51)$$

If moreover J_A is m -strongly convex on a convex $U \subseteq P$ containing $p_A \in \arg \min_P J_A$ and $p_B \in \arg \min_P J_B$, then

$$\|p_B - p_A\| \leq \sqrt{\frac{2}{m} (\Delta_{A,B}(p_A) + \Delta_{A,B}(p_B))} \leq \sqrt{\frac{4}{m} \sup_{q \in U} \Delta_{A,B}(q)}, \quad (52)$$

and if instead the hypotheses of Corollary 5 hold with J_A, J_B and $\Delta_{A,B}$ in place of $J_K, \hat{J}_K$ and Δ_K , the minimiser of J_B over $\bar{B}(p_A, r)$ is interior to that ball and satisfies $\|\cdot - p_A\| \leq 2\sqrt{\bar{\Delta}/m} < r$.

Proof. $A \subseteq C$ and $B \subseteq C$, so A is obtained from the fine partition C by removing the nodes $C \setminus A$, and likewise for B ; neither removed set contains the endpoints. Corollary 4 applied twice with fine partition C gives $|J_A - J_C| \leq \Delta^{C \rightarrow A}$ and $|J_C - J_B| \leq \Delta^{C \rightarrow B}$, where ΔT_2 for the removal $C \rightarrow S$ is at most ΔT^S , because the nearest retained node to the left of any removed node lies in S and successive S -nodes are at most ΔT^S apart. The triangle inequality gives the first claim; (51) is Lemma 4(iv). For (52), run the proof of Theorem 1 with $(J_K, \hat{J}_K, \Delta_K)$ replaced by $(J_A, J_B, \Delta_{A,B})$: the only properties used are the pointwise two-sided bound, $J_B(p_B) \leq J_B(p_A)$, and Lemma 2; monotonicity of each $\bar{\rho}$ in δ (Lemma 4(iii)) gives the supremum form. The last claim is Corollary 5 verbatim with the same substitution. $\square$

Read this: (50) inherits the corrected step of Proposition 3: the two residual envelopes are kept *separately* in the bracket, and the leading factor is $n_{\max}^C$, not $2n_{\max}^C$. A previous revision carried

the discarded relaxation $2n_{\max}^C |C \setminus S| e^{\mu^+ \Delta T_1^C} \bar{\rho}_{\Delta T^S} \bar{\rho}_{\Delta T_1^C + \Delta T^S}$, which is still a valid upper bound — the bracket is at most twice its longer envelope — but strictly weaker, and inconsistent with the tightened Proposition 3 it is supposed to inherit. Every downstream constant (the displacement (52), and the (B2) threshold when Corollary 5 is applied with $\Delta_{A,B}$) is computed from (50) as displayed.

Remark 4 (the re-partition bound on this algorithm). For the FULL schedule’s stage $\kappa = 2 \rightarrow \kappa = 3$ (a stage that is *not* a node removal, see §2.3), with $N = 101$ and the common terminal node t_{100} belonging to every partition:

$$A = \{0, 2, 4, \dots, 98\} \cup \{100\} \text{ (51 nodes),} \quad B = \{0, 3, 6, \dots, 99\} \cup \{100\} \text{ (35 nodes),}$$

$$A \cap B = \{0, 6, \dots, 96\} \cup \{100\} \text{ (18 nodes),} \quad C = A \cup B \text{ (68 nodes),}$$

so $|C \setminus A| = 17$, $|C \setminus B| = 33$. The gaps of C are 1 or 2 data intervals (the pattern 0, 2, 3, 4, 6 repeats with period 6; the tail is 96, 98, 99, 100), giving $\Delta T_1^C = 2\Delta t$ and $n_{\max}^C = 2$; the gaps of A are all $2\Delta t$ (including $98 \rightarrow 100$) and the largest gap of B is $3\Delta t$, so $\Delta T^A = 2\Delta t$ and $\Delta T^B = 3\Delta t$. The bound is therefore driven by $|C \setminus A| + |C \setminus B| = 50$ removed-node terms — more than for a single clean removal. Corollary 6 restores a theorem for the algorithm as run; it does not make its stages cheap. **Read this:** every count above *includes the terminal node* t_{100} , which is a node of every partition. A previous version of this remark reported 50, 34, 17 and 67, which are the *launch-node* counts; mixing the two conventions in one example is what the indexing paragraph of §2 exists to prevent. The removed-node counts 17 and 33 are the same under either convention, since $|C \setminus S| = |C| - |S|$ and both counts shift by the same endpoint.

5 Refinement: trajectory-dependent exponents and a weighted norm

The bounds above are a priori: they hold for every right-hand side with the same constants and are attained by the linear system $\dot{x} = \mu x$, so they are necessarily exponential. Along an attracting orbit the actual growth is far smaller, and the same proof yields a sharper, trajectory-dependent statement — but, as the remark after it records, one that may *not* simply be substituted back into the propositions.

Corollary 7 (per-window and per-subinterval exponents). Scope (S1). *Assume (A0) and let $\mu(t) = \lambda_{\max}(\frac{1}{2}(A(t) + A(t)^\top))$ for the mean-value matrix $A(t)$ of (28) associated with the specific pair of trajectories and the specific parameter under consideration, on an interval where that pair is confined in the sense of Definition 1. Then*

(i) *(state sensitivity, local form) for $0 \leq a \leq b$,*

$$\|\delta(b)\| \leq \exp\left(\int_a^b \mu(r) dr\right) \|\delta(a)\|, \quad \delta = \varphi_f(\cdot; p, x_1) - \varphi_f(\cdot; p, x_2); \quad (53)$$

(ii) *(parameter sensitivity, local form) with $\delta = \varphi_f(\cdot; p_1, x_0) - \varphi_f(\cdot; p_2, x_0)$ and $\delta(0) = 0$,*

$$\|\delta(t)\| \leq \tilde{L} \|p_1 - p_2\| \int_0^t \exp\left(\int_s^t \mu(r) dr\right) ds. \quad (54)$$

Both are realisation-wise and pair-specific: $\mu(\cdot)$ is built from the two trajectories and the parameter named in the statement, and from no others.

Proof. (i) is (28) integrated between a and b instead of 0 and t . (ii) is the Dini-derivative inequality $D^+\|\delta\| \leq \mu(t)\|\delta\| + \tilde{L}\|p_1 - p_2\|$ of Lemma 6 integrated by variation of constants. $\square$

Remark 5 (what would be needed to substitute these into the propositions). Corollary 7 does *not* license replacing e^{μ^+s} by e^{Λ_k} , or by M_k of (20), anywhere in §3–§4. What would license a substitution is a pair of *deterministic envelopes* taken over the class of pairs the proofs actually use. Call a pair (ξ_1, ξ_2) of solutions of $\dot{\xi} = f(\xi; p_j)$ on $[a, b] \subseteq [t_0, t_{N-1}]$, with $p_1, p_2 \in P$, *confined* if $[\xi_1(s), \xi_2(s)] \subset X$ for every $s \in [a, b]$, and write $\mu_\xi(s) = \lambda_{\max}(\frac{1}{2}(A_\xi + A_\xi^\top))$ for its mean-value matrix $A_\xi(s) = \int_0^1 J_f(\xi_2(s) + \theta(\xi_1(s) - \xi_2(s)); p_1) d\theta$. For $h > 0$ put

$$\mathcal{M}(h) = \sup \left\{ \exp \left(\int_a^b \mu_\xi(r) dr \right) : \text{confined pairs with } p_1 = p_2 \in P, 0 \leq b - a \leq h \right\}, \quad (55)$$

$$\mathcal{Q}(h) = \sup \left\{ \tilde{L} \int_a^b \exp \left(\int_s^b \mu_\xi(r) dr \right) ds : \text{confined pairs with } p_1, p_2 \in P, 0 \leq b - a \leq h \right\}. \quad (56)$$

With these, Lemma 8, Propositions 1 and 3, Theorem 1 and Corollaries 5–6 hold with $e^{\mu^+s} \rightsquigarrow \mathcal{M}(s)$ and $Q(s) \rightsquigarrow \mathcal{Q}(s)$, and nothing is lost, since $\mu_\xi(r) \leq \mu$ pointwise on X gives $\mathcal{M}(h) \leq e^{\mu^+h}$ and $\mathcal{Q}(h) \leq Q(h)$. **Read this:** *why the class must be “confined”, not “launched within r' of the orbit”.* On E_X every shooting node satisfies $\|y_\tau - x^*(\tau)\| \leq r'$, so Lemma 7 confines the trajectory launched from *each* such node — separately, under its own parameter — to $\mathcal{T} \subseteq X$, and convexity of X places every joining chord in X . Every pair used in §3–§4 is a pair of such trajectories, so every such pair is confined; that, and not any statement about launches from reachable states, is what licenses the substitution. Note that the pairs are *not* all launched at a common time or from states within r' of the orbit at the comparison time: the coarse predictor of Proposition 3 is compared with the fine one at τ_k , where it has already run for w_k and is known only to be within r_X of the orbit, and the pairs of Corollary 6 are launched from the nodes of two different partitions. A previous version of this remark defined the suprema over pairs “launched anywhere within r' of the orbit, at any $p \in P$ ”, which excludes those pairs and left the advertised substitution unlicensed in precisely the place it was advertised; and “at any $p \in P$ ” is ambiguous for $\mathcal{Q}$, whose pairs carry *two* parameters. Both are fixed above. *Four reasons the per-window quantities cannot play this role*, each sufficient on its own. (1) *Pair dependence.* $\mu(\cdot)$ is attached to one pair and one parameter, while the proofs use at least five: noisy versus clean launch, p versus p^* , coarse versus fine prediction, different $q \in U$, and windows of two partitions and their common refinement. (2) *Interval spanning.* A coarse interval crosses several fine windows and $\exp(\int_a^b \mu)$ is then a *product* of per-window factors, which $\max_k M_k$ does not bound. (3) *The forcing integral.* $\int_0^t \exp(\int_s^t \mu) ds \leq M_k t$ requires $[s, t]$ to lie inside window k for every $s \leq t$, which fails as soon as t is in a later window than 0. (4) *Randomness.* An a posteriori M_k computed along a realised noisy pair is a random variable and cannot be factored out of $\mathbb{E}\|\eta\|^2$. And what §6 reports is neither $\mathcal{M}$ nor M_k : Table 2 gives *linearised endpoint* exponents e^{Λ_k} computed from $J_f(x^*(t); p^*)$ along the *true* orbit at the *true* parameter with windows aligned to t_0 — the infinitesimal-perturbation limit of one pair. These are not suprema over a tube of trajectories, not valid at another p , and not sub-interval bounds ($M_k \geq \max\{1, e^{\Lambda_k}\}$, strictly wherever $\mu(\cdot)$ changes sign inside the window, which on FHN happens twice per period, Figure 7c). $\mathcal{M}(h)$ is not computed anywhere in this report. The node-placement rule below is therefore a *heuristic* read off a measurement, and a previous version of Corollary 7 carried it as a licensed substitution (an item (iii) asserting that every proposition holds with $e^{\mu\Delta T}$ replaced by $\max_k e^{\Lambda_k}$), which is withdrawn.

For FHN the heuristic reads: keep nodes just before the fast jumps, where $\int \mu$ accumulates, and remove them freely on the slow branches, where the linearised window exponents are negative. What

no Grönwall-type argument can deliver is the observed *plateau* of the amplification (Figure 7a): that is a property of the attractor and would require a contraction metric (impossible globally on a limit cycle, whose phase direction is neutral), a Lyapunov function, or a Floquet analysis of the variational equation.

Remark 6 (weighted norms: what actually transfers). Let $\|x\|_D = \|Dx\|$ with D invertible, μ_D the logarithmic norm of $DJ_f D^{-1}$ over $X \times P$, and $\text{cond}(D) = \|D\| \|D^{-1}\|$. Lemma 5 holds verbatim in $\|\cdot\|_D$ with μ_D in place of μ , and converting to the Euclidean norm costs one factor of the condition number: $\|\delta(t)\| \leq \text{cond}(D) e^{\mu_D t} \|\delta(0)\|$. *Nothing else transfers for free*: Lemma 6 requires the weighted parameter constant $\tilde{L}_D = \sup_{X \times P} \|D \partial f / \partial p\|_{2 \leftarrow 2}$; Lemma 8 requires the weighted noise $\|D\eta_i\|$ with $\mathbb{E}\|D\eta_i\|^2 = \text{tr}(D\Sigma D^\top)$; a cost built from D -weighted residuals has a *different* *minimiser* from (13); and a D -norm residual bound converted back into a statement about the Euclidean *squared* cost picks up the conversion factor *squared*. We therefore make no weighted claim about J_K , $\hat{J}_K$, $p^{(K)}$ or $\hat{p}^{(K)}$. The weighted entries of Table 2 are reported as what they are: linearised endpoint amplification factors in $\|\cdot\|_D$ along the true orbit, whose Euclidean counterpart is larger by at most $\text{cond}(D) = 3.54$. On FHN, $D = \text{diag}(1, s)$ with $s = \sqrt{|J_{12}|/|J_{21}|} = 3.54$ makes the off-diagonal entries of $DJ_f D^{-1}$ equal and opposite, so its symmetric part is diagonal and $\mu_D(x) = \max(1 - v^2, -0.064)$ — negative whenever $|v| > 1$, i.e. on both slow branches of the relaxation oscillation, with global supremum 1 *at* p^* .

6 Numerical results

All results in this section were computed in report R002 (frozen code snapshot, 522 optimisation runs, 18 gates) and R003 (bound evaluation, 8 gates); the figures and tables are copied here byte for byte and every number quoted in the text is regenerated from the copied result files (§8). The protocol is that of §2.3. **Read this:** the constants reported here are *measurements along the true orbit at p^* over R002's probe box*. They are not the suprema over $X \times P$ that (A0) defines, because $X \times P$ has never been instantiated; and the optimisers are unconstrained, so the minimisers reported here are the objects of the theory only under hypothesis (R) of Remark 1. Both gaps are recorded as open premise checks P1 and P2 in §8.

6.1 The bounds against measurement

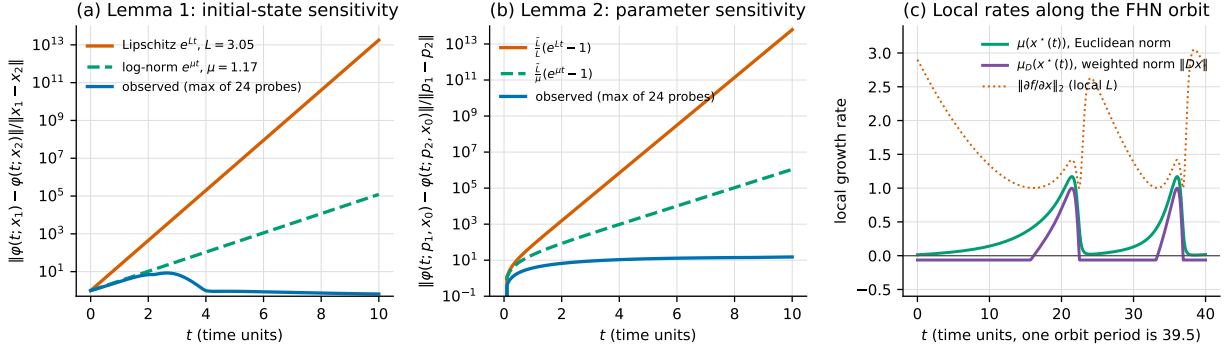


Figure 7: Lemmas 5 and 6 on FHN. (a) Amplification $\|\varphi(t; p^*, x_1) - \varphi(t; p^*, x_2)\|/\|x_1 - x_2\|$, largest value over 24 probes (x_1 a datum at four phases of the orbit, $x_2 = x_1 + 10^{-3}u$ for six random unit directions; blue), against the draft’s bound e^{Lt} (vermilion) and the revised $e^{\mu t}$ (green dashed). (b) The same for the parameter sensitivity $\|\varphi(t; p_1, x_0) - \varphi(t; p_2, x_0)\|/\|p_1 - p_2\|$ ($p_2 = p^* + 10^{-3}u$, $u \in \mathbb{R}^{20}$) against $\frac{\tilde{L}}{L}(e^{Lt} - 1)$ and $Q(t) = \frac{\tilde{L}}{\mu}(e^{\mu t} - 1)$. Both bounds hold at every probe; the revised one is 10^8 closer at $t = 10$ but still grows, while the measurement peaks at 8.5 near the fast jump and then contracts because the orbit is attracting. (c) Local rates along one FHN period (39.5 time units): the local Lipschitz constant $\|\partial f/\partial x\|_2$ (dotted) never drops below 1; the Euclidean logarithmic norm $\mu(x^*(t))$ (green) is near zero on the slow branches and reaches 1.17 only during the two fast jumps; in the weighted norm the local μ_D (purple) is negative for 75% of the period. All panels are evaluated at p^* along the true orbit, i.e. they probe the constants, they do not certify the suprema of (A0). Inherited from R003 Figure 1 (figures/R003_fig01_bounds.pdf).

Table 1: Draft versus revised bounds on FHN, evaluated at $t = \Delta T$, against the largest value observed over the 24 probes of Figure 7 for $t \leq \Delta T$. The two bound columns are the Lemma 5 and Lemma 6 factors with the orbit-measured constants $L = 3.05$, $\mu = 1.17$, $\tilde{L} = 10.56$; the “observed peak” column is a finite sample of probes, not a supremum. Inherited from R003 Table 1.

$t = \Delta T$	Lemma 1: $\ \varphi(t; x_1) - \varphi(t; x_2)\ /\ x_1 - x_2\ $			Lemma 2: $\ \varphi(t; p_1, x_0) - \varphi(t; p_2, x_0)\ /\ p_1 - p_2\ $		
	original $e^{L\Delta T}$	proposed $e^{\mu\Delta T}$	observed peak	original $\frac{\tilde{L}}{L}(e^{L\Delta T} - 1)$	proposed $\frac{\tilde{L}}{\mu}(e^{\mu\Delta T} - 1)$	observed peak
1	21.1	3.23	2.93	69.8	20.1	3.7
2	447	10.4	6.92	1.5×10^3	84.9	6.7
5	4.2×10^6	349	8.48	1.5×10^7	3.1×10^3	12.0
10	1.8×10^{13}	1.2×10^5	8.48	6.2×10^{13}	1.1×10^6	15.1

Table 2: Per-window *linearised endpoint* exponents e^{Λ_k} computed from $J_f(x^*(t); p^*)$ along the true FHN orbit ($\Delta t = 1$, windows aligned to t_0), Corollary 7: worst and median window, in the Euclidean norm and in the weighted norm $\|Dx\|$; “contracting” is the share of windows with $\Lambda_k^D < 0$. **Read this:** these are *not* rigorous bounds for the propositions. They are endpoint exponents of one linearised pair, so the transition bound of (20) satisfies $M_k \geq \max\{1, e^{\Lambda_k}\}$, strictly wherever $\mu(\cdot)$ changes sign inside the window, and Proposition 3 needs M_k — in fact the envelope $\mathcal{M}$ of (55) — not e^{Λ_k} ; see Remark 5 for the four independent reasons the substitution is not licensed. Weighted values are statements in $\|\cdot\|_D$; the Euclidean counterpart is larger by at most $\text{cond}(D) = 3.54$ (Remark 6). Inherited from R003 Table 2.

ΔT	windows	a priori	per-window e^{Λ_k} , Euclidean		per-window $e^{\Lambda_k^D}$, weighted norm		
		$e^{\mu\Delta T}$	worst	median	worst	median	contracting windows
1	100	3.23	3.10	1.16	2.60	0.94	74 %
2	50	10.4	8.56	1.41	5.89	0.88	70 %
5	20	349	34.78	3.75	8.04	0.73	55 %
10	10	1.2×10^5	138.79	29.25	5.89	3.20	30 %

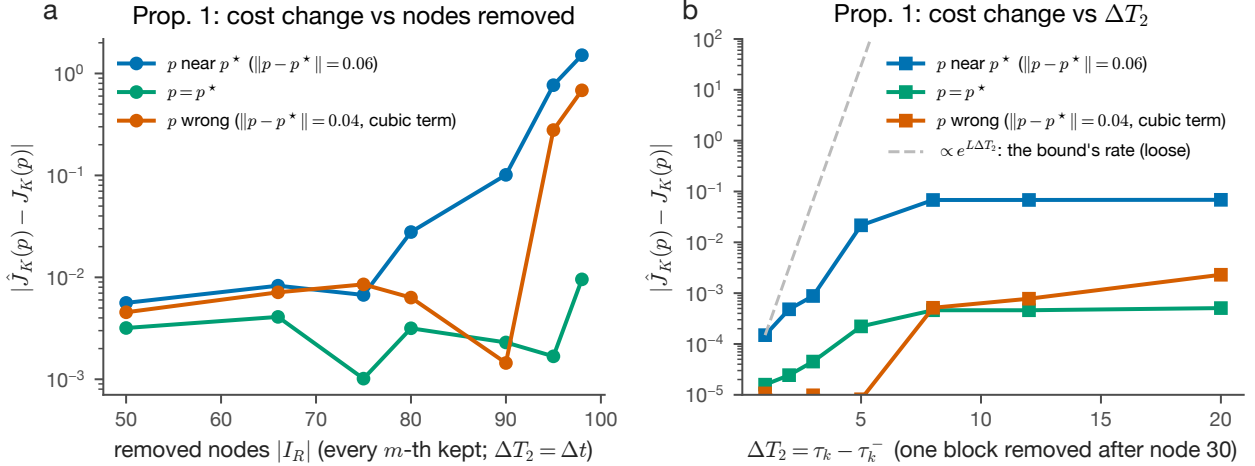


Figure 8: Proposition 3: node removal changes the cost by a bounded amount. Both panels start from the finest partition ($\kappa = 1$, every datum a node, $K = 100$, $\gamma = 0$) and *remove* nodes, so both are genuine sub-partitions and both are within the scope of Proposition 3 — unlike the stages of the κ -sweep (§2.3). (a) Keep every m -th node, so $|I_R| = 100 - \lceil 100/m \rceil$ nodes are removed; the fine window length is $\Delta T_1 = \Delta t$ *throughout*, and the quantity that grows with m is the removed-node offset $\Delta T_2 = (m - 1)\Delta t$. (b) Remove one contiguous block of b nodes after node 30, so $|I_R| = b$, $\Delta T_1 = \Delta t$ and $\Delta T_2 = b\Delta t$ grows. Shown for $p = p^*$ (green), a vector 0.06 from p^* on six coefficients (blue) and p_{wrong} of Figure 2 (vermillion). $|\hat{J} - J|$ grows with $|I_R|$ and with ΔT_2 , and much faster away from p^* , as the proposition says; the rate $e^{L\Delta T_2}$ of the draft (grey dashed in (b)) is far steeper than the observed growth, which saturates once the removed block spans a full fast excursion. **Read this:** the producing script `02_concept.jl` and the R002 caption label the stride experiment’s growing quantity $\Delta T_1 = m\Delta t$ and its fixed quantity $\Delta T_2 = \Delta t$; under the definitions of §2 those two are *swapped*, and the CSV column `DeltaT1` holds the *coarse* window length ($m\Delta t$ in panel (a), $(b + 1)\Delta t$ in panel (b)), not ΔT_1 . The caption above uses the manuscript’s definitions; the CSV column names are not corrected in place, since the result file is frozen. Inherited from R002 Figure 10.

6.2 The cost landscape as the window grows

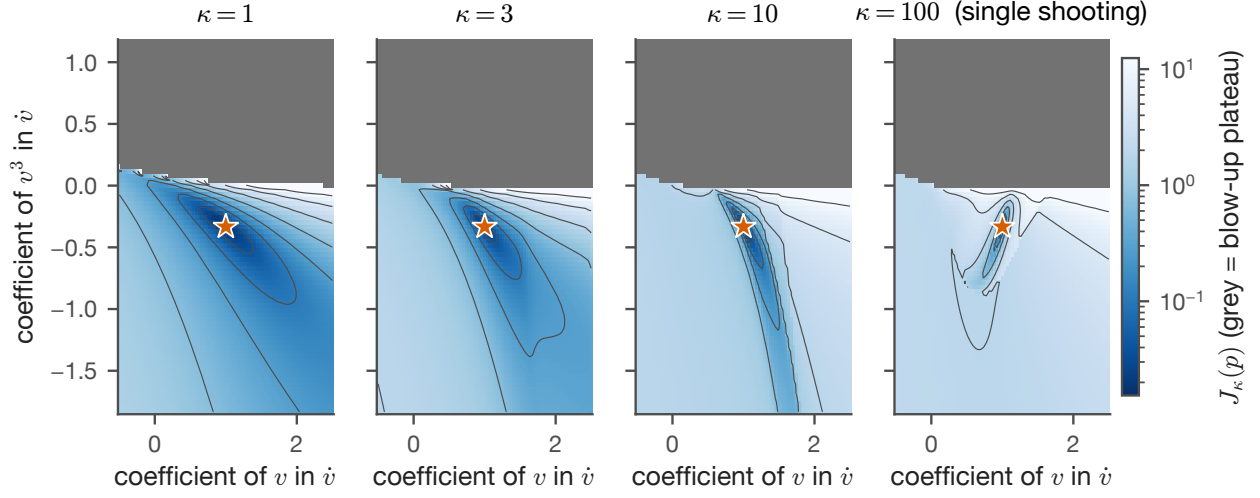


Figure 9: Two-dimensional slices of J_κ in the (v, v^3) coefficient plane of the $\dot{v}$ equation. All other coefficients at their true values, $x_0 = y_0$, $\gamma = 0.05$; star = truth $(1, -1/3)$; grey = blow-up plateau; colour is $\log J$. As κ grows from 1 to 100 the basin around the truth narrows by an order of magnitude and the plateau expands: any positive cubic coefficient makes $\dot{v} \sim cv^3$ blow up in finite time, and at $\kappa = 100$ even v -coefficients above 1.6 do. This is the geometric reason the control arm fails at large κ . The grey region is where the *implemented* loss returns its flat penalty; no inference about P or ∂P is drawn from it (Remark 3(iv)). Inherited from R002 Figure 11.

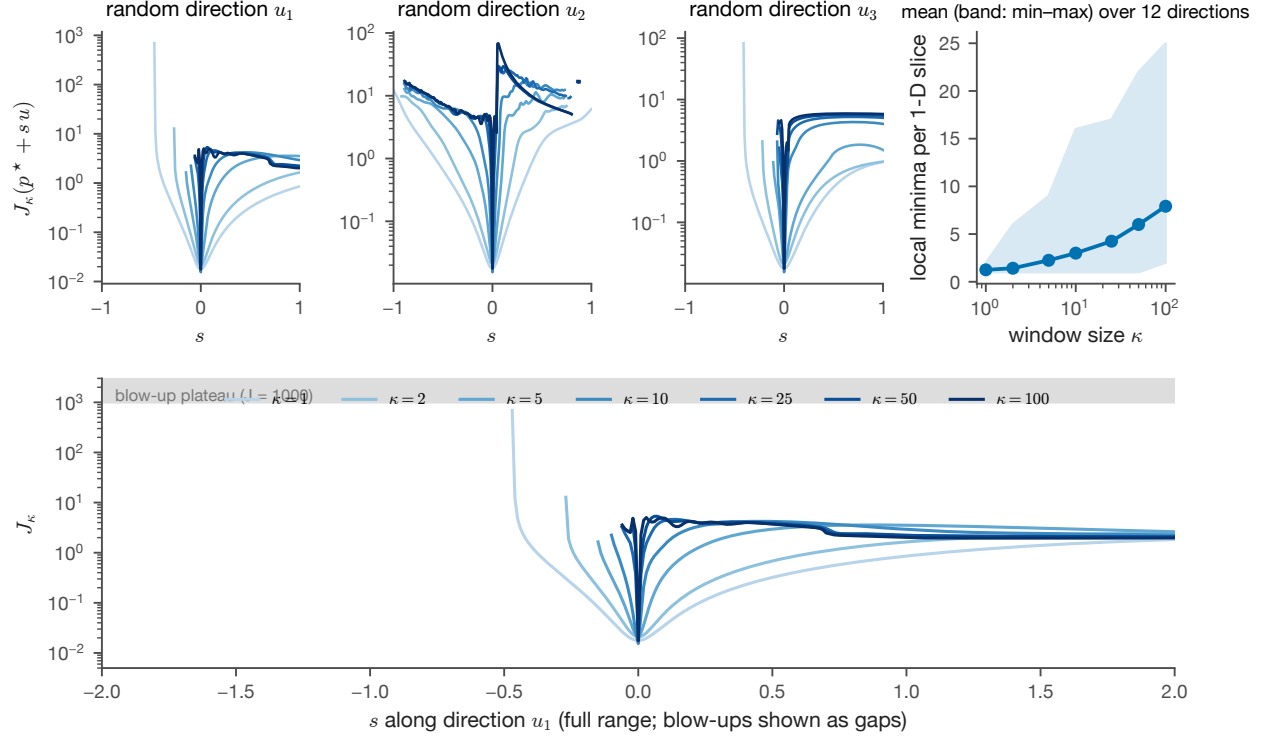


Figure 10: One-dimensional slices along random directions and the number of local minima. Top: $J_\kappa(p^* + s u_d)$ for three random unit directions $u_d \in \mathbb{R}^{20}$ (light to dark = $\kappa = 1, 2, 5, 10, 25, 50, 100$; gaps = plateau); top right: local minima per slice, mean over 12 directions (band = min-max), rising from 1.25 at $\kappa = 1$ to 7.9 at $\kappa = 100$. Bottom: the full $s \in [-2, 2]$ range for direction 1. Larger windows sharpen the minimum but add spurious minima and cut the finite region down to $|s| \lesssim 0.5$. Inherited from R002 Figure 13.

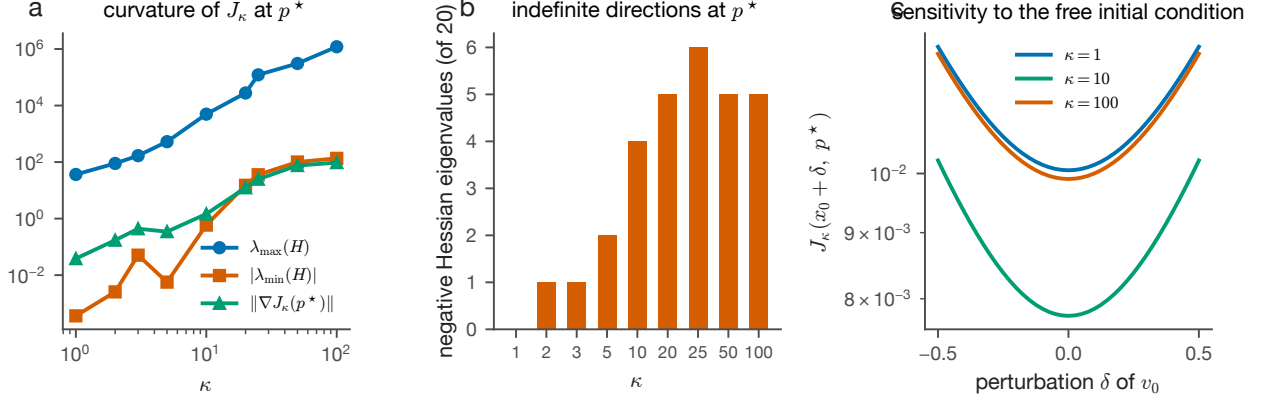


Figure 11: Curvature of the data term at p^* versus window size. (a) Largest and smallest eigenvalues of the Hessian of the *data term* ($\gamma = 0$) at p^* in the $n_p = 20$ parameters, and the data-term gradient norm: $\lambda_{\max}$ grows from 36 ($\kappa = 1$) to 1.2×10^6 ($\kappa = 100$) while $\|\nabla J_\kappa(p^*)\|$ grows from 0.04 to 96; with noisy data the truth is not a stationary point and drifts away from the minimiser as κ grows. (b) Number of negative eigenvalues: the noisy *data term* is indefinite at p^* for every $\kappa \geq 2$ (up to 6 of 20 directions), so the strong-convexity premise of Proposition 2 and Theorem 1 cannot hold near p^* ; note that the optimised objective additionally carries the $\gamma = 0.05$ penalty, whose Hessian is not included here (Remark 3(iii)). (c) The dependence on the free variable x_0 : by the dummy-variable argument of §2.3 this is exactly the parabola $\|x_0 - y_0\|^2/N$, identical at every κ ; it is a property of the objective’s parametrisation, not of the landscape. Inherited from R002 Figure 14.

6.3 Guess propagation versus no propagation: the headline experiment

Protocol: 8 optimiser seeds $\times$ 3 arms $\times$ the 16-stage FULL schedule, Nelder–Mead with 2500 iterations per stage, $\gamma = 0.05$, $S = 10$, flat penalty. Table 3 gives the seed medians and the paired statistics per stage; Figures 12–15 show them.

Table 3: Headline sweep, seed medians per stage: cost J , parameter error $\|p - p^*\|$ and recovery score for GP, best-GP and the control; the mean paired difference GP – control in $\|p - p^*\|$ with its 95 % bootstrap CI over the 8 seeds (resampling unit: the optimiser seed); the number of seeds in which GP is strictly better; the number of GP seeds on the plateau. Inherited from R002 Table T1 (tables/T1_headline.tex).

κ	J GP	J best-GP	J control	$\ p - p^*\ $ GP	$\ p - p^*\ $ best-GP	$\ p - p^*\ $ control	score GP	score control	GP–control	CI lo	CI hi	#GP better	#GP on plateau
1	0.02569	0.02569	0.02569	0.8641	0.8641	0.8641	0.4	0.4	0	0	0	0	3
2	0.02081	0.02081	0.03726	0.3803	0.3803	1.314	0.75	0.275	-0.7724	-1.112	-0.4196	7	1
3	0.01864	0.019	0.05955	0.3748	0.3893	1.466	0.75	0.225	-0.8679	-1.309	-0.4865	7	1
4	0.01549	0.01577	0.09718	0.3325	0.3615	1.51	0.775	0.25	-0.9232	-1.248	-0.5652	7	1
5	0.01906	0.01889	0.1259	0.2218	0.221	1.543	0.775	0.2	-1.133	-1.654	-0.6627	7	1
6	0.02114	0.02054	0.2428	0.3441	0.4522	1.716	0.55	0.2	-1.159	-1.526	-0.7401	7	1
8	0.0161	0.0154	0.3185	0.3593	0.3964	1.59	0.675	0.275	-1.113	-1.36	-0.7559	7	1
10	0.01406	0.01438	0.3086	0.2995	0.3347	1.737	0.8	0.2	-1.234	-1.634	-0.7787	7	1
12	0.01556	0.0155	0.6561	0.278	0.3312	1.622	0.8	0.25	-1.163	-1.415	-0.7988	7	1
15	0.01414	0.01409	0.5398	0.3348	0.3402	1.574	0.725	0.275	-1.109	-1.327	-0.7739	7	1
20	0.01436	0.01459	0.5358	0.2425	0.2709	1.612	0.8	0.25	-1.236	-1.547	-0.8214	7	1
25	0.01345	0.01362	0.861	0.3113	0.3455	1.534	0.85	0.25	-1.111	-1.401	-0.7455	7	1
33	0.01375	0.01373	0.8233	0.3372	0.3758	1.498	0.825	0.225	-0.9927	-1.358	-0.5627	7	1
50	0.01327	0.01334	1.334	0.3353	0.3393	1.807	0.8	0.2	-1.63	-2.426	-0.9189	7	1
75	0.0128	0.01289	1.392	0.3223	0.3179	1.793	0.8	0.225	-1.167	-1.522	-0.7642	7	1
100	0.01273	0.01288	1.506	0.3216	0.3105	1.649	0.8	0.25	-1.138	-1.426	-0.7719	7	1

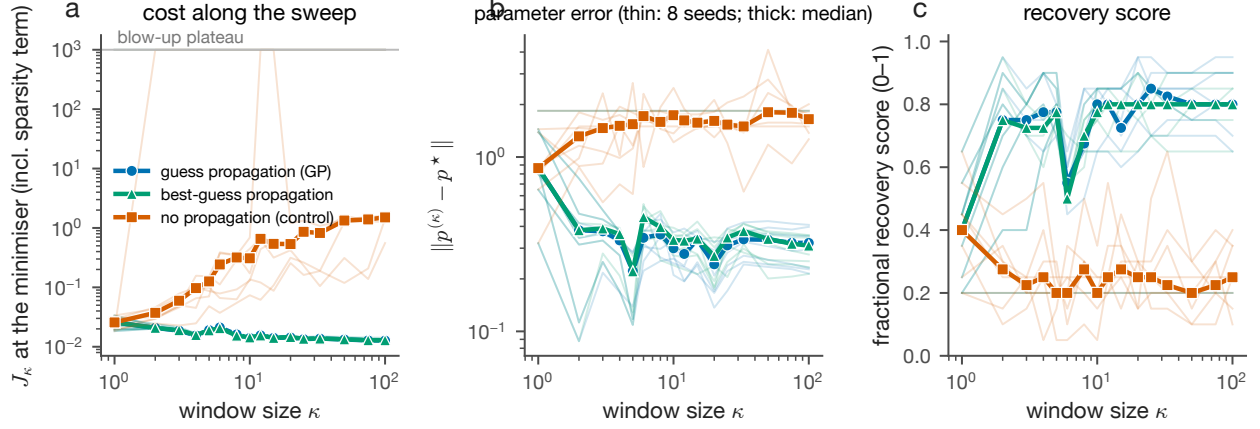


Figure 12: Guess propagation versus no propagation, 8 seeds. (a) Cost, (b) parameter error and (c) recovery score at the minimiser of each stage; thin lines = seeds, thick = median; blue circles GP, green triangles best-GP, vermilion squares control. All arms coincide at $\kappa = 1$ (same start). From $\kappa = 2$ on, GP drops to $\|p - p^*\| \approx 0.22$ – 0.38 and ends single shooting at 0.32 (score 0.80), whereas the control climbs back to the seed distance (1.65 at $\kappa = 100$; 25% of its seeds end on the plateau, the others in spurious minima). One GP seed starts on the plateau at $\kappa = 1$ and never moves (flat line at 1.84). The parameter error is not monotone: it is smallest at $\kappa = 5$ (0.22) and drifts up to 0.32 while the cost keeps falling, the noise-induced bias of Figure 11. Inherited from R002 Figure 15.

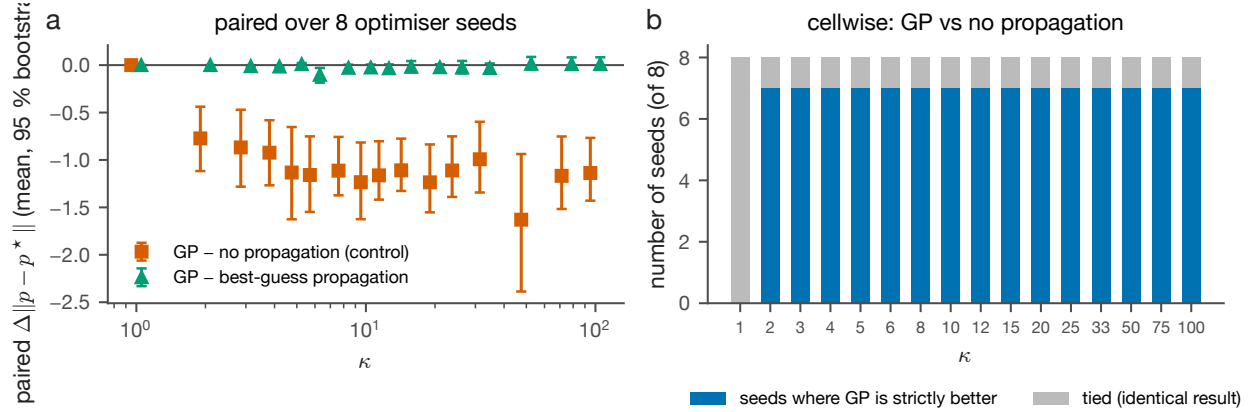


Figure 13: Paired comparison. (a) Mean paired difference in $\|p - p^*\|$ (GP minus control, vermilion; GP minus best-GP, green) with 95 % bootstrap CIs over the 8 seeds: the GP–control difference is -1.14 $[-1.43, -0.77]$ at $\kappa = 100$ and its CI excludes zero at every $\kappa \geq 2$; GP and best-GP are statistically indistinguishable. (b) Cellwise counts: GP is strictly better than the control in 7 of 8 seeds at every $\kappa \geq 2$ and tied in the eighth (the plateau seed). Inherited from R002 Figure 16.

single-shooting simulation from the minimiser found at each window size (optimiser seed 2); grey: truth and data

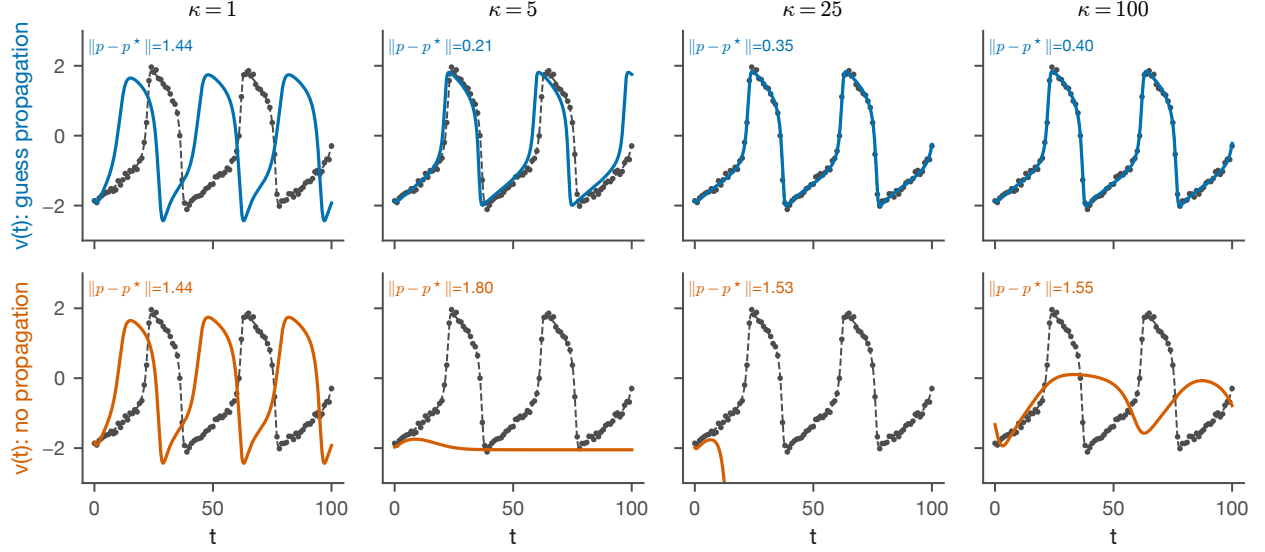


Figure 14: Filmstrip of the sweep (seed 2). Single-shooting simulation of $v(t)$ from the minimiser returned at $\kappa = 1, 5, 25, 100$ (columns) for GP (top, blue) and the control (bottom, vermilion); grey dashed = truth, dots = data; the parameter error is printed in each panel. The GP fit tracks the data at every stage; the control loses the trajectory as soon as the windows are long. Inherited from R002 Figure 18.

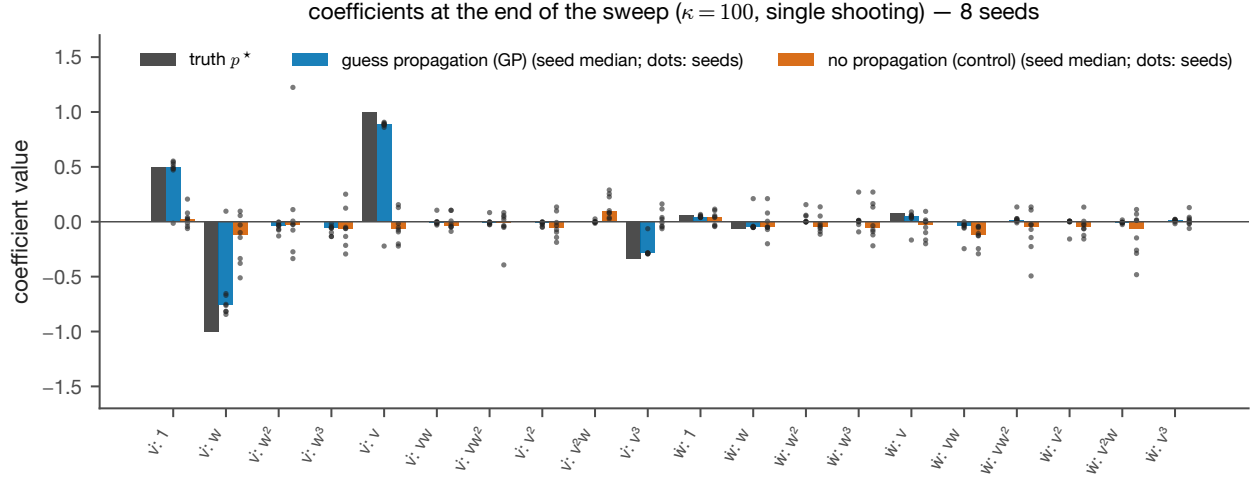


Figure 15: Coefficients at the end of the sweep ($\kappa = 100$). Truth (grey), GP seed-median (blue) and control seed-median (vermilion) with the 8 seeds as dots, in library order. GP recovers the support (7 non-zeros, no false positive in the median) with a shrinkage of the w coefficient of $\dot{v}$ due to the smooth- ℓ_1 term at $\gamma = 0.05$ — the bias bounded, under (C1)–(C2), by (25); the control's median is near zero everywhere because its seeds scatter. Inherited from R002 Figure 20.

6.4 Variations

Table 4 summarises the variation experiments of R002 (SHORT schedule unless stated; the schedule rows change the number of stages by design, and the optimiser rows do different work per iteration,

so those comparisons are not at equal budget). Three readings matter for the method. Removing few nodes per step matters: DENSE ends at 0.30, COARSE at 0.52 and JUMP at 0.76 (Figure 16), which is the design rule of Proposition 3 in practice. The graded blow-up penalty removes the “stuck on the plateau” failure (GP 0.27, no seed on the plateau). With the exact gradient, BFGS with guess propagation is the best and cheapest arm (0.21, score 0.95; Figure 17), while BFGS without propagation fails because the gradient vanishes on the plateau. GP is robust up to 10 % noise (0.31) and fails at 20 % (1.88); the control fails even at zero noise.

Table 4: Selected variation experiments (seed medians at $\kappa = 100$). Derived from R002 Table T2 (`external_data/tables/T2_variations.csv`); the full table has 55 rows. Schedule and optimiser rows are *not* at equal budget, as the text says. Inherited from R002.

variation (arm)	seeds	$\ p - p^*\ $ at $\kappa = 100$	score	seeds on plateau
schedule DENSE (44 stages)	8	0.30	0.80	12 %
schedule COARSE (4 stages)	8	0.52	0.55	12 %
schedule JUMP (2 stages)	8	0.76	0.42	12 %
graded blow-up penalty, GP	8	0.27	0.82	0 %
graded blow-up penalty, control	8	1.63	0.30	0 %
noise 0 %, GP	6	0.23	0.80	17 %
noise 10 %, GP	6	0.31	0.62	17 %
noise 20 %, GP	6	1.88	0.23	17 %
noise 0 %, control	6	1.63	0.25	33 %
$\gamma = 0$, GP	6	0.81	0.38	17 %
$\gamma = 0.05$, GP	6	0.22	0.80	17 %
$\gamma = 1$, GP	6	1.01	0.60	17 %
BFGS, GP	8	0.21	0.95	12 %
BFGS, control	8	1.65	0.23	62 %
Nelder–Mead 1000 iterations, GP	4	0.61	0.57	25 %
Nelder–Mead 5000 iterations, GP	4	0.20	0.90	25 %
integrator sub-steps $S = 1$, GP	4	0.50	0.75	0 %
integrator sub-steps $S = 5$, GP	4	0.29	0.82	25 %

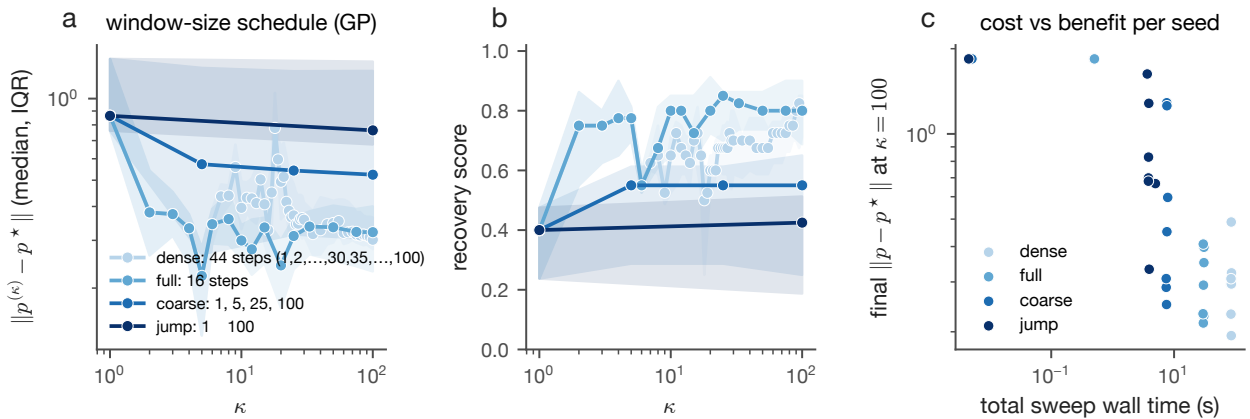


Figure 16: Window-size schedule. Parameter error along the sweep for the DENSE (44 stages), FULL (16), COARSE (4) and JUMP (2) schedules, 8 seeds; removing few nodes per step matters. Note that the fully nested schedules (COARSE, JUMP) are the ones Proposition 3 covers stage by stage, and they are the ones that perform *worst*; the schedules that perform best are covered only through the common refinement of Corollary 6 (§2.3). Inherited from R002 Figure 21.

optimiser: derivative-free vs AD-gradient (solid: GP, dashed: no propagation)

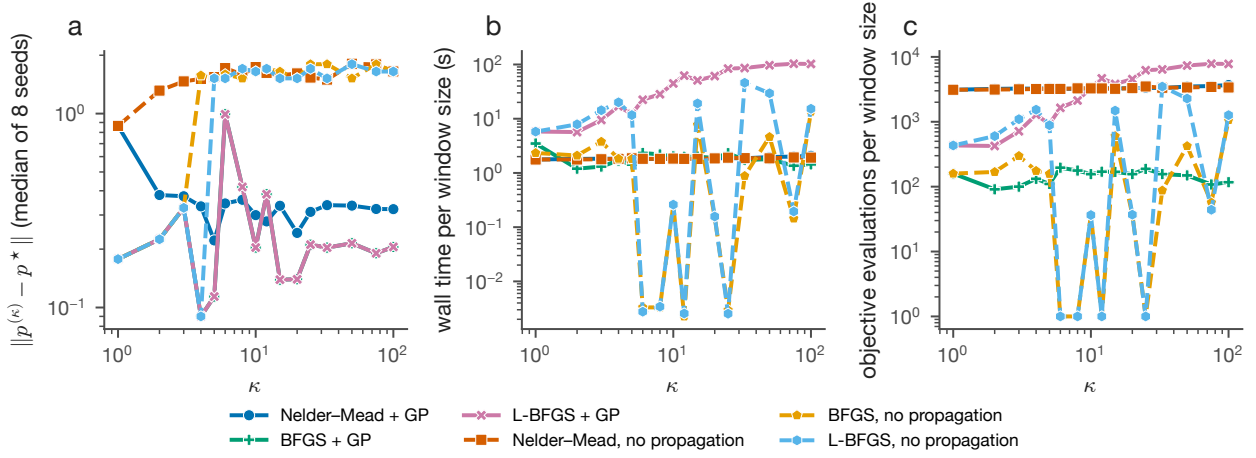


Figure 17: Optimisers. Nelder–Mead versus BFGS and L-BFGS (automatic-differentiation gradient through the windowed loss), with and without guess propagation; wall time per stage is machine-dependent and only comparative. BFGS with propagation is the best and cheapest arm; without propagation the gradient methods die on the plateau. Inherited from R002 Figure 25.

6.5 Other systems

The implementation is dimension- and degree-agnostic; Table 5 and Figures 18–19 apply the same sweep to Lotka–Volterra ($\dot{x} = x - 0.5xy$, $\dot{y} = -0.8y + 0.3xy$; quadratic library, 12 coefficients, $N = 101$, $\Delta t = 0.25$, 5% noise, Nelder–Mead, 6 seeds) and Lorenz ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$; quadratic library, 30 coefficients, $N = 101$, $\Delta t = 0.02$, 2% noise, 4 seeds; BFGS and Nelder–Mead). GP recovers Lotka–Volterra (0.29, score 0.83; the control ends at 1.41 with 83% of its seeds on the plateau) and, with BFGS, Lorenz (3.9 on coefficients of size up to 28, score 0.83); GP with Nelder–Mead does not converge in 33 dimensions (18.1, score 0.23), so a derivative-free sweep does not scale.

Table 5: Other systems, seed medians at $\kappa = 100$. Inherited from R002 Table T5.

system	optimizer	arm	n_seeds	$\ p - p^*\ $ ($\kappa=100$)	score	plateau frac
lorenz	bfgs	propagate	4	3.852	0.8333	0
lorenz	bfgs	reset	4	31.95	0.03333	1
lorenz	nm	propagate	4	18.11	0.2333	0
lorenz	nm	reset	4	31.95	0.03333	1
lv	nm	propagate	6	0.29	0.8333	0
lv	nm	reset	6	1.408	0.2917	0.8333

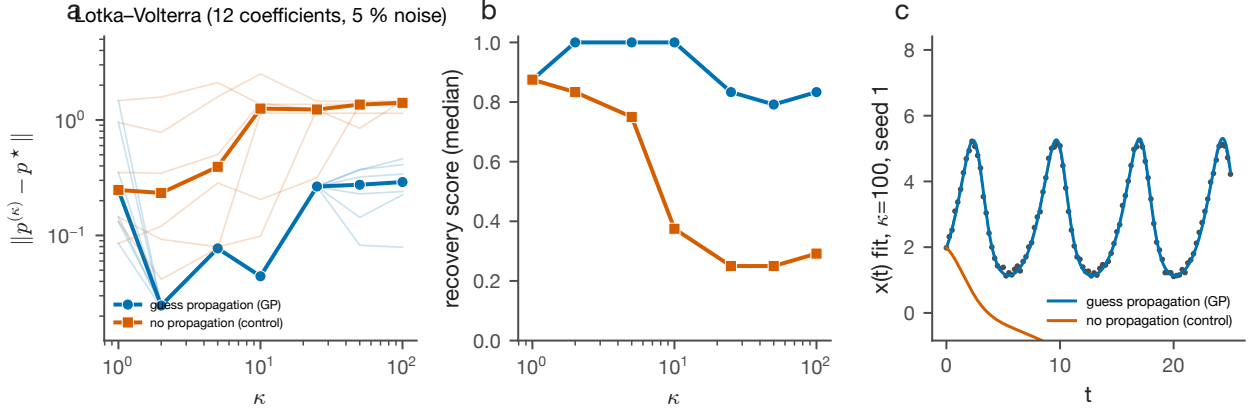


Figure 18: Lotka–Volterra. (a) Parameter error along the sweep (thin: 6 seeds, thick: median), (b) score, (c) the $x(t)$ fit at $\kappa = 100$ for seed 1. Inherited from R002 Figure 35.

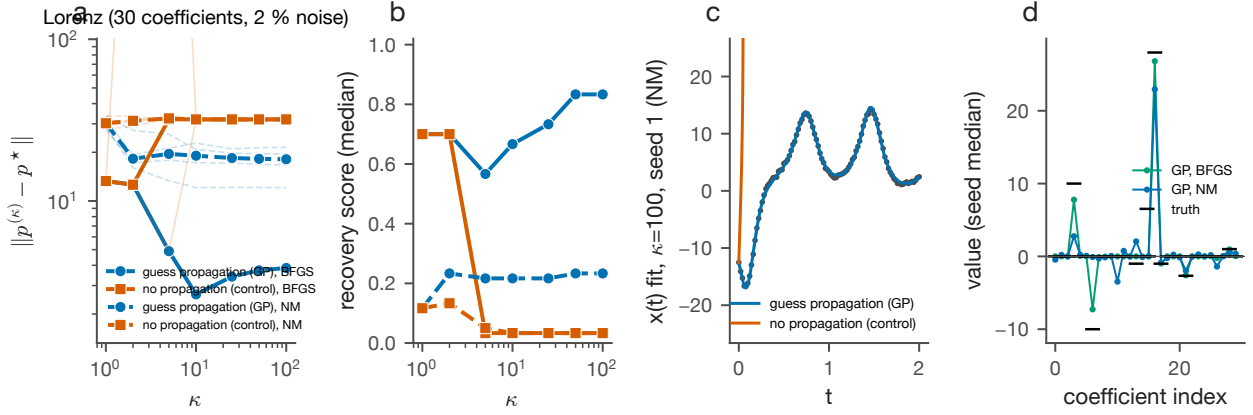


Figure 19: Lorenz (chaotic, 30 coefficients). (a) Parameter error along the sweep for BFGS (solid) and Nelder–Mead (dashed), GP (blue) and control (vermillion), 4 seeds; (b) score; (c) the $x(t)$ fit at $\kappa = 100$, seed 1; (d) seed-median GP coefficients versus truth. Inherited from R002 Figure 36.

6.6 Empirical diagnostics for the premises of the theory

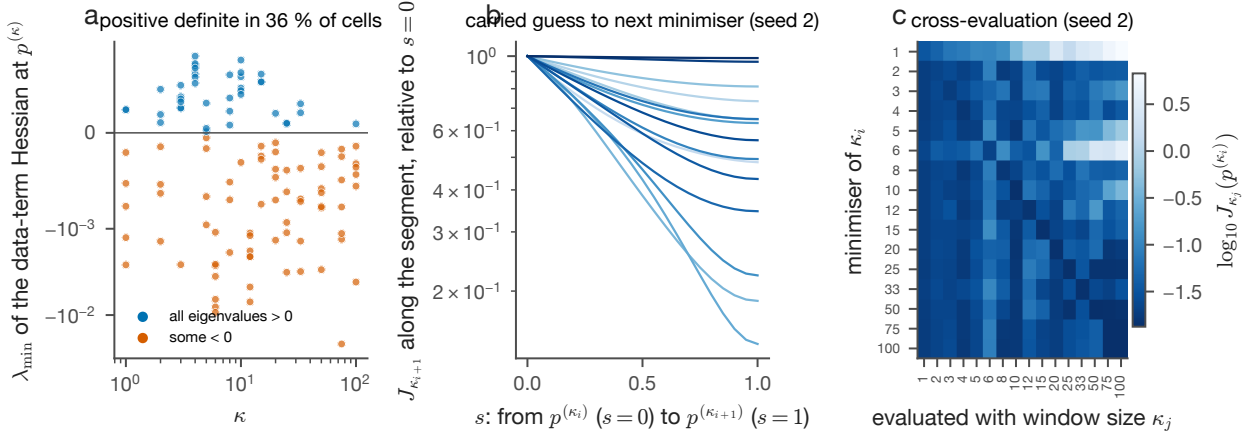


Figure 20: Empirical diagnostics at the minimisers. (a) Smallest eigenvalue of the *data-term* Hessian ($\gamma = 0$) at the GP minimiser of the *penalised* objective ($\gamma = 0.05$) returned by Nelder–Mead at every stage, 7 seeds (plateau seed excluded; blue = positive definite, vermilion = at least one negative eigenvalue): the returned point is positive definite in only 36% of the (seed, κ) cells, with on average less than one of 20 eigenvalues slightly negative elsewhere. This is an *empirical diagnostic*: it does not check strong convexity of the optimised objective on a neighbourhood, since it evaluates a different function (no penalty), at a single point rather than on a neighbourhood, at an iterate whose stationarity is not verified. The premise (A3) of Proposition 2 and the strong-convexity hypothesis of Theorem 1 are therefore *not verified* at these iterates (the BFGS iterates would be the place to test them). (b) The next stage’s cost along the straight chord from $p^{(\kappa_i)}$ to $p^{(\kappa_{i+1})}$ (seed 2, 21 sampled points), relative to its value at the carried guess: it decreases monotonically along every sampled chord, which is a *necessary condition for*, not a proof of, the carried guess lying in the basin of the next minimiser. (c) Cross-evaluation: the minimiser of κ_i evaluated at every other κ_j ; the near-diagonal structure shows that a minimiser is good only for nearby window sizes, which is why the schedule must be gradual. Inherited from R002 Figure 37.

7 Discussion

The analysis and the experiments agree on the mechanism, with the analysis conditional throughout. Short windows bound the amplification of the initial-condition error by $e^{\mu\Delta T}$ (Lemma 5, Proposition 1), which keeps the landscape smooth (Figures 5, 10) at the price of weak discrimination between nearby parameters (Figure 4c); removing few nodes at a time moves the cost, and hence the minimiser, by a bounded amount (Proposition 3, Theorem 1, Figures 8, 20b), so — under the hypotheses of Corollary 5 — the minimiser of the restricted coarse problem stays near the carried guess while the windows grow into single shooting (Figures 12–13). The revised constants are of the right kind (they can saturate, and for the true FHN field at p^* the upper one is global in x), and along the orbit the linearised per-window endpoint exponents of Corollary 7 are within a factor of the measurement; the remaining gap to the observed plateau is intrinsic to worst-case analysis.

Four limitations are visible in the data and in the hypotheses. The strong-convexity premise is not verified at the iterates a derivative-free optimiser returns (Figure 20a), so Theorem 1 explains the observed behaviour rather than certifying it; with noisy data the truth is not a stationary point of any J_κ and the parameter error is non-monotone along the sweep (Figures 11, 12); the flat blow-

up penalty can trap a start on the plateau at the very first stage, a defect of the penalty rather than of the propagation that the graded penalty removes (Table 4); and, most consequentially, the two premise checks of §8 are unevaluated, so every statement of §3–§4 is conditional on an admissible region and an event this report has not instantiated, and on a reconnection hypothesis relating the constrained minimisers of the theory to the unconstrained iterates reported here. Gradient optimisers with guess propagation are the recommended configuration (Table 4, Figure 19).

8 Provenance and gates

This report contains no new numerics. Its 20 figures are byte-identical copies of R002 Figures 1–6, 10, 11, 13–16, 18, 20, 21, 25, 35–37 and R003 Figure 1 (prefixed by their source report in `figures/`); the three hand-drawn figures of the draft (`shooting_partition.png`, `multiple_shooting.png`, `comparison_shooting_nodes.png`) and R002’s schematic Figure 7 (drawn with the draft’s e^{Lt} envelope) are retired. Tables 1, 2, 3 and 5 are verbatim copies; Table 4 and every number in the prose are regenerated by `analysis/make_tables.py` from the R002/R003 result files copied into `external_data/` (SHA-256 in `INPUTS.md`). `analysis/verify_R004.py` writes `analysis/results/gates_summary.json`:

- G1: every file in `external_data/` matches its hash in `INPUTS.md`, and every hash equals that of the file in the source report’s folder (byte identity with R002/R003).
- G2: every figure in `figures/` is byte-identical to its source figure in R002/R003, and there are exactly 20 of them.
- G3: every figure referenced by `report.tex` exists, the references are 20 distinct files, and no retired figure is referenced.
- G4: the verbatim tables in `tables/` are byte-identical to their sources.
- G5: `tables/numbers.tex`, `numbers.json` and `T2_compact.tex` regenerate byte-identically from `external_data/`.
- G6: the prose numbers agree with R002’s own `table_numbers.json` and R003’s `summary.json` to the printed precision.
- G7: the revised *lemma* constants never exceed the draft’s with the FHN numbers: $\mu \leq L$, and $e^{\mu s} \leq e^{Ls}$ and $\frac{\tilde{L}}{\mu}(e^{\mu s} - 1) \leq \frac{\tilde{L}}{L}(e^{Ls} - 1)$ at $s \in \{1, 2, 5, 10\}$. **Read this:** this gate compares Lemma 5 and Lemma 6 factors only. It does *not* compare the Proposition 3 product over $(\Delta T_1, \Delta T_2)$ pairs; an earlier version of this bullet said it did, which was a misdescription of the gate that exists.
- G8: no probe violates the revised Lemma 5/6 bounds at any of the 2×2424 probe points of R002 (recomputed from the copied CSVs): the state ratios lie in $[e^{\mu_- t}, e^{\mu t}]$ and the parameter ratios below $Q(t)$. **Read this:** this is a finite check, not a verification of the suprema in (A0); the lower-bound half uses the orbit-restricted $\mu_- = -2.96$ and the probes all start on the orbit.
- G9: `report.md` contains every section heading, table and figure caption of `report.tex`.
- G10: `report.pdf` exists, is newer than `report.tex`, and its page count is recorded.

Outcome: see `tables/gates_status.tex` (inlined at build time: 10 of 10 gates pass). Regeneration: `REPRODUCE.md`.

Open theoretical-premise checks. The following two items are *not* gates and are not counted among G1–G10: those ten are mechanical inheritance checks on files, hashes, figures and regenerated numbers, all of which pass, and they are the ones recorded in `gates_summary.json`. The items

below are premises of the *theory* of §2–§4 that this report has not evaluated. They are listed separately so that an unevaluated premise is not mistaken for a broken inheritance check, and so that no future revision “repairs” one by weakening it.

- **P1 (premise check — *not evaluated*).** The closure condition (A1), (9)–(10), has not been evaluated for any κ . Evaluating it requires instantiating a tube radius r_X , recomputing μ and $\tilde{L}$ as suprema over the resulting $X \times P$, fixing $\text{diam}(P)$, and exhibiting $r' > 0$ satisfying both (A1-i) and (A1-ii). None of these has been instantiated numerically. The values $\mu = 1.17$, $\tilde{L} = 10.56$ quoted in §2 were measured over the probe box of R002/R003 at p^* , which is *not* the $X \times P$ of (A0); they are therefore *indicative*, and the order-of-magnitude table computed from them is not a check of (A1). What inserting them into the binding condition (A1-i), $Q(\Delta T_{\max}) \text{diam}(P) < r_X$, shows is that the probe-region constants produce enormous factors for long windows — which *motivates*, but does not *establish*, the infeasibility of a useful admissible region at those window lengths. *Status: unverified, not failed* — a condition evaluated with constants from the wrong region has not been evaluated. Consequently every statement of §3–§4 is conditional on an event, E_X , whose probability this report does not bound numerically, at *every* stage of the schedule.
- **P2 (premise check — *not evaluated*).** The reconnection hypothesis (R) of Remark 1 — the existence of a compact convex P satisfying (A0)–(A1), a convex $U \subseteq P$ containing the reported optimum in its interior, stationarity of J_K there, and m -strong convexity of J_K on U (together with $p^* \in U$ for the variant (R^{p^*}) that Proposition 2 needs) — has not been evaluated. P , X , r_X and $\text{diam}(P)$ are not instantiated anywhere in this report, the optimisers of §6 are unconstrained, and the Hessian panels of Figures 11 and 20 are measurements at particular points, not a strong-convexity certificate on a named convex set. Note also that (R) speaks of the unpenalised, exact-flow J_K , whereas the runs minimise the penalised, discretised (24): stationarity of the penalised numerical objective does not imply the stationarity (R) asserts. Until (R), the event E_X and those two qualifications are established, the displacement, basin-retention and penalty-bias statements apply to the *constrained* minimisers of (16) and not to the numbers reported in §6.